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Analysis of quantitative reconstruction methods for solution to acoustic coefficient inverse problem

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Abstract

This work deals with the resolution of a standard inverse problem, the acoustic coefficient inverse problem. This problem consists in the reconstruction of a spatially distributed wave speed function using measurements of the solution to the acoustic wave equation. This typical problem have been greatly studied in literature with many theoretical results and even more practical techniques of resolution. However, quantitative and qualitative comparisons of the numerical methods to solve the acoustic coefficient inverse problem have not been really considered before work made by L. Beilina, Y. G. Gleichmann and M. J. Grote [8].

In this work, we try to complete and to improve the work done by L. Beilina, Y. G. Gleichmann and M. J. Grote. We present and numerically study the first two presented methods from the article [8], the reconstruction using internal measurements and the optimisation approach of reconstruction using boundary measurements. We quantitatively and qualitatively analyse and compare the various trials to improve the numerical methods.

On the one hand, we consider an analytical reconstruction using internal measurements of the solution to the wave equation. This method rests on the Laplace transform in time of the wave equation to transform the problem into a stationary problem. Then using Finite Element Method, we approximate the acoustic coefficient using the scattered data inside the domain. However, contrary to the presented situation in [8], we do not limit our study to the model with Engquist-Majda first-order absorbing boundary condition on the top and the bottom side of the domain but we propose two other models by changing the boundary conditions. The first model consists in considering homogeneous Neumann boundary conditions and the last one considers absorbing conditions on the whole boundary.

On the other hand, we present an optimisation approach for the reconstruction of the acoustic coefficient using a single measurement on a part of the boundary. To overcome the ill-posedness of the optimisation problem, we use the Tikhonov regularisation term that has been introduced in the reference work. This term is based on an initial guess of the shape of the acoustic coefficient. Thus, we apply a gradient algorithm to numerically solve the inverse problem. As a consequence of a lack of time, the numerical implementation of this method has only been done on the first model but it can be easily transposed to the others.

In addition to the two new models of boundary conditions, this work differs from the original one by a more complete analysis of numerical techniques to implement the methods in practical situations. We present some numerical improvements such as post-processing techniques or smoothing approaches to improve the results. Furthermore, this work presents an adaptation of the proof of a theorem on the well-posedness of the inverse problem to reconstruct the acoustic coefficient using a single measurement on the whole boundary, which has been initially proven for an analogous problem by M. Bellassoued [10].

In the first part, we have been able to effectively reproduce similar results as in [8] and propose a global analysis of the selection of the parameters. In the second part, despite the failure to obtain such satisfying results, we have studied some ideas to potentially improve the method of reconstruction. We hope that this work and the additional fully documented free-access Github [15] made during this internship could open some perspectives to greatly improve this approach.

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Introduction

The development of non-invasive and non-destructive imaging is one of the most challenging issue in many fields. Some applications could be for detection of ground inclusions for geological or archaeological purposes, to detect landmines or for medical imaging. This work has been motivated by this last application. Although there already exists efficacious non-invasive devices such as MRI, X-Ray or Ultrasound imaging, medical imaging can still benefit from the development of more precise imaging.

In a forthcoming article [7], L. Beilina, A. Eriksson and N. Neittaanmäki present a new device for detection of cancerous melanomas. In that article, they develop a quantitative method to reconstruct the dielectric permittivity and the electric conductivity in a domain using scattered data. That work does not only consist in a detection of the presence and the shape of a melanoma but also of the quantitative values of the dielectric permittivity inside the melanoma in order to distinguish malignant melanomas from benign ones. The goal of our work is inspired by that article [7].

In this work, we consider a similar problem of reconstruction of a spatially distributed coefficient in a partial differential equation. We consider more precisely the typical acoustic coefficient inverse problem. In this problem, we consider the 2-dimensional acoustic wave equation with an inhomogeneous wave speed function. The main objective of this work is to quantitatively reconstruct the acoustic coefficient from this equation using scattered data of the solution to the wave equation. The acoustic coefficient is directly linked to the wave speed function as defined in the first part of this article.

In this work, we present two methods of reconstruction of the acoustic coefficient using scattered data. These methods have been developed by Larisa Beilina, Yannick Gleichmann and Marcus Grote in [8]. The first method consists in reconstructing the coefficient using internal measurements of the solution to the acoustic wave equation whereas the second method only requires measurements on a part of the boundary of the domain. The effectiveness of these methods as well as the convergence of the numerical algorithm have been described in several articles [6, 8, 9].

The goal of this paper is to reproduce similar results on these two methods and propose further numerical analysis of the methods including a quantitative selection of the parameters and a new proposals to improve the methods. Furthermore, we study three different boundary conditions and we compare the efficacy of the methods for these models. In [8], the authors have studied the acoustic wave equation with mixed boundary conditions. On the top and the bottom of the square domain, they have considered first order Engquist-Majda boundary conditions and homogeneous Neuman boundary conditions on the other sides. However, in this paper, we also consider a model with only homogeneous Neumann boundary conditions and a model with absorbing boundary conditions on the whole boundary. From our knowledge, such numerical analyses and the new models have not been already studied.

To overcome the lack of real data to test the methods, we study the reconstruction of two test functions as it has been done in [8]. Thus, we generate simulated measurements by solving the forward problems for these test functions and applying the numerical methods. We quantitatively study the error on the reconstructed functions by computing relative quadratic and maximal errors. Additionally, in this work, we consider an implementation of the numerical methods using Matlab language which differs from the original implementation using C++/PETSc in [8]. All the algorithms are available on the Github repository [15].

Moreover, the parameters for each simulated figure are available in Appendix K and can be reached clicking on the figure.

In Section 1, we describe the formalisation of the three studied models in the reconstruction part. The assumptions made in this part are directly linked to similar assumptions made in [8]. We also present common results on the well-posedness of the forward problems whose proofs are inspired by techniques from [5, 14, 16] and are available in Appendix B. In Appendix A, we prove Trace Theorem [1] to ensure that the problems are well-defined. We also present in this section the numerical method of generation of the data for simulations of reconstruction. Partial differential equations are solved using Finite Element Method in space [3]. Finite Element Method is implemented in Matlab using the library Gypsilab made by Matthieu Aussal [4]. Complements on the numerical resolution of the forward problems are available in Appendix C.

In Section 2, we describe the reconstruction method using internal data. This method relies on a Laplace transform of the wave equation to effectively recover the acoustic coefficient. We then analyse the efficacy of the numerical methods and the parameters. Complements to this analysis are also presented in Appendix E. A first approach that has been studied is also described in Appendix D. In Section 2, we also present an effective method to reconstruct the coefficient without the knowledge of the pseudo-frequency, with complements in Appendix F. We conclude this section with tests on the robustness of the method to the noise with more simulations in Appendix G.

In Section 3, we analyse the inverse problem of reconstructing the coefficient using data on a part of the boundary. We first present a theorem on the well-posedness of such inverse problem whose proof is adapted from [10] and is done in Appendix H. Next, we present the optimisation method proposed in [8] to numerically reconstruct the coefficient. Some calculation to formalise the method can be found in Appendix I. Finally, we present some numerical results and improvements of the method. Ineffective trials are also described in Appendix J.

Keywords: *Acoustic Wave Equation, Coefficient Inverse Problem, Finite Element Method, Tikhonov Functional, Gradient Algorithm, Matlab*

MSC Codes: 35L05, 44A10, 46E35, 65K10, 65M32, 65M60

1 Mathematical models and resolution of the forward problems

In this section, we first present the mathematical models studied in this paper. These three models differ by the boundary conditions so we can compare the efficacy of each boundary condition in the case of the reconstruction of a coefficient in the acoustic wave equation. We also present in this section some theorems on the well-posedness of the forward problems in a weak formulation. Finally, we describe a numerical implementation of forward solvers in Matlab to generate data to apply methods of reconstruction below.

1.1 Mathematical models

In this work, we consider the problem of reconstructing a spatially distributed coefficient of the 2-dimensional acoustic wave equation. Usually, we write the non-damped acoustic wave equation as

$$\frac{\partial^2 u}{\partial t^2} - c^2(x)\Delta u = 0.$$

To formalise our model, we make some assumptions and set some notations according to the model in [8]. Let $T > 0$ (eventually $T = +\infty$) and $\Omega =]x_1, x_2[\times]y_1, y_2[$ be an open bounded rectangular domain. The boundary of Ω is divided in four parts as explained in Figure 1. Moreover, let Ω_1 be another open bounded rectangular domain such that $\overline{\Omega_1} \subset \Omega$.

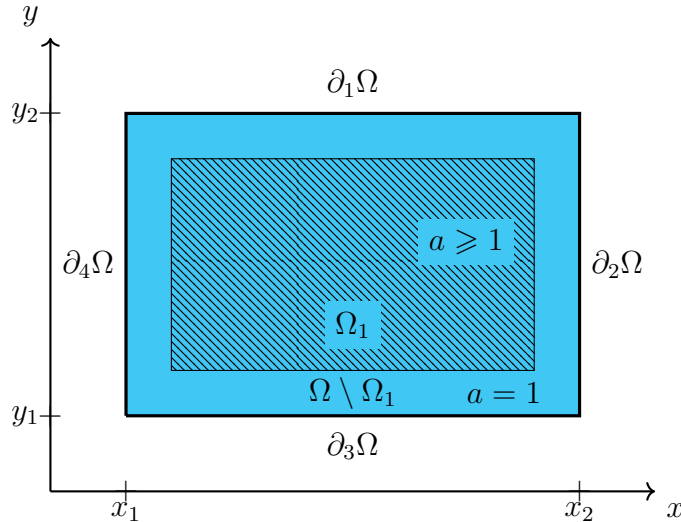


Figure 1: Notations for the domain Ω . Ω is the domain in blue and Ω_1 is the hatched domain.

Let $c(x)$ be the spatially distributed acoustic wave speed function. Let $a(x) := \frac{1}{c^2(x)}$ be the acoustic coefficient of the wave. We consider that $c = 1$ is the reference speed of the material without inclusion. As an inclusion results in a decrease of the wave speed [8], we make the following assumption:

Assumption 1. *We assume that $c \in \mathcal{C}^1(\overline{\Omega})$ (ie. $\mathcal{C}^1$ on a neighbourhood of $\overline{\Omega}$) and it verifies*

$$\forall x \in \overline{\Omega}, \quad 0 < c_0 \leq c(x) \leq 1$$

$$\forall x \in \overline{\Omega} \setminus \Omega_1, \quad c(x) = 1$$

Thus, for all $x \in \overline{\Omega}$, $1 \leq a(x) \leq M$, where $M = \frac{1}{c_0^2}$, and, for all $x \in \overline{\Omega} \setminus \Omega_1$, $a(x) = 1$.

We assume that the material has no internal source. However, we consider an external source coming from the top boundary. In the numerical results, this source is taken as a plane wave. In the general approach of this part, we make the following assumption:

Assumption 2. Let $p \in \mathcal{C}(\mathbb{R}_+)$ be an external source function and $t_1 > 0$. We assume that $p(t) = 0$ for $t = 0$ and for $t \geq t_1$.

Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$ be the initial conditions. In numerical analyses, we take homogeneous initial conditions. We state our three mathematical models which differ from each other by the boundary conditions:

Model 1:

$$\begin{cases} a(x) \frac{\partial^2 u}{\partial t^2}(x, t) - \Delta u(x, t) = 0 & \text{in } \Omega \times]0, T[, \\ u(x, 0) = u_0(x), \quad \frac{\partial u}{\partial t}(x, 0) = v_0(x) & \text{in } \Omega, \\ \frac{\partial u}{\partial n}(x, t) = p(t) & \text{on } \partial_1 \Omega \times]0, t_1[, \\ \frac{\partial u}{\partial n}(x, t) = 0 & \text{on } \partial_1 \Omega \times]t_1, T[, \\ \frac{\partial u}{\partial n}(x, t) = 0 & \text{on } \partial \Omega \setminus \partial_1 \Omega \times]0, T[. \end{cases} \quad (1)$$

Model 2:

$$\begin{cases} a(x) \frac{\partial^2 u}{\partial t^2}(x, t) - \Delta u(x, t) = 0 & \text{in } \Omega \times]0, T[, \\ u(x, 0) = u_0(x), \quad \frac{\partial u}{\partial t}(x, 0) = v_0(x) & \text{in } \Omega, \\ \frac{\partial u}{\partial n}(x, t) = p(t) & \text{on } \partial_1 \Omega \times]0, t_1[, \\ \frac{\partial u}{\partial n}(x, t) = -\frac{\partial u}{\partial t}(x, t) & \text{on } \partial_1 \Omega \times]t_1, T[, \\ \frac{\partial u}{\partial n}(x, t) = -\frac{\partial u}{\partial t}(x, t) & \text{on } \partial_3 \Omega \times]0, T[, \\ \frac{\partial u}{\partial n}(x, t) = 0 & \text{on } \partial \Omega \setminus (\partial_1 \Omega \cup \partial_3 \Omega) \times]0, T[. \end{cases} \quad (2)$$

Model 3:

$$\begin{cases} a(x) \frac{\partial^2 u}{\partial t^2}(x, t) - \Delta u(x, t) = 0 & \text{in } \Omega \times]0, T[, \\ u(x, 0) = u_0(x), \quad \frac{\partial u}{\partial t}(x, 0) = v_0(x) & \text{in } \Omega, \\ \frac{\partial u}{\partial n}(x, t) = p(t) & \text{on } \partial_1 \Omega \times]0, t_1[, \\ \frac{\partial u}{\partial n}(x, t) = -\frac{\partial u}{\partial t}(x, t) & \text{on } \partial_1 \Omega \times]t_1, T[, \\ \frac{\partial u}{\partial n}(x, t) = -\frac{\partial u}{\partial t}(x, t) & \text{on } \partial \Omega \setminus \partial_1 \Omega \times]0, T[. \end{cases} \quad (3)$$

In these problems, we consider two types of boundary conditions. Neumann boundary condition corresponds to a reflective boundary on which a plane wave is reflected. The other condition is called a first-order Engquist-Majda absorbing condition [13] which is commonly used to numerically solve hyperbolic equations when $p(t)$ corresponds to a plane wave [5]. This condition corresponds to an exact absorbing boundary condition in the case of a plane wave with a normal incidence on the boundary (as it is the case for the top and bottom edges in our numerical simulations).

1.2 Well-posedness of the forward problems and Energy estimates

In this paragraph, we state some theorems concerning the well-posedness of the previous models. First, we state an existence-uniqueness theorem for solutions to Model 1 under an additional assumption. Then, we present an energy estimate for solutions to Models 1-3 when $p \equiv 0$ which ensures uniqueness of the solution.

On the one hand, the proof of existence and uniqueness of solutions to Problem (1) relies on Hille-Yosida Theorem and Semi-Group Theory which can be found in [14]. On the other hand, the method used to establish the energy estimate is explained in [5] and relies on the principle presented by O. A. Ladyzhenskaya in [16]. Proofs can be found in Appendix B.

We first add another assumption on p :

Assumption 3. *We assume that there exists a function $v \in \mathcal{C}^2(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^2(\Omega))$ such that*

$$\frac{\partial v}{\partial n}(x, t) = \begin{cases} p(t) & \forall (x, t) \in \partial_1 \Omega \times \mathbb{R}_+, \\ 0 & \forall (x, t) \in \partial \Omega \setminus \partial_1 \Omega \times \mathbb{R}_+. \end{cases}$$

The normal derivative in Assumption 3 exists according to the results from Appendix A. This assumption requires some smoothness on p . However, we have not been able to find sufficient assumptions on p to ensure this condition in practical situations.

Thus, from these assumptions, we can state the existence and uniqueness of a weak solution to Problem (1).

Theorem 1 (Existence-Uniqueness of a weak solution to Problem (1)). *Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$.*

Assuming that Assumptions 1, 2 and 3 hold, then there exists a unique $u \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$ such that

$$u(0) = u_0, \quad \frac{\partial u}{\partial t}(0) = v_0,$$

$$\forall t > 0, \quad \Delta u(t) \in L^2(\Omega),$$

$$\forall t > 0, \forall v \in H^1(\Omega), \quad \int_{\Omega} \Delta u(x, t) v(x) dx = - \int_{\Omega} \nabla u(x, t) \cdot \nabla v(x) dx + \int_{\partial_1 \Omega} p(t) v(x) dx, \quad (*)$$

and such that in $\mathcal{D}'(\mathbb{R}_+^ \times \Omega)$*

$$a(x) \frac{\partial^2 u}{\partial t^2}(t, x) - \Delta u(t, x) = 0.$$

The equation (*) is the weak formulation of the boundary condition from (1). Indeed, if $\forall t \geq 0, u(t) \in H^2(\Omega)$, then it verifies the right boundary condition. Thus, Theorem 1 states the existence and uniqueness of a weak solution to Problem (1).

Finally, we state an energy estimate for solutions to Models 1, 2 and 3 when $p \equiv 0$.

Theorem 2 (Energy estimate). *Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$.*

Let $T > 0$ and $u \in H^2(\Omega \times]0, T[)$ be a weak solution to Problem (1), (2) or (3) for $p \equiv 0$ under Assumption 1. Then the following energy estimate holds

$$\forall t \in]0, T[, \quad \int_{\Omega} a(x) \left| \frac{\partial u}{\partial t} \right|^2 (x, t) dx + \int_{\Omega} |\nabla u|^2 (x, t) dx \leq \int_{\Omega} a(x) |v_0|^2 (x) dx + \int_{\Omega} |\nabla u_0|^2 (x) dx.$$

From this energy estimate, we can directly deduce uniqueness of solution in $H^2(\Omega \times]0, T[)$ to Problems (2) and (3).

Corollary 3 (Uniqueness of solution to Problems (2) and (3)). *Under Assumptions 1 and 2, there exists at most one $u \in H^2(\Omega \times]0, T[)$ solution to (2) (respectively (3)).*

1.3 Numerical Method for solving forward problems

In this subsection, we present a numerical integration method to effectively solve Problems (1), (2) and (3). Here we use Finite Element Method (FEM) in space to solve these initial-boundary value problems. For the discretisation in time, we use the effective Newmark numerical integration.

1.3.1 Numerical parameters for simulations

In this numerical part and for the other simulations, we consider, without a lack of generality, that Ω and Ω_1 are squares and more precisely $\Omega = \left] -\frac{1}{2}, \frac{3}{2} \right[$ and $\Omega_1 =]0, 1[$. Moreover, we solve the previous forward problems over $[0, T]$ where T is assumed to be large enough.

We take, as previously mentioned, $u_0 = 0$ and $v_0 = 0$. Moreover, we consider a monochromatic plane wave external source

$$p(t) = \sin(\omega t), \quad \text{where } \omega = 80 \text{ and } t_1 = \frac{2\pi}{\omega}.$$

In every numerical parts, we will consider two tests consisting in reconstruction of two acoustic coefficients:

Test 1: one Gaussian inclusion

$$a(x, y) = 1 + 2e^{-\frac{(x-0.5)^2 + (y-0.7)^2}{0.001}}$$

Test 2: two Gaussian inclusions

$$a(x, y) = 1 + 2e^{-\frac{(x-0.5)^2 + (y-0.7)^2}{0.001}} + 3e^{-\frac{(x-0.2)^2 + (y-0.6)^2}{0.001}}$$

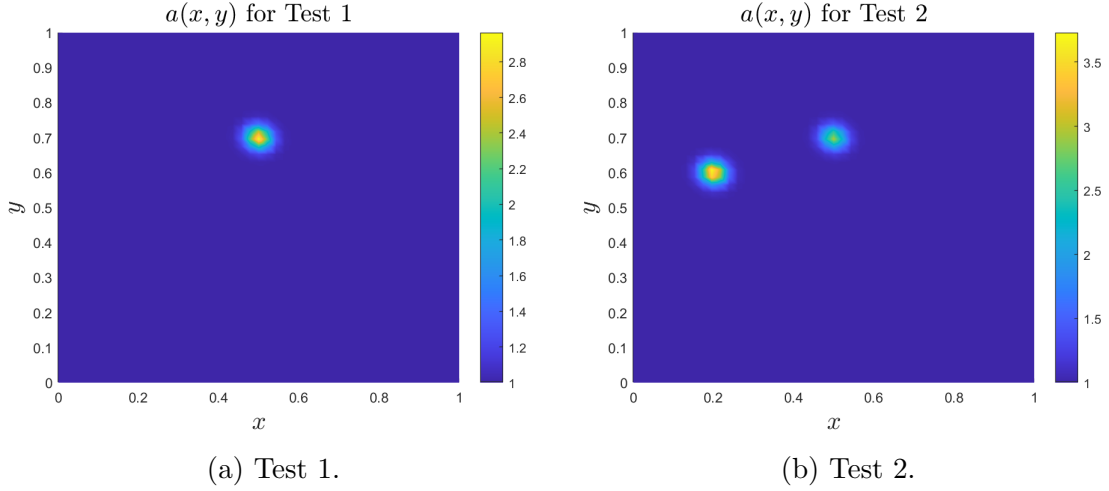


Figure 2: Test functions to reconstruct in further simulations. (a) One Gaussian inclusion. (b) Two Gaussian inclusions.

1.3.2 Discretisation of Problem (1)

We recall the result given by Theorem 1 in the particular case of homogeneous initial conditions. There exists $u \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$ such that

$$a(x) \frac{\partial^2 u}{\partial t^2}(t, x) - \Delta u(t, x) = 0 \quad \text{in } \mathcal{D}'(\mathbb{R}_+^* \times \Omega),$$

$$u(0) = 0, \quad \frac{\partial u}{\partial t}(0) = 0.$$

Furthermore, we have the weak formulation of the boundary condition

$$\forall t \geq 0, \forall v \in H^1(\Omega), \quad \int_{\Omega} \Delta u(x, t) v(x) \, dx = - \int_{\Omega} \nabla u(x, t) \cdot \nabla v(x) \, dx + \int_{\partial_1 \Omega} p(t) v(x) \, dx.$$

As for all $t \geq 0$, $\Delta u(t)$ is a $L^2(\Omega)$ -function and using the fact that a is bounded (Assumption 1), we obtain that $\frac{\partial^2 u}{\partial t^2}(t)$ is a $L^2(\Omega)$ -function. Thus, by multiplying by $H^1(\Omega)$ -functions and integrating, it yields

$$\forall t \geq 0, \forall v \in H^1(\Omega), \quad \frac{d^2}{dt^2} \left(\int_{\Omega} a(x) u(x, t) v(x) \, dx \right) + \int_{\Omega} \nabla u(x, t) \cdot \nabla v(x) \, dx = p(t) \int_{\partial_1 \Omega} v(x) \, dx. \quad (4)$$

Equation (4) is the weak formulation of Problem (1). This weak formulation is the key equation to spatially discretise the problem. We now consider a mesh on $\bar{\Omega}$. Let $h > 0$ be a parameter of the mesh $\mathcal{K}_h$. We only consider uniform triangular meshes over the domain $\bar{\Omega}$ as in Figure 3 where h is the maximal diameter of the finite elements.

Let $\mathcal{V}_h$ be the span of the $\mathbb{P}_1$ -finite elements over $\mathcal{K}_h$. We denote by $(\varphi_j)_{0 \leq j \leq n}$ the associated $\mathbb{P}_1$ -finite element Lagrange basis. We also denote by $(x_j)_{0 \leq j \leq n}$ the associated points. We obtain the discretised problem:

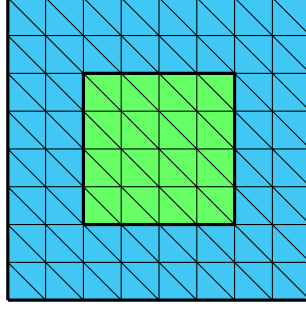


Figure 3: Example of a uniform triangular mesh for $\bar{\Omega}$. Ω is the larger square and Ω_1 is the green square. The mesh must be coincident with the boundary of Ω_1 .

Discretised problem Find $u_h \in \mathcal{C}^2(\mathbb{R}_+, \mathcal{V}_h)$ such that

$$\forall t \geq 0, \forall v \in \mathcal{V}_h, \quad \frac{d^2}{dt^2} \left(\int_{\Omega} a(x) u_h(x, t) v(x) dx \right) + \int_{\Omega} \nabla u_h(x, t) \cdot \nabla v(x) dx = p(t) \int_{\partial_1 \Omega} v(x) dx, \quad (5)$$

and

$$u_h(0) = \frac{\partial u_h}{\partial t}(0) = 0.$$

We decompose u_h over the basis such that

$$\forall t \geq 0, \quad u_h(x, t) = \sum_{k=0}^n u_{h,k}(t) \varphi_k(x),$$

where $u_{h,k}(t) = u_h(x_k, t)$.

Equation (5) is equivalent to verify the equality on the basis $(\varphi_j)_{0 \leq j \leq n}$

$$\forall t \geq 0, \forall j \in \llbracket 0, n \rrbracket, \quad \sum_{k=0}^n u_{h,k}''(t) \int_{\Omega} a(x) \varphi_k(x) \varphi_j(x) dx + \sum_{k=0}^n u_{h,k}(t) \int_{\Omega} \nabla \varphi_k(x) \cdot \nabla \varphi_j(x) dx = p(t) \int_{\partial_1 \Omega} \varphi_j(x) dx$$

We define the following matrices

$$\begin{aligned} K_h &= \left(\int_{\Omega} \nabla \varphi_k(x) \cdot \nabla \varphi_j(x) dx \right)_{0 \leq j, k \leq n} \quad \text{the stiffness matrix,} \\ M_h &= \left(\int_{\Omega} a(x) \varphi_k(x) \varphi_j(x) dx \right)_{0 \leq j, k \leq n} \quad \text{the anisotropic mass matrix,} \\ B_h(t) &= p(t) \left(\int_{\partial_1 \Omega} \varphi_j(x) dx \right)_{0 \leq j \leq n} \quad \text{the boundary vector,} \\ U_h(t) &= (u_{h,j}(t))_{0 \leq j \leq n} \quad \text{the unknown vector.} \end{aligned}$$

The problem can be easily rewritten as

$$\begin{cases} M_h \frac{d^2 U_h}{dt^2}(t) + K_h U_h(t) = B_h(t), \\ U_h(0) = \frac{dU_h}{dt}(0) = 0. \end{cases} \quad (6)$$

Now, we also discretise in time the previous problem using finite differences method. Let $0 = t_0 < t_1 < \dots < t_N = T$ with $t_k = k \, dt = k \frac{T}{N}$. To numerically solve Equation (6), we use the Newmark numerical integration. Details on the Newmark method and on the discretisation of Problems (2) and (3) can be found in Appendix C.

In practice, we implement the presented numerical methods in Matlab using Matlab R2024a version. The generation of the meshes and of the Finite Element Methods and matrices is done using the effective implementation made by Matthieu Aussal with *Gypsilab* [2, 4]. Details on the practical implementation and figures to compare the results can be found in Appendix C. All the files used for simulations and numerical analyses can be found on my Github [15].

2 Reconstruction using scattered internal measurements

In this section, we consider the inverse problem of reconstructing a using internal measurements. As a is known in $\Omega \setminus \Omega_1$, we only need to reconstruct a in Ω_1 . Such inverse problem can be formulated as:

Inverse Problem 1. *Assume that Ω , Ω_1 , T , u_0 and v_0 are known. Let a such that Assumption 1 holds, be an unknown acoustic coefficient. Let p be such that Assumption 2 holds.*

Let u be a solution to Problem (1) (respectively Problem (2) and Problem (3)) in $\Omega \times]0, T[$ for those parameters.

Assuming u is known over $\Omega \times]0, T[$, reconstruct a in Ω_1 .

On a theoretical aspect, the knowledge of u in $\Omega \times]0, T[$ is sufficient to calculate Δu and $\frac{\partial^2 u}{\partial t^2}$. Thus, using the wave equation, we can easily obtain a from this knowledge. Nonetheless, the problem still remains when we consider only a knowledge of scattered data of u over the mesh $\mathcal{K}_h$ and discrete time moments t_k .

In this section, we present a theoretical approach of the problem using Laplace transform in time of the wave equation. We discuss a quantitative selection of parameters for a better reconstruction and verify the qualitative reconstruction of the tested acoustic coefficient. We also verify the robustness to noise and to the lack of knowledge of the precise pseudo-frequency used in Laplace transform.

2.1 Reconstruction method using scattered data

In this subsection, we consider a solution u to either Problem (1), (2) or (3). We assume $u \in H^2(\Omega \times]0, T[)$ and that T is large enough to ensure that u and its derivatives are insignificant at $t = T$.

Let $s > 0$ be a pseudo-frequency. We consider the Laplace transform in time of $u \mathbb{1}_{[0, T]}$ and we define

$$w(x, s) := \mathcal{L}(u(x, \cdot) \mathbb{1}_{[0, T]})(s) = \int_0^T u(x, t) e^{-st} dt.$$

This Laplace transform is well-defined as we assume that $u \in H^2(\Omega \times]0, T[)$. We apply this transform to the acoustic wave equation. We obtain after integrations by part

$$a(x) \left[\frac{\partial u}{\partial t}(x, t) e^{-st} \right]_0^T + sa(x) [u(x, t) e^{-st}]_0^T + s^2 a(x) w(x, s) - \Delta w(x, s) = 0.$$

According to the assumption of insignificance of u and its derivatives at $t = T$, we have the approximation

$$-\Delta w(x, s) + s^2 a(x) w(x, s) = a(x)(v_0 + s u_0).$$

For the sake of simplicity and as it is the case in numerical analyses, we take $u_0 = v_0 = 0$ but the following method can be adapted to inhomogeneous initial conditions. Thus, we obtain the following Poisson problem in Ω_1 .

$$\forall s > 0, \quad \begin{cases} -\Delta w(x, s) + s^2 a(x) w(x, s) = 0 & \text{in } \Omega_1, \\ \frac{\partial w}{\partial n} = f & \text{on } \partial\Omega_1, \end{cases} \quad (7)$$

where $f = \frac{\partial w}{\partial n}$ is the boundary condition.

We apply the same Finite Element Method as in Subsection 1.3 and it yields

$$M_h A_h = \frac{1}{s^2} (F_h - K_h W_h), \quad (8)$$

where (notations do not refer to Subsection 1.3)

$$\begin{aligned} W_h &= (w(x_j))_{0 \leq j \leq n} \text{ are the known values of } w(x, s) \text{ over the mesh,} \\ K_h &= \left(\int_{\Omega_1} \nabla \varphi_k(x) \cdot \nabla \varphi_j(x) \, dx \right)_{0 \leq j, k \leq n} \text{ is the stiffness matrix,} \\ M_h &= \left(w(x_k) \int_{\Omega_1} \varphi_k(x) \varphi_j(x) \, dx \right)_{0 \leq j, k \leq n} \text{ is the weighted mass matrix,} \\ F_h &= \left(\int_{\partial\Omega_1} f(x) \varphi_j(x) \, dx \right)_{0 \leq j \leq n} \text{ is the boundary vector,} \\ A_h &= (a_{h,j})_{0 \leq j \leq n} \text{ is the unknown vector.} \end{aligned}$$

As in Subsection 1.3, we can assemble these matrices using scattered data of $w(x, s)$ at a pseudo-frequency s . Indeed, the only element which might be calculated from the scattered measurements is $f(x)$. To calculate this function for the integral, we choose to use symmetric finite differences over the boundary. A difficulty remains in the calculation near the vertices of the square which can be avoid by putting the value of the normal derivative which is at the nearest point. However, some artifacts are still present after the matrix inversion on the vertices. These artifacts can be easily suppressed applying a post-processing on the boundary to set the values to 1 as ensured by Assumption 1. See details and other tried methods in Appendix D.

Finally, to calculate $w(x, s)$ over the nodes of the mesh using scattered measurements of $u(x, t)$, we apply a quadrature method to approximate the integral. Such method is done in `discrete_laplace_transform`.

2.2 Quantitative analysis of the numerical method

In this subsection, we present some results of the numerical method of reconstruction using internal measurements. For each result, we compute two types of error denoted by

$$e_{quad} = \frac{\|a_h - a\|_2}{\|a\|_2},$$

$$e_{max} = \|a_h - a\|_{\infty},$$

where a_h denotes the vector of the reconstructed values of a over the nodes of the mesh and a is the vector of the exact values of the input test acoustic coefficient. In this subsection and for the rest of the paper, the norm $\|\cdot\|_2$ denotes the Euclidean norm on $\mathbb{R}^n$ and $\|\cdot\|_{\infty}$ denotes the infinity norm of a vector in $\mathbb{R}^n$.

For each test function a , we generate artificial measurements over a mesh $\mathcal{K}_h$ using a forward solver as explained in Subsection 1.3. Thus, several parameters are tested as the choice of the model, the mesh size h , the time step dt and the duration of the observation T . Furthermore, we also study the effect of the pseudo-frequency s used to calculate the discrete Laplace transform and solve the inverse problem. Other parameters linked to Newmark numerical integration can be tested and are discussed in Appendix E.

First, we analyse the convergence in h for Model 1 and Test 1. Similar results have been found for the other models and Test 2. Figure 4b shows that the convergence in h occurs with an order $a = 1.1503$ which is near to 1. Thus, we can conjecture about a linear convergence of the reconstruction according to the mesh size.

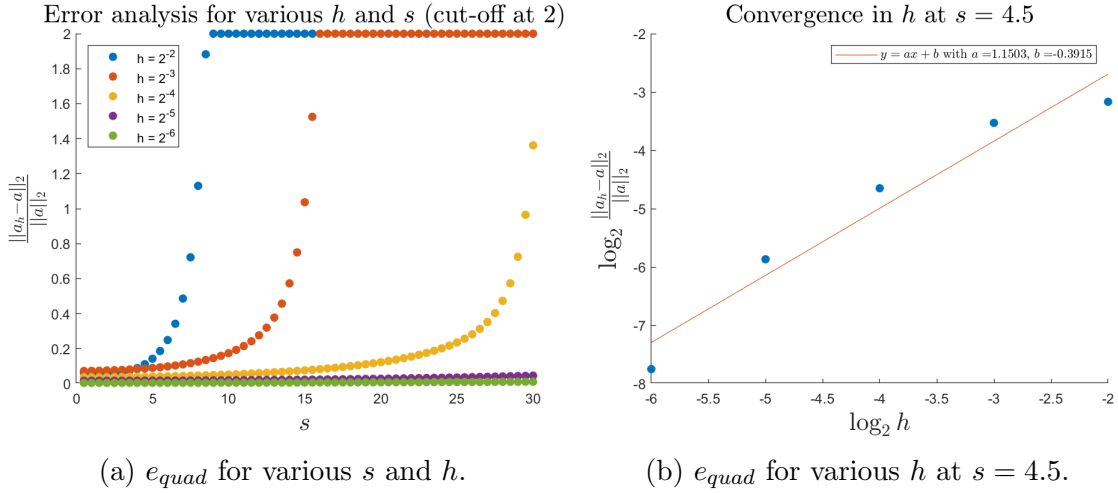


Figure 4: Analysis of the convergence in h for various s . (a) Values of e_{quad} for various mesh sizes and pseudo-frequencies. Values greater than 2 are set to 2. (b) Values of e_{quad} in logarithmic scale for various mesh sizes at $s = 4.5$. The red line is the simple linear regression. Coefficients of the linear function are at the top.

Figure 5 shows the relative quadratic error e_{quad} and the maximal error e_{max} for each model and for various values of s . Figures 5a and 5b show that the errors are increasing for pseudo-frequencies higher than 6. This phenomenon is the result of the decreasing of the amplitude of the values of w as the exponential term in the Laplace transform flatten them. Thus, computational errors in the matrix inversion become significant at large pseudo-frequencies. Moreover, Model 1 and 3 are not stable to small pseudo-frequencies. Thus, we should use the inversion method in a range of pseudo-frequencies from $s = 2$ (only for Model 1 and 2 and $s = 5$ for Model 3) to $s = 7$. According to Figure 5c, over this range of pseudo-frequencies, e_{quad} is minimal when we use data generated using Model 1. In the case of e_{max} , it is minimal for data generated using Model 2 until $s = 4.4$ and then for Model 3. Thereby, every model is interesting to study in simulations.

However, on a Physics aspect, Model 2 seems to well model a situation of a homogeneous domain that is larger than Ω as it is the case of a ground analysis or a local analysis of a

body for medical imaging. Nonetheless, such hypothesis remains to be tested in practical situations.

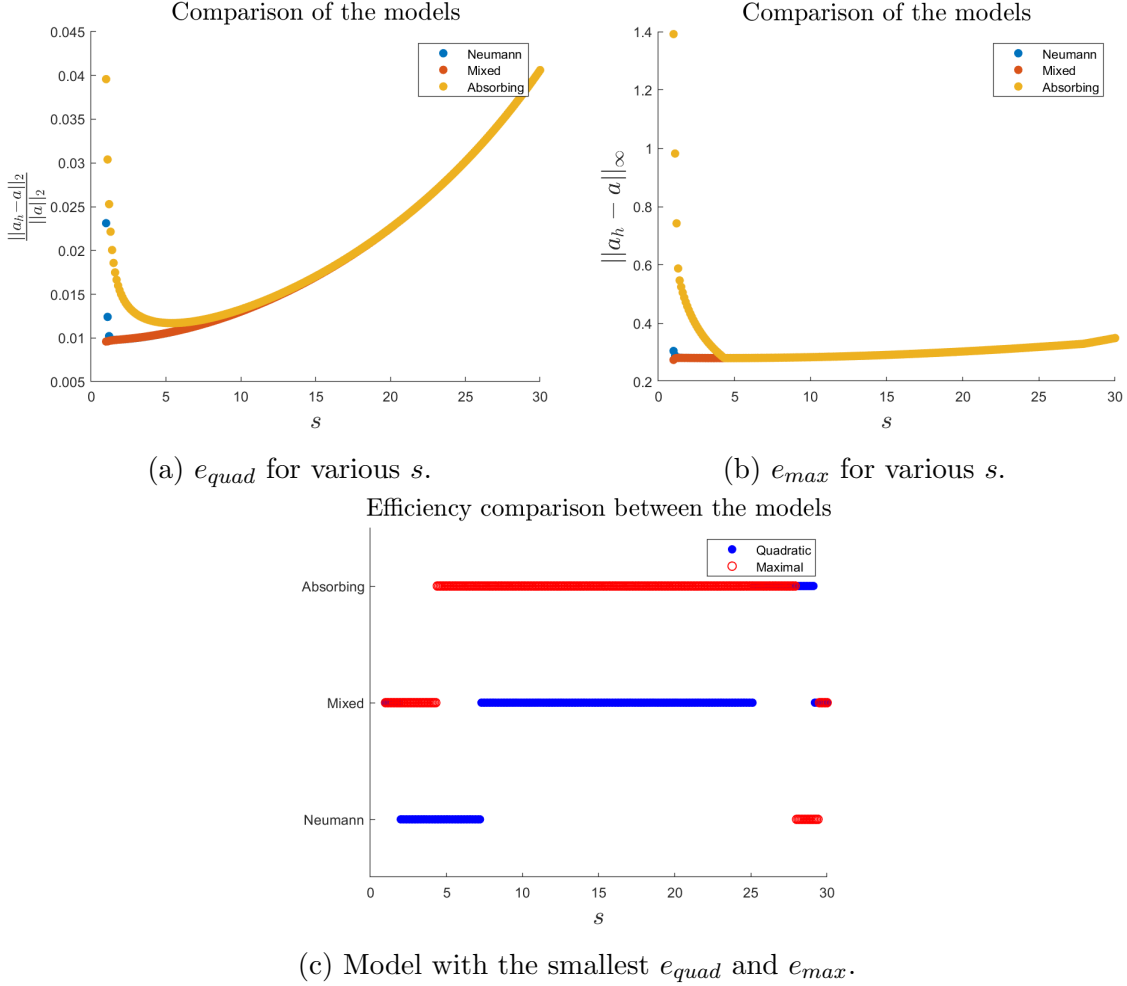


Figure 5: Comparison of the models for Test 1. (a) Values of e_{quad} for each model for various pseudo-frequencies. (b) Values of e_{max} for each model for various pseudo-frequencies. (c) Selection of the model with the smallest e_{quad} and e_{max} for each pseudo-frequency. Neumann refers to Model 1, Mixed to Model 2 and Absorbing to 3.

Figure 6 presents a quantitative analysis of the effect of the time parameters. We generate measurements using a small time step and a large value of T to ensure precision. Then, for the calculation of the discrete Laplace transform, we either change the value of the time step or of T by extracting only data over this new time discretisation. Reconstruction of the inclusion does not depend on such parameter as long as dt is sufficiently small ($dt < 0.027$ s) and T is sufficiently large ($T > 2.7$ s). The limit on dt comes from the discrete computation of the integral of the Laplace transform that presents significant errors for large time steps. Concerning T , the result seems to come from the time that the plane wave takes to reach the bottom side which is about 2.3 s.

To conclude this subsection, we can select sufficiently optimal parameters to reconstruct the test functions. Figure 7 presents an effective reconstruction using data from Model 3. Reconstruction for other models are available in Appendix E.

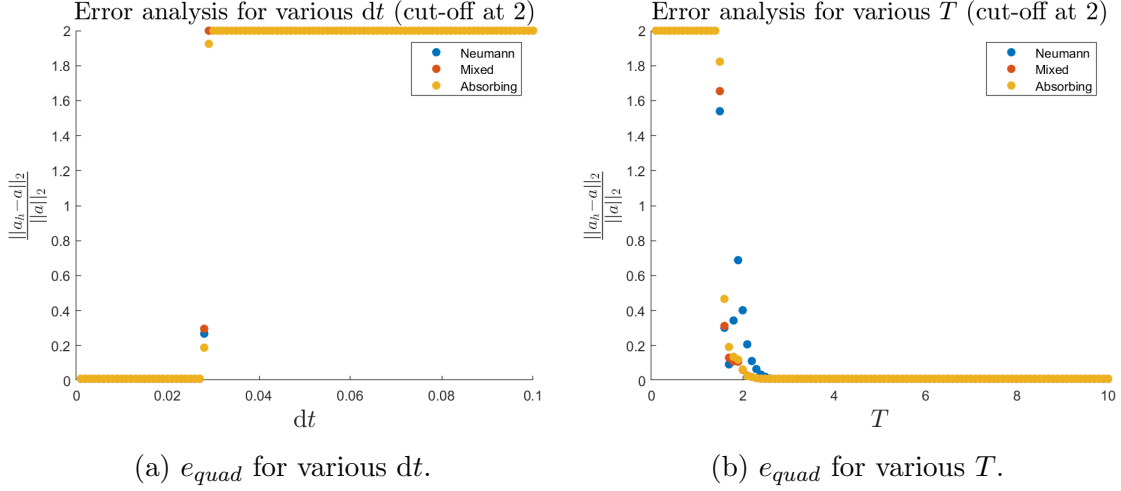


Figure 6: Analysis of the time discretisation and the duration of observation. (a) Values of e_{quad} for various time steps. Values greater than 2 are set to 2. (b) Values of e_{quad} for various time steps. Values greater than 2 are set to 2. Measurements have been calculated for $dt = 0.001$ s and for $T = 10$ s.

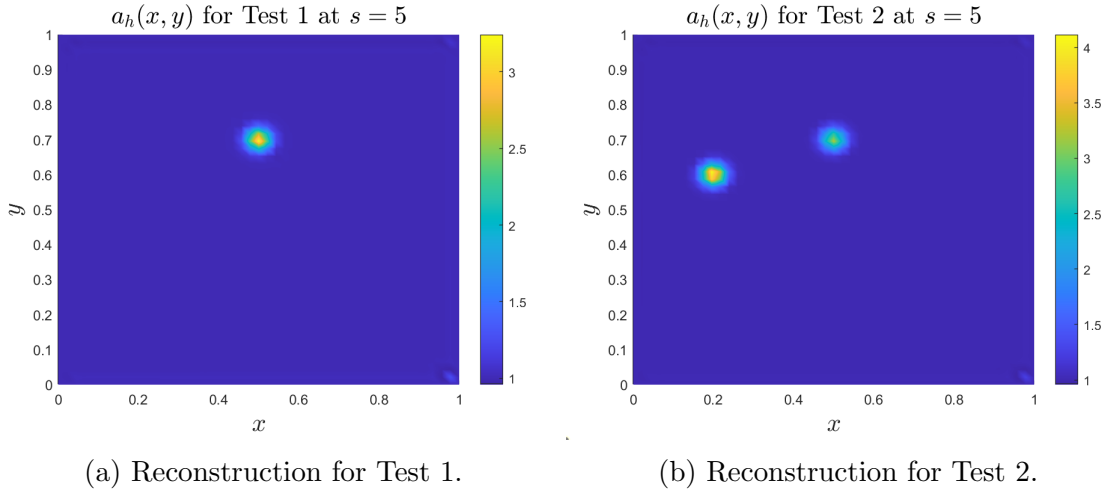


Figure 7: Reconstruction of the test functions using Model 3. (a) Reconstruction of a for Test 1. $e_{quad} = 0.0117$ and $e_{max} = 0.281$. (b) Reconstruction of a for Test 2. $e_{quad} = 0.0179$ and $e_{max} = 0.385$.

2.3 Reconstruction for an unknown pseudo-frequency

In this subsection, we present a method of reconstruction which does not require to know the pseudo-frequency. In this situation, we assume that we have access to scattered measurements $(w(x_k, s_*))_k$ on the nodes of the mesh but the pseudo-frequency is unknown or known with a lack of precision. Such situation could occur in the case of the device that measures the data have access to only the values at one frequency (by adapting the previous method to the Fourier transform instead of the Laplace transform).

The first idea is that we know s_0 which is an approximation of the value of s_* . Then, we simply apply the inverse solver at the pseudo-frequency s_0 and calculate e_{quad} . Figure 8 presents this method of reconstruction using various s_0 , with $s_0 \in [s_* - 10\%, s_* + 10\%]$ in Figure 8b. We note that the result is the same for each model. In the case where s_* is known with a good precision, we observe that the error is not significantly higher. However, even for s_* known with a 10% error, the quadratic error becomes significant (reaching $e_{quad} > 20\%$).

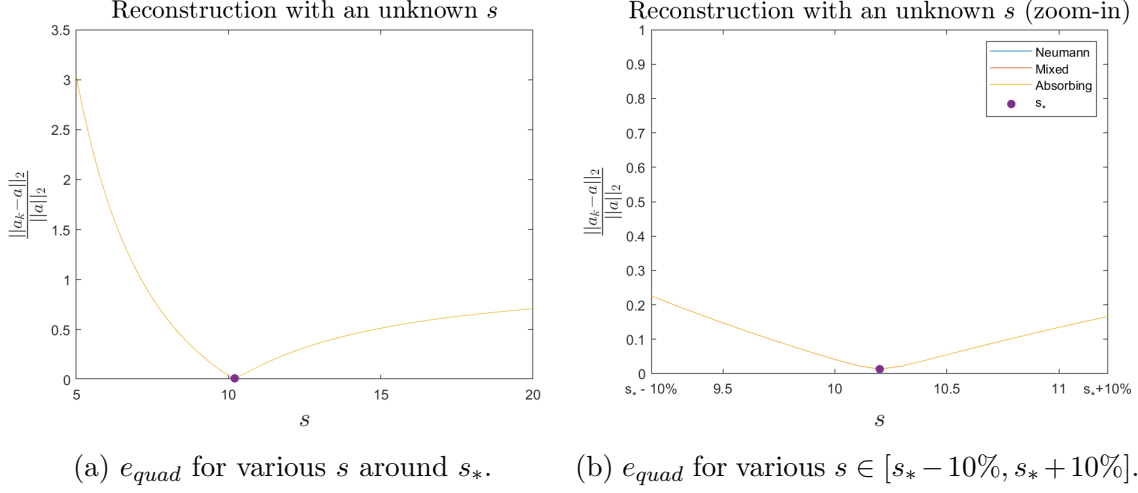


Figure 8: Analysis of relative quadratic error for a reconstruction at an incorrect pseudo-frequency. (a) Values of e_{quad} for various pseudo-frequencies. (b) Zoom-in of the results for pseudo-frequencies in $[s_* - 10\%, s_* + 10\%]$.

Thus, it seems necessary to develop a method of reconstruction without the knowledge of s_* . Figure 9 presents the relative quadratic error on w defined as

$$e_{quad, w} = \frac{\|w(\cdot, s) - w(\cdot, s_*)\|_2}{\|w(\cdot, s_*)\|_2},$$

where $w(\cdot, s_*)$ are the measurements at s_* .

$w(\cdot, s)$ is calculated using the following protocol. First, we apply the inverse solver on $(w(x_k, s_*))_k$ at the pseudo-frequency s to obtain the scattered values $(a_h(x_k))_k$ of the reconstructed coefficient. Then these values are used into the forward solver as an acoustic coefficient (by interpolation by the nearest point) to get a solution u_s . Finally, $w(\cdot, s)$ is the discrete Laplace transform of u_s at s .

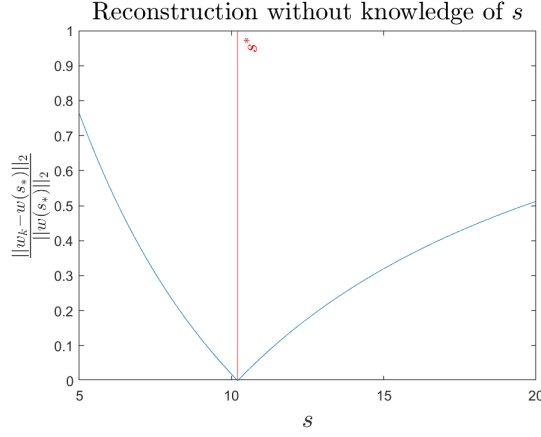


Figure 9: Analysis of the error on w using the protocol of blind reconstruction.

In Figure 9, we note that $e_{quad, w}$ is minimal for $s = s_*$. Thus, we use this protocol to implement a blind inverse solver in pseudo-frequency. We successively apply the protocol for various pseudo-frequencies $s_0 < s_1 < \dots < s_n$ given by the user and we stop the algorithm when $e_{quad, w}$ is greater for s_{k+1} than for s_k . Thereby, we return the selected pseudo-frequency $s_{reconstructed}$ as an approximation of s_* and the associated reconstructed acoustic coefficient a_h . As it can be seen in Figure 10, the method of blind reconstruction is effective to recover the pseudo-frequency and reconstruct the acoustic coefficient.

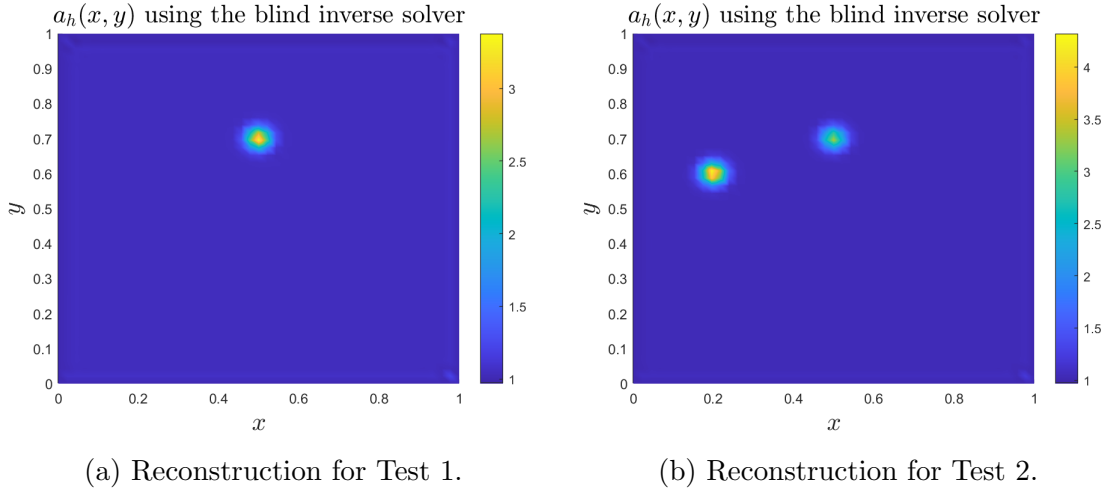


Figure 10: Reconstruction of the test functions using Model 3 and the blind inverse solver. $s_* = 10.2$ (a) Reconstruction of a for Test 1. $s_{reconstructed} = 10$, $e_{quad} = 0.0416$ and $e_{max} = 0.281$. (b) Reconstruction of a for Test 2. $s_{reconstructed} = 10$, $e_{quad} = 0.0179$ and $e_{max} = 0.385$.

Nonetheless, to counterbalance the lack of knowledge of the pseudo-frequency, the blind inverse solver requires the knowledge of the model (and so the type of boundary conditions) as well as the external source p , which has not been needed in the previous inverse solver. Hence, we have proposed a further analysis of these requirements and some techniques to overcome them in Appendix F.

2.4 Reconstruction using noised data

In this subsection, we shortly discuss the robustness of the method to the noise on the measurements. If it seems obvious that the method can not deal with noised data without an increase of the error on the reconstructed acoustic coefficient, we present in this part a first example of trial to smooth the data to obtain a satisfying result. Of course, some better techniques for deblurring the measurements to obtain better results could have been tried. However, we show in this part that the simplest method of smoothing is already effective to obtain qualitatively acceptable results.

In this paragraph, we consider a noise level δ_{noise} . In practical situation, we take this noise level at $\delta_{noise} = 3\%$. We consider a set of measurements with a random additive noise on each measurement such as

$$\forall j, k, \quad u_{noise}(x_j, t_k) = u(x_j, t_k) (1 + \delta_{noise} X_{j,k}),$$

where $(X_{j,k})_{j,k}$ are independent random variables such that $X_{j,k} \sim \mathcal{U}([-1, 1])$.

Figure 11 presents some reconstructed acoustic coefficients using various methods. In Figure 11a, we can see the reference reconstruction using unnoised data. Figure 11b is the result of an application of the inverse solver without any processing on the data. As we can see, the method is not robust to the noise by itself. In this figure, there is no reconstruction of the inclusion at all. Figure 11c presents the result of the reconstruction after applying a pre-smoothing on the data. More precisely, we compute the discrete Laplace transform $w(x, s)$ and then we apply a 2D-Gaussian-smoothing on the scattered data before using the smoothed data in the inverse solver. We notice a qualitatively acceptable reconstruction of the acoustic coefficient. Indeed, the presence of the inclusion as well as its location are correctly reconstructed. However, the size of the inclusion and its peak value are not reconstructed at all. Even if we apply another 2D-polynomial-smoothing on the reconstructed coefficient as in Figure 11d, the result is not correctly improved.

Despite the quantitative failure to reconstruct the function, we observe that the noise can be easily removed applying the appropriate smoothing method. Thus, a further analysis to try several deblurring techniques could be done to ensure that a pre-processing of the measurements is sufficient to obtain sufficiently acceptable results. Appendix G presents some additional results in the case of Test 2.

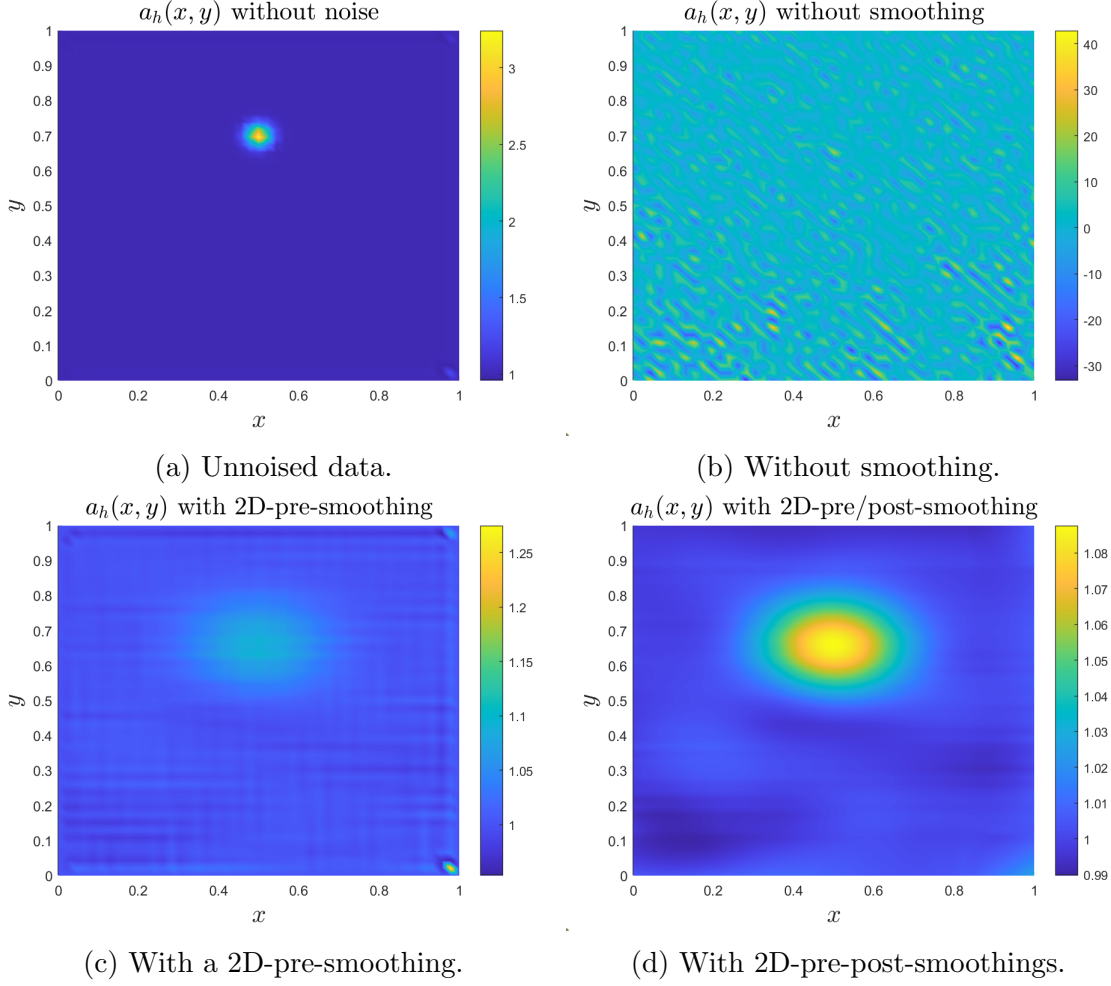


Figure 11: Reconstruction of the test function for Test 1 using noised data. (a) Reconstruction of a with unnoised data. $e_{quad} = 0.0117$. (b) Reconstruction of a with noised data and without smoothing methods. $e_{quad} = 7.57$. (c) Reconstruction of a with noised data and with a smoothing before inversion. $e_{quad} = 0.0731$. (d) Reconstruction of a with noised data and with smoothings before and after inversion. $e_{quad} = 0.0727$.

3 Reconstruction using scattered boundary measurements

In this section, we consider the inverse problem of reconstructing a using measurements on a part of the boundary. In this problem, let $\Gamma \subset \partial\Omega$. We formulate the inverse problem as:

Inverse Problem 2. Assume that Ω , Ω_1 , T , u_0 , v_0 and p are known. We assume that p is such that Assumption 2 holds. Let a such that Assumption 1 holds, be an unknown acoustic coefficient.

Let u be a solution to Problem (1) (respectively Problem (2) and Problem (3)) in $\Omega \times]0, T[$ for those parameters.

Assuming u is known over $\Gamma \times]0, T[$, reconstruct a in Ω_1 .

Even if Γ could be the whole boundary, the main objective is to reconstruct a using data only from the top side. Thus, in simulation, we consider that $\Gamma = \partial_1\Omega$.

On a theoretical aspect, by assumption, there exists a that is solution to Inverse Problem 2. However, the uniqueness and the well-posedness of Inverse Problem 2 are not clear. Thus, in a first subsection, we present an adaptation of a theorem [10] which ensures this well-posedness under additional assumptions. Then, we explain a formal optimisation approach to establish a numerical method to reconstruct a . Finally, we present some simulations and numerical methods to improve the results.

3.1 Uniqueness and well-posedness of the inverse problem

In this subsection, we state a theorem that ensure the uniqueness of the solution to Inverse Problem 2. Moreover, the theorem states a Lipschitz inequality which ensures a continuous dependency of the solution according to the known data. Nonetheless, this theorem requires additional assumptions, such as non homogeneous initial conditions, which do not hold in simulations. It is well-known that such assumption is usual to obtain uniqueness theorems.

Theorem 4 has been proved by Mourad Bellassoued in [10]. However, the proof of this theorem has been done in the case of homogeneous Neumann boundary condition. If the result is easily adapted in case of Model 1, it requires to adapt a little more the proof for Models 2 and 3. The proof is founded on a Carleman estimate for the acoustic wave equation whose proof is admitted in this report. The proof of Theorem 4 can be found in Appendix H.

3.1.1 Assumptions

Let $x_0 \notin \overline{\Omega}$, $\varepsilon_0 > 0$ and $0 < r_0 < \frac{2}{3}$ be selected parameters. We define

$$T_0 = \sup_{x \in \overline{\Omega}} \left(\frac{\|x - x_0\|_2}{\sqrt{c_0 \left(1 - \frac{3r_0}{2}\right)}} \right).$$

We also define the set of the admissible wave speed functions

$$\mathcal{C} = \left\{ c \in \mathcal{C}^1(\overline{\Omega}) \left| c_0 \leq c(x) \leq 1, \forall x \in \overline{\Omega} \text{ and } \sup_{x \in \overline{\Omega}} \frac{|\nabla c(x)| \|x - x_0\|_2}{c(x)} \leq r_0 \right. \right\},$$

where c_0 is defined in Assumption 1.

3.1.2 Uniqueness and Lipschitz stability

Through these notations, the following theorem holds:

Theorem 4 (Uniqueness and Lipschitz stability). *Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$ such that*

$$|\Delta u_0(x)| \geq \varepsilon_0, \quad \text{for almost every } x \in \overline{\Omega}.$$

Let $T > T_0$. Let $c, c_ \in \mathcal{C}$ and u, u_* be the associated solutions to Problem (1) (respectively Problem (2) or (3)) on $\Omega \times]0, T[$.*

Assuming $u, u_* \in H^2(\Omega \times]0, T[) \cap W^{3,\infty}(\Omega \times]0, T[)$, there exists $C > 0$ (independent from u, c and c_*) such that

$$\|c - c_*\|_{L^2(\Omega)} \leq C \left(\left\| \frac{\partial u}{\partial t} - \frac{\partial u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)} + \left\| \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u_*}{\partial t^2} \right\|_{L^2(\partial\Omega \times]0, T[)} + \left\| \frac{\partial \nabla u}{\partial t} - \frac{\partial \nabla u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)} \right). \quad (9)$$

Theorem 4 states that the values of c inside Ω are controlled by the values of u on the boundary $\partial\Omega$. Thus, the knowledge of u on $\partial\Omega \times [0, T]$ is sufficient to ensure the uniqueness of c (and so of a) in some class of smooth solutions such that the assumptions hold. Indeed, the derivatives in time of u only requires data of u on $\partial\Omega \times [0, T]$. Furthermore, to obtain ∇u on the boundary, we only need to calculate the normal derivative on each side and the tangential derivative. On the one hand, the first one is given by the boundary conditions for each model. On the other hand, the tangential derivative can be calculated using the knowledge of u on the side.

In this theorem, the constant C depends on $c_0, r_0, \varepsilon_0, x_0, T, \Omega$ and u_0 (and also the parameters γ, δ and β defined in Appendix H). However, C depends also on u_* . This dependency does not refute the uniqueness. Furthermore, we can assume that we know u_* on $\partial\Omega \times [0, T]$ and we want to reconstruct c_* . Thus, C does not depend on u and c and so the Lipschitz inequality ensures a continuous dependency on the parameters.

Theorem 4 ensures the well-posedness of Inverse Problem 2 in the case of $\Gamma = \partial\Omega$. For the case of $\Gamma = \partial_1\Omega$, a similar result have been proven by M. Bellassoued and M. Yamamoto in [11]. The proof of this theorem must also easily be adapted to our models and could be an extension of this work.

3.2 Formal optimisation approach

In this subsection, we present the method of reconstruction of the acoustic coefficient using boundary measurements. This method is based on the optimisation approach as described in [6, 8]. Even if the approach is purely formal, we present a gradient descent method whose convergence should be tested in simulation in the following subsection. In this part, we only present the method of reconstruction using measurements generated from Model 1. However, this method could be adapted for the other models and might be an extension of the present work.

Let $\Gamma \subset \partial\Omega$ be a part of the boundary as defined in Inverse Problem 2. $\tilde{u}$ denotes the known data on $\Gamma \times]0, T[$. We define the following sets:

$$H_u^1 = \{u \in H^1(\Omega \times]0, T[) \mid u(x, 0) = u_0(x) \text{ a.e. } \Omega\},$$

$$H_\lambda^1 = \{\lambda \in H^1(\Omega \times]0, T[) \mid \lambda(x, T) = 0 \text{ a.e. } \Omega\}.$$

We define some functionals

$$\begin{aligned} \mathcal{J} : H_u^1 \times \mathcal{C}^2(\overline{\Omega}, [1, M]) &\longrightarrow \mathbb{R} \\ (u, a) &\longmapsto \frac{1}{2} \int_0^T \int_\Gamma |u - \tilde{u}|^2 dx dt \end{aligned}$$

$$\begin{aligned}
\mathcal{T} : H_u^1 \times \mathcal{C}^2(\overline{\Omega}, [1, M]) &\longrightarrow \mathbb{R} \\
(u, a) &\longmapsto \frac{1}{2} \int_{\Omega} |a - a_0|^2 dx, \\
\mathcal{G} : H_u^1 \times \mathcal{C}^2(\overline{\Omega}, [1, M]) &\longrightarrow H_{\lambda}^{1*}
\end{aligned}$$

such that

$$\begin{aligned}
\mathcal{G}(u, a)(\lambda) = & \int_0^T \int_{\Omega} \nabla u \cdot \nabla \lambda dx dt - \int_0^T \int_{\Omega} a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt \\
& - \int_{\Omega} a(x) v_0(x) \lambda(x, 0) dx - \int_0^T \int_{\partial_1 \Omega} p(t) \lambda(x, t) d\sigma(x) dt.
\end{aligned}$$

The spaces are set here to ensure the initial conditions for some initial-boundary value problems which are described below. $\mathcal{J}$ is the cost functional for the minimisation problem. This functional forces u to be near enough to the data $\tilde{u}$. $\mathcal{T}$ is the Tikhonov functional whose role is presented below. $\mathcal{G}$ is a functional that encodes the weak formulation of Problem (1) to ensure that u is solution to Problem (1) for the acoustic coefficient a when we impose $\mathcal{G}(u, a) = 0$. The proof of this weak formulation is done in Appendix I.

Using these notations, Inverse Problem 2 can be reformulate as an optimisation problem:

Optimisation Problem 1. Find $(u, a) \in H_u^1 \times \mathcal{C}^2(\overline{\Omega}, [1, M])$ solution to

$$\min_{\mathcal{G}(u, a)=0} \mathcal{J}(u, a).$$

By assumption in Inverse Problem 2, Optimisation Problem 1 has a solution. However, it is well-known that such problem is ill-posed. Thus, we introduce a Tikhonov functional with a regularisation parameter $\gamma > 0$ to ensure the well-posedness.

Optimisation Problem 2. Find $(u, a) \in H_u^1 \times \mathcal{C}^2(\overline{\Omega}, [1, M])$ solution to

$$\min_{\mathcal{G}(u, a)=0} \mathcal{J}(u, a) + \gamma \mathcal{T}(u, a).$$

The well-posedness of Optimisation Problem 2 has been studied in [9]. In fact, the functional $\mathcal{J}(u, a) + \gamma \mathcal{T}(u, a)$ is locally strongly convex. However, the solution to Optimisation Problem 2 might not be exactly the solution to Optimisation Problem 1 and so of the inverse problem. Nonetheless, a control of the error on the solution has been also established.

The efficacy of the resolution of Inverse Problem 2 by solving Optimisation Problem 2 rests upon the selection of the initial guess a_0 and the regularisation parameter γ . The initial guess might be chosen using an initial idea of the solution a to ensure the smoothness of the reconstructed acoustic coefficient. As we assume that we do not know more than $\tilde{u}$, we have chosen to take $a_0 \equiv 1$. For the selection of the regularisation parameter, a discussion is done below.

To solve Optimisation Problem 2, we apply a formal Lagrangian approach. In this approach, we consider the Lagrangian for Optimisation Problem 2:

$$\mathcal{L}(u, a, \lambda) = \mathcal{J}(u, a) + \gamma \mathcal{T}(u, a) + \mathcal{G}(u, a)(\lambda)$$

Then, we search (u_*, a_*, λ_*) such that

$$\frac{\partial \mathcal{L}}{\partial u}(u_*, a_*, \lambda_*) = \frac{\partial \mathcal{L}}{\partial a}(u_*, a_*, \lambda_*) = \frac{\partial \mathcal{L}}{\partial \lambda}(u_*, a_*, \lambda_*) = 0.$$

The derivation according to λ yields to the weak formulation of Problem (1) which is here the principal problem. Thus u_* must be solution to (1) for a_* . The derivation according to u yields for λ_* to be a weak solution to the adjoint problem

$$\begin{cases} a_*(x) \frac{\partial^2 \lambda}{\partial t^2}(x, t) - \Delta \lambda(x, t) = 0 & \text{in } \Omega \times]0, T[, \\ \lambda(x, T) = 0, \quad \frac{\partial \lambda}{\partial t}(x, T) = 0 & \text{in } \Omega, \\ \frac{\partial \lambda}{\partial n}(x, t) = \tilde{u}(x, t) - u_*(x, t) & \text{on } \Gamma \times]0, T[, \\ \frac{\partial \lambda}{\partial n}(x, t) = 0 & \text{on } \partial\Omega \setminus \Gamma \times]0, T[. \end{cases} \quad (10)$$

Finally, the derivation of the Lagrangian according to a gives the formulation of the gradient according to a

$$\nabla_a L(u, a, \lambda)(x) = \gamma(a(x) - a_0(x)) - \int_0^T \frac{\partial u}{\partial t}(x, t) \frac{\partial \lambda}{\partial t}(x, t) dt - v_0(x) \lambda(x, 0).$$

Thus, from this approach and since $v_0 \equiv 0$ in simulations, we can formulate the following gradient algorithm:

Algorithm 1: Simple gradient algorithm

Initialisation: $a_0 \equiv 1$

for $n \leftarrow 0$ **to** $N - 1$ **do**

 Solve (1) with $a_n \rightarrow u_n$,

 Solve (10) with a_n, u_n and $\tilde{u} \rightarrow \lambda_n$,

 Compute the gradient $\rightarrow g_n = \gamma_n(a_n - a_0) - \int_0^T \frac{\partial u_n}{\partial t} \frac{\partial \lambda_n}{\partial t} dt$,

$a_{n+1} = a_n - \frac{g_n}{\gamma_n}$.

In this algorithm, N denotes the number of iterations we set (as we have not selected a stopping criterion). We also consider $\gamma_n = \frac{\gamma_0}{(n+1)^q}$ with $\gamma_0 > 0$ and $q \in]0, 1]$. This choice for γ is described in [8] and ensures that the regularisation term in the optimisation problem decreases in significance to tend to Optimisation Problem 1. To select γ_0 and q , we have done a numerical study that can be found in Appendix J. We describe more the numerical implementation of this algorithm in the following subsection.

3.3 Numerical results and trials for improving the method

In this last subsection, we present some numerical results on the reconstruction using boundary measurements. Similarly to Subsection 2.2, we generate the measurements by solving Problem (1) in time and extracting the scattered values over $\partial_1\Omega \times]0, T[$. Then we apply Algorithm 1 to reconstruct a over multiple iterations. According to Assumption 1, we know that $a \equiv 1$ over $\bar{\Omega} \setminus \Omega_1$, so we apply a post-processing after each iteration to set the value 1 outside Ω_1 .

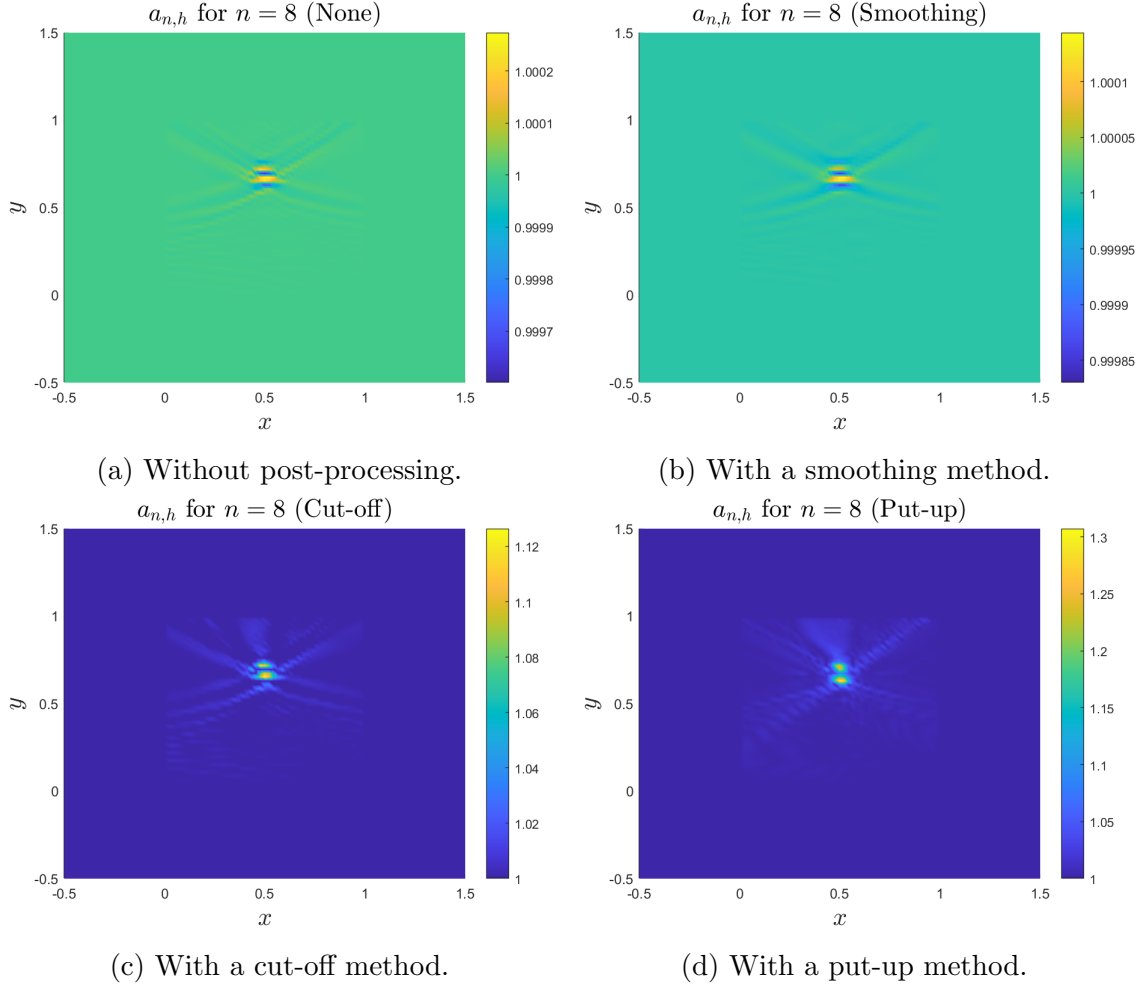


Figure 12: Reconstruction for Test 1 using data on $\partial_1\Omega$ and the gradient method with various post-processing methods. Results correspond to the 8th iteration of the gradient descent. (a) Reconstruction without post-processing. $e_{quad} = 0.0769$. (b) Reconstruction with a smoothing post-processing. $e_{quad} = 0.0769$. (c) Reconstruction with a cut-off post-processing. $e_{quad} = 0.0733$. (d) Reconstruction with a put-up post-processing. $e_{quad} = 0.0673$.

Figure 12 presents some results of reconstruction using the gradient method. Figure 12a presents the result without post-processing at each iteration. We note that the method is qualitatively acceptable since the inclusion can be located at the correct spot and with a radius that is approximately correct. However, the height of the peak is very far from the exact value. Moreover, the peak could be considered as an artifact since it is a very small peak and thus the relative quadratic error is not acceptable. Finally, we notice that the peak

seems to be a sinusoidal peak with values lower than 1. As a consequence, the reconstruction is not quantitatively satisfying.

Consequently, it is necessary to apply a post-processing method after each iteration to improve the results. The first idea is to smooth the values as in Subsection 2.4. Figure 12b presents the result of this method but it is not better than the reconstruction without post-processing. Thus, in Figure 12c, we have applied a Cut-off post-processing in which we just cut the values lower than 1 to set them at 1. We observe that the results are greatly improved. Indeed, the inclusion has a more correct shape and a higher peak value. Furthermore, e_{quad} is lower. In Figure 12d, we have applied a Put-up post-processing which consists in putting up the values lower than 1 by replacing them by $1 + |a_h(x_k) - 1|$. We note that the results are again better with this post-processing method. Thus, the post-processing method is an essential element of the gradient method to obtain better results of reconstruction.

Figure 13 illustrates a comparison over the iterations between the presented post-processing techniques. We can notice that the put-up method gives clearly quantitatively better reconstructions. However, this method is not robust over the iterations and some artifacts grow after some iterations. Thus, there is a lack of stability for the put-up post-processing. A similar observation can be done for the cut-off method since the error skyrockets after 17 iterations. Nonetheless, this method is more stable but converges slower. Finally, without post-processing or with a smoothing, it seems that the gradient method does not converge to the solution and e_{quad} seems to be stable and even to increase. The same comments could be done in the case of Test 2 but we can also note that the convergence seems slower for the cut-off and the put-up methods than in the case of Test 1, which is coherent.

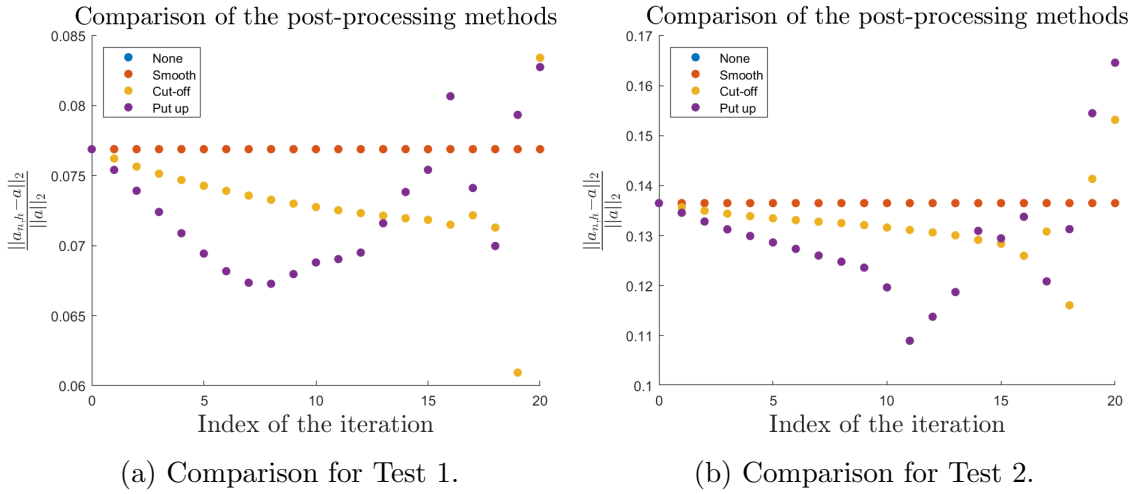


Figure 13: Comparison between the post-processing methods over the iterations for the reconstruction using measurements on $\partial_1\Omega$. (a) Values of e_{quad} over the iterations for various post-processing methods for Test 1. (b) Values of e_{quad} over the iterations for various post-processing methods for Test 2.

As a consequence, we conclude that the cut-off and the put-up post-processing methods are two interesting and useful methods to obtain qualitatively satisfying results as well as quantitatively better reconstructions. Then, we have tried during this internship to apply some additional post-processing techniques to improve again the results in these two methods. The first idea is to put a tolerance $tol \in [0, 1]$ such that all the values of a_h which are such that $|a_{h,k} - 1|$ is lower than $tol \max_k |a_{h,k} - 1|$ are set to 1. We expected that the artifacts that

can be observed in Figure 12 could be reduced with this method. The other idea is called the mean post-processing. It consists in taking the average of the values of the three nodes linked to a finite element for each element to obtain a smooth coefficient a_h .

Figure 14 illustrates the effect of these two post-processing methods. We have taken for tol the value of the optimal tolerance after an analysis for each post-processing method. This selection can be found in Appendix J. The tolerance parameter does not seem to improve the artifacts in the case of the cut-off but greatly improve the reconstruction for the put-up method. Moreover, the shape of the inclusion is strongly improved thanks to the mean post-processing. Contrary to the two distinct inclusion wrongly reconstructed in Figure 12, we note that the method seems to tend to one inclusion in Figure 14. Thus these post-processing methods are effective to get better results.

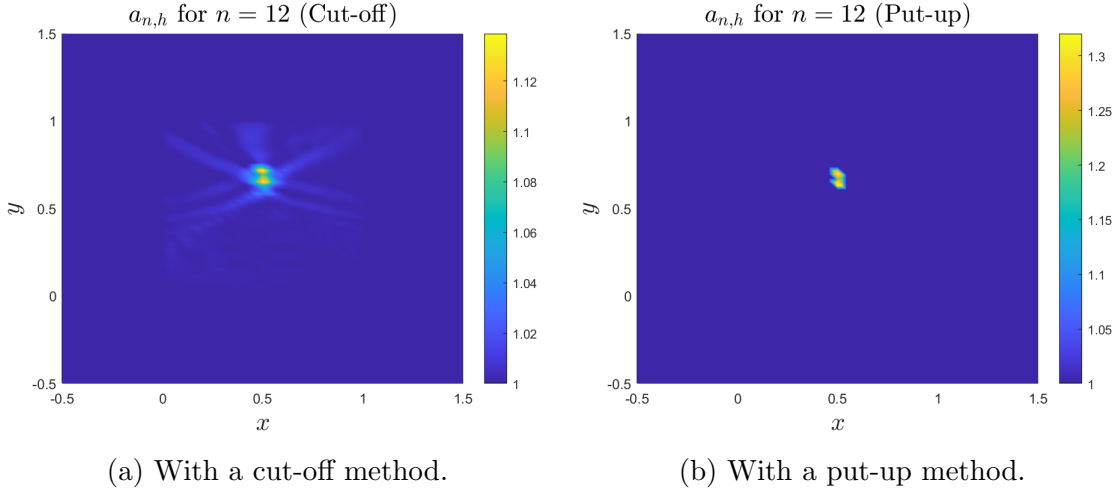


Figure 14: Reconstruction for Test 1 using data on $\partial_1\Omega$ and the gradient method with various post-processing methods and with a tolerance and the mean post-processing. (a) Reconstruction with a cut-off post-processing. $e_{quad} = 0.0707$. (b) Reconstruction with a put-up post-processing. $e_{quad} = 0.0695$.

To conclude this subsection, we just list some trials to improve the gradient method. Indeed, the proposed gradient method takes a lot of time to be applied even for 20 iterations. The analysis of each parameter have taken several days to compute before to be analysed. Thus, the main issue of this method that has limited our analysis is the computational time. Therefore, we have tried two ideas to reduce the computational time and do better analyses. The first idea was to apply the conjugate gradient algorithm instead of the simple one. A presentation of the results on the conjugate gradient is available in Appendix J. However, this method was not efficient at all for the cut-off and the put-up post-processings.

The second idea was to do an adaptive gradient method. In this method, we generate some measurements with a very fine mesh but we work with a coarse mesh ($h = 2^{-3}$) and we refine the coarse mesh after each iteration at the points where the values are such that $|a_{h,k} - 1|$ is greater than $tol \max_k |a_{h,k} - 1|$ where tol is another tolerance parameter which is commonly taken at 0.95 but can be analysed. Figure 15 shows a reconstruction using this adaptive method. We observe that the method is ineffective at all. After some trials, we have concluded that the main issue is the refinement method. Indeed, the refinement of the mesh is done with the library Gypsilab [4] and consists in a midpoint algorithm. The main issue relies

in the angles of the triangular elements after some refinements. In Figure 15b, we can note that some triangles have very small angles and are almost flat. However, this phenomenon is well-known as a variational crime since Cea's Lemma does not hold in this case. Thus, the adaptive method could be a very effective method to reduce the computational time and so allow us to analyse more the method but we have to find another refinement method to counterbalance the variational crimes.

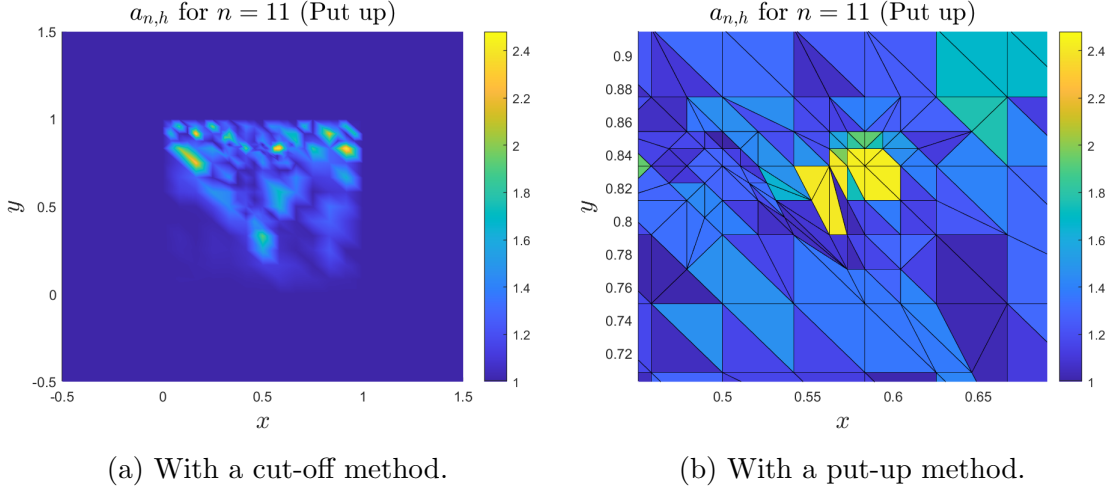


Figure 15: Reconstruction for Test 1 using data on $\partial_1\Omega$ and the adaptive gradient method with a put-up post-processing. (a) Reconstruction of the acoustic coefficient at the 11th iteration. (b) Zoom-in of the reconstructed acoustic coefficient.

To conclude this section, we have been able to effectively reconstruct the acoustic coefficient using scattered boundary measurements in a qualitative aspect. We have proposed many trails to improve the method with several post-processing techniques. Furthermore, in Appendix J, a complete analysis of the selection of the parameters as well as the effect of other parameters can be found. However, the large computational time has limited our analyses and we have not been able to obtain a precise determination of the parameter or to study several parameters together. Despite our trials to reduce the computational time, we have not been able to propose a more efficacious method. Therefore, many trails of improvements remain for the future of this project. We hope that the Github repository [15] will help future researchers to pursue the work done in this report.

Conclusion and perspectives

In this work, we have presented two inverse problems of the reconstruction of a spatially-distributed acoustic coefficient in the acoustic wave equation. We have presented two numerical methods of reconstruction, one using internal measurements and the other one using boundary measurements. The selection of the parameters as well as the effectiveness of the methods have been qualitatively and quantitatively analysed and discussed.

On the one hand, we have successfully reproduced similar and further numerical analyses on the method using internal measurements. We have proposed a complete study of the method including a testing of the robustness to the noise. Furthermore, we have proposed a new approach of the reconstruction without the knowledge of the pseudo-frequency. However, the robustness to the noise still requires some improvements and better smoothing methods to obtain more satisfying results.

On the other hand, we have quantitatively compared several proposals of amelioration of the gradient method for the reconstruction using boundary data. Despite the inability to reproduce satisfying results in this case, we have proposed multiple trails of improvement that should be more analysed in a future work. In addition, some methods could be developed to reduce the calculation time of the algorithm including a conjugate gradient method or an adaptive mesh algorithm to work on coarser meshes. Finally, further to the lack of time, we have not been able to apply the proposed methods on Models 2 and 3. However, we have proposed an interesting adaptation of a theorem concerning the well-posedness of Inverse Problem 2 which could be improved by an adaptation of the theorem from [11].

Finally, many extensions to this study could be imagined. First, it might be useful to test the methods using real data from a real situation (as the detection of ground inclusions or of melanomas). Then, one could think about some modifications of the models such as considering more homogeneous layers or more complex boundary conditions like Engquist-Majda second-order absorbing conditions. The idea of considering several sets of measurements for two external sources (two frequencies for instance) has been raised during this internship. Finally, the main objective of this project is to develop some methods to adapt them to Maxwell's equations for many applications in medical imaging. These are some ideas to pursue this work.

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Appendices

A Trace Theorem for a rectangular domain

In this subsection, we state fundamental theorems on trace and Green's formula. These theorems allow us to give meaning to Assumption 3 and to study the previous problems. If these theorems are well-known on $\mathcal{C}^1$ open sets, we must state them on rectangular open sets.

In this section, $d \in \mathbb{N}^*$ denotes the dimension of the space and $n \in \mathbb{N}^*$ is an arbitrary integer. We denote by

$$Q = \{(x, y) \in \mathbb{R}^{d-1} \times \mathbb{R} \mid \|x\|_2 < 1 \text{ and } |y| < 1\}.$$

First, we define the notion of an open set with Lipschitz boundary.

Definition 1. Let $\Omega \subset \mathbb{R}^d$ be a bounded open set. Ω has a Lipschitz boundary if for all $x \in \partial\Omega$, there exists an open neighbourhood V in $\mathbb{R}^d$ and there exists a bijection $\varphi : V \rightarrow Q$ such that:

- φ and φ^{-1} are Lipschitz continuous functions.
- $\varphi(V \cap \Omega) = Q \cap (\mathbb{R}^{d-1} \times \mathbb{R}_+^*)$.
- $\varphi(V \cap \partial\Omega) = Q \cap (\mathbb{R}^{d-1} \times \{0\})$.

This definition is introduced in [12] to state an extension theorem. Here we present Stein's Theorem whose proof is done in [1].

Theorem 5 (Stein's Theorem). *Let $\Omega \subset \mathbb{R}^d$ be a bounded open set with Lipschitz boundary. Let $n \in \mathbb{N}^*$.*

Then there exists a bounded linear operator $P : H^n(\Omega) \rightarrow H^n(\mathbb{R}^d)$ such that

$$\forall u \in H^n(\Omega), \quad Pu|_{\Omega} = u.$$

This extension theorem is essential to demonstrate the density of $\mathcal{C}_c^\infty(\overline{\Omega})$ -functions in $H^n(\Omega)$. Before proving this result, we show that a rectangular domain (for $d = 2$) has a Lipschitz boundary.

Lemma 6. *Let $\Omega =]x_1, x_2[\times]y_1, y_2[\subset \mathbb{R}^2$ be an open bounded rectangular domain.*

Then Ω is an open set with Lipschitz boundary.

Proof. It is clear that we only have two cases for x to show that Ω verifies the property of the definition.

Case of an edge: x is not a vertex

In this case, we consider V as presented in Figure 16 for $\varepsilon > 0$ sufficiently small to avoid a vertex being in V . Then, it is sufficient to make a translation and a homothety to transform V into $Q =]-1, 1[^2$

$$\forall y \in V, \quad \varphi(y) = \frac{y - x}{\varepsilon}.$$

It is clear that φ and φ^{-1} are Lipschitz continuous functions and the other conditions hold.

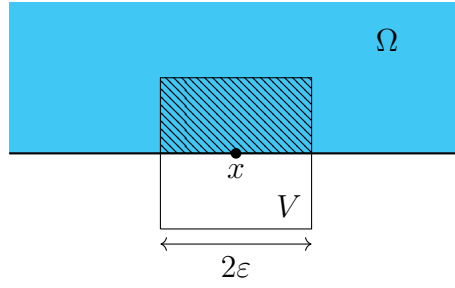


Figure 16: Neighbourhood around x . A part of Ω is represented by the domain in blue and V is the square of side 2ε centered in x . The hatched part is the intersection of V and Ω .

Case of a vertex:

From the previous explanation, it is clear that composition of rotations, translations and homotheties, do not change the Lipschitz property. So, after such operations, we can obtain the situation as in Figure 17.

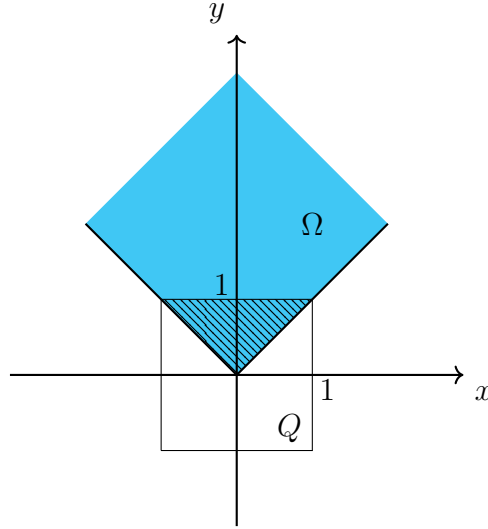


Figure 17: Neighbourhood around x after rotation, translation and homothety. A part of the transformed Ω is represented by the domain in blue and $V = Q$ is the square $] - 1, 1[^2$. The hatched part is the intersection of Q and Ω .

We observe that

$$\begin{aligned} (x, y) \in Q \cap \Omega &\iff y > 0 \text{ and } y > x \text{ and } y > -x \\ &\iff y - |x| > 0 \end{aligned}$$

In this case, we can define

$$\varphi(x, y) = (x, y - |x|) \quad \text{and so} \quad \varphi^{-1}(x, y) = (x, y + |x|).$$

Thanks to the previous observation, it is clear that φ verifies the second and the third points of the definition. It remains to show that they are Lipschitz continuous functions:

$$\begin{aligned}
\|\varphi(x, y) - \varphi(x', y')\|_2^2 &= |x - x'|^2 + |y - y' + |x'| - |x||^2 \\
&= |x - x'|^2 + |y - y'|^2 + ||x'| - |x||^2 + 2|y - y'| ||x'| - |x|| \\
&\leq 2|x - x'|^2 + |y - y'|^2 + 2|y - y'| |x - x'| \\
&\leq 4\|(x, y) - (x', y')\|_2^2.
\end{aligned}$$

Thus, φ is a Lipschitz continuous function and we can do the analogue for φ^{-1} . □

Thanks to Stein's Theorem, we can prove the three main theorems. First, we state a density result on $H^n(\Omega)$ for a rectangular domain.

Theorem 7 (Density of $\mathcal{C}_c^\infty(\overline{\Omega})$). *Let $\Omega \subset \mathbb{R}^2$ be an open bounded rectangular domain and $n \in \mathbb{N}^*$. $\mathcal{C}_c^\infty(\overline{\Omega})$ denotes the set of $\mathcal{C}^\infty$ -compactly supported functions on $\mathbb{R}^2$, restricted to Ω . Then $\mathcal{C}_c^\infty(\overline{\Omega})$ is dense in $H^n(\Omega)$.*

Proof. Let $u \in H^n(\Omega)$ and $\varepsilon > 0$.

Thanks to Lemma 6, we have shown that Ω is a bounded open set with Lipschitz boundary. Thus, by Stein's Theorem (Theorem 5), there exists a bounded linear extension operator $P : H^n(\Omega) \rightarrow H^n(\mathbb{R}^2)$.

Then $Pu \in H^n(\mathbb{R}^2)$ and, by density of $\mathcal{C}_c^\infty(\mathbb{R}^2)$ in $H^n(\mathbb{R}^2)$, there exists $v \in \mathcal{C}_c^\infty(\mathbb{R}^2)$ such that

$$\|Pu - v\|_{H^n(\mathbb{R}^2)} \leq \varepsilon.$$

We still denote by v the restriction of v to Ω . Then $v \in \mathcal{C}_c^\infty(\overline{\Omega})$ and we note that

$$\|Pu - v\|_{H^n(\Omega)} \leq \|Pu - v\|_{H^n(\mathbb{R}^2)} \leq \varepsilon.$$

By definition of P , $Pu|_\Omega = u$ and finally:

$$\|u - v\|_{H^n(\Omega)} \leq \varepsilon.$$

□

Theorem 7 is an essential theorem required to prove Trace Theorem and Green's formula in the case of a rectangular domain. Before presenting these theorems, we remind that we can define a measure on the boundary of Ω . Indeed, the superficial measure is just the 1-dimensional Hausdorff measure on the boundary.

Theorem 8 (Trace Theorem). *Let $\Omega \subset \mathbb{R}^2$ be an open bounded rectangular domain. σ denotes the superficial measure on $\partial\Omega$.*

We define

$$\begin{aligned}
\gamma_0 : \mathcal{C}_c^\infty(\overline{\Omega}) &\longrightarrow L^2(\partial\Omega, \sigma) \\
u &\longmapsto u|_{\partial\Omega}
\end{aligned}$$

Then γ_0 can be extended to $H^1(\Omega)$ into a continuous linear map called Trace on the boundary.

Proof. As γ_0 is linear, it is sufficient to state the continuity inequality. We first prove such inequality on smooth functions.

Let $u \in \mathcal{C}_c^\infty(\overline{\Omega})$. We want to prove that there exists $C > 0$ such that

$$\|u\|_{L^2(\partial\Omega)} \leq C \|u\|_{H^1(\Omega)}. \quad (11)$$

It is sufficient to prove such inequality on a side, as it is the same on the others. We consider the side $\{x_1\} \times]y_1, y_2[$. We take $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^2)$ such that $\chi \geq 0$, $\chi = 1$ on the side $\{x_1\} \times]y_1, y_2[$ and $\chi = 0$ on $\{x_2\} \times]y_1, y_2[$.

$$\begin{aligned} \int_{y_1}^{y_2} (u(x_1, y))^2 dy &= \int_{y_1}^{y_2} \chi(x_1, y) u(x_1, y)^2 dy \\ &= - \int_{x_1}^{x_2} \int_{y_1}^{y_2} \frac{\partial}{\partial x} [\chi(x, y) u(x, y)^2] dx dy \\ &= - \int_{x_1}^{x_2} \int_{y_1}^{y_2} \left[\frac{\partial \chi}{\partial x}(x, y) u(x, y)^2 + 2u(x, y) \chi(x, y) \frac{\partial u}{\partial x}(x, y) \right] dx dy \\ &\leq \max_{\Omega} \left(\left| \frac{\partial \chi}{\partial x} \right|, \chi \right) \left(\|u\|_{L^2(\Omega)}^2 + 2 \|u\|_{L^2(\Omega)} \left\| \frac{\partial u}{\partial x} \right\|_{L^2(\Omega)} \right) \end{aligned}$$

So, it states (11) with $C = \sqrt{3 \max_{\Omega} \left(\left| \frac{\partial \chi}{\partial x} \right|, \chi \right)}$.

From Theorem 7, we have that for every $u \in H^1(\Omega)$, there exists a sequence $(u_n)_{n \in \mathbb{N}}$ in $\mathcal{C}_c^\infty(\overline{\Omega})$ which tends in $H^1(\Omega)$ to u . Using (11) and that $(u_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $H^1(\Omega)$, we obtain that it is a Cauchy sequence in $L^2(\partial\Omega)$. By completeness of $L^2(\partial\Omega)$, $(\gamma_0(u_n))_{n \in \mathbb{N}}$ tends to a function denoted by $\gamma_0(u)$ in $L^2(\partial\Omega)$. Thus, we can extend γ_0 to $H^1(\Omega)$. Moreover, Inequality (11) holds at the limit which ensures the continuity of the extension of γ_0 . $\square$

Trace Theorem gives a meaning to Assumption 3. Indeed, we actually assume that there exists $v \in \mathcal{C}^2(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^2(\Omega))$ such that

$$\gamma_0(\nabla v)(x, t) \cdot \vec{n}(x) = \begin{cases} p(t) & \forall (x, t) \in \partial_1 \Omega \times [0, T], \\ 0 & \forall (x, t) \in \partial \Omega \setminus \partial_1 \Omega \times [0, T], \end{cases}$$

where $\vec{n}$ denotes the exterior unit normal vector.

Trace Theorem is also an essential component of the proof of Green's formula.

Theorem 9 (Green's formula). *Green's formula for the Laplacian holds for an open bounded rectangular domain $\Omega \subset \mathbb{R}^2$:*

$$\forall u \in H^2(\Omega), \forall v \in H^1(\Omega), \quad \int_{\Omega} \Delta u(x) v(x) dx = \int_{\partial\Omega} \frac{\partial u}{\partial n}(x) v(x) d\sigma(x) - \int_{\Omega} \nabla u(x) \cdot \nabla v(x) dx,$$

where we improperly write $\int_{\partial\Omega} \frac{\partial u}{\partial n}(x) v(x) d\sigma(x)$ for $\int_{\partial\Omega} \gamma_0(\nabla u)(x) \cdot \vec{n}(x) \gamma_0(v)(x) d\sigma(x)$.

Proof. First, we note that Green's formula can be easily proved for $u, v \in \mathcal{C}_c^\infty(\overline{\Omega})$ by making the various terms of the equation explicit. The formula is just a consequence of two integrations by part according to x and y and the definition of the normal derivative on each edge.

Then, it is sufficient to show that the equation holds at the limit. The first term can be seen as an inner product of Δu and v . Thus, if $(u_n)_{n \in \mathbb{N}}$ (respectively $(v_n)_{n \in \mathbb{N}}$) is a sequence of $\mathcal{C}_c^\infty(\overline{\Omega})$ -functions which tends to u in $H^2(\Omega)$ (respectively to v in $H^1(\Omega)$), $(\Delta u_n)_{n \in \mathbb{N}}$ tends to Δu and $(v_n)_{n \in \mathbb{N}}$ to v in $L^2(\Omega)$. So the first term holds at the limit. An analogous reasoning can be done on the last term.

It remains to analyse the second term. According to Trace Theorem 8, $(\gamma_0(v_n))_{n \in \mathbb{N}}$ tends to $\gamma_0(v)$ in $L^2(\partial\Omega)$ by continuity of γ_0 . Similarly, as $(\nabla u_n)_{n \in \mathbb{N}}$ tends to ∇u in $H^1(\Omega)$ since $(u_n)_{n \in \mathbb{N}}$ tends to u in $H^2(\Omega)$, then $(\gamma_0(\nabla u_n))_{n \in \mathbb{N}}$ tends to $\gamma_0(\nabla u)$ in $L^2(\partial\Omega)$.

As a conclusion, every term holds to the limit and Green's formula holds for $u \in H^2(\Omega)$ and for $v \in H^1(\Omega)$. $\square$

B Proof of well-posedness and Energy estimates

In this appendix, we present the proofs of the theorems and the corollary from Subsection 1.2. First, we develop the proof of the existence and the uniqueness of a weak solution to Problem (1) using Hille-Yosida Theory. Then we make the proofs of the energy estimate and of the uniqueness of the weak solution explicit.

B.1 Proof of Theorem 1

In this subsection, we consider that Assumption 1 holds without recalling it at each theorem. In the case of Assumptions 2 and 3, we refer to these hypotheses when it is required. To apply Hille-Yosida Theorem, we first need to define a Hilbert space.

Lemma 10. *Let $\mathcal{H} = H^1(\Omega) \times L^2(\Omega)$ with the inner product*

$$\left\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \middle| \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \right\rangle = \langle u_1 \mid v_1 \rangle_{L^2(\Omega)} + \langle c^2 \nabla u_1 \mid \nabla v_1 \rangle_{L^2(\Omega)} + \langle u_2 \mid v_2 \rangle_{L^2(\Omega)}.$$

Then $\mathcal{H}$ is a Hilbert space.

Proof. First, it is clear that the operation defined is a sesquilinear form which verifies the conjugate symmetry and the positiveness, using the properties of the inner product in $L^2(\Omega)$. It remains to show the definiteness and completeness for the associated norm.

First, using Assumption 1, it is clear that

$$\forall (u_1, u_2) \in \mathcal{H}, \quad c_0^2 \left(\|u_1\|_{H^1(\Omega)}^2 + \|u_2\|_{L^2(\Omega)}^2 \right) \leq \left\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \middle| \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \right\rangle \leq \|u_1\|_{H^1(\Omega)}^2 + \|u_2\|_{L^2(\Omega)}^2. \quad (12)$$

Thus, definiteness of the product is ensured by definiteness of the inner products on $L^2(\Omega)$ and $H^1(\Omega)$ and the second inequality.

Finally, first inequality in (12) ensures that any Cauchy sequence for $\mathcal{H}$ $(u_{1,n}, u_{2,n})_{n \in \mathbb{N}}$ gives two Cauchy sequences $(u_{1,n})_{n \in \mathbb{N}}$ in $H^1(\Omega)$ and $(u_{2,n})_{n \in \mathbb{N}}$ in $L^2(\Omega)$ for their common norm. Thus, by completeness of these Hilbert spaces, it yields that $(u_{1,n}, u_{2,n})_{n \in \mathbb{N}}$ is convergent in $H^1(\Omega) \times L^2(\Omega)$.

This ends the proof. $\square$

Using the Hilbertian property of $\mathcal{H}$, we can easily state that the inhomogeneous Neumann wave operator generates a strongly continuous semi-group. Indeed, we can use here the Hilbertian form of Hille-Yosida Theorem and only demonstrate that the operator is (up to a constant) a maximal accretive operator.

Theorem 11. *We define a linear operator on $\mathcal{H}$ as*

$$A = \begin{pmatrix} 0 & -\text{Id} \\ -c^2 \Delta & 0 \end{pmatrix},$$

with the domain of definition

$$D(A) = D(\Delta_N) \times H^1(\Omega),$$

where $D(\Delta_N) = \left\{ u \in H^1(\Omega) \mid \Delta u \in L^2(\Omega) \text{ and } \forall v \in H^1(\Omega), \int_{\Omega} \Delta u v = - \int_{\Omega} \nabla u \cdot \nabla v \right\}$ denotes the domain of definition of the Neumann Laplacian.

Then A generates a strongly continuous semi-group $S(t)$ on $\mathcal{H}$.

Proof. As $\mathcal{H}$ is a Hilbert space, it is sufficient to prove that there exists $\omega \geq 0$ such that $A + \omega$ is a maximal accretive operate (or conversely that $-A - \omega$ is maximal dissipative). First, we study the property of accretivity.

Let $(u, v) \in D(A)$. We calculate using that $u \in D(\Delta_N)$

$$\begin{aligned} \left\langle A \begin{pmatrix} u \\ v \end{pmatrix} \middle| \begin{pmatrix} u \\ v \end{pmatrix} \right\rangle &= -\langle v | u \rangle_{L^2(\Omega)} - \langle c^2 \nabla v | \nabla u \rangle_{L^2(\Omega)} - \langle c^2 \Delta u | v \rangle_{L^2(\Omega)} \\ &= -\langle v | u \rangle_{L^2(\Omega)} - \langle c^2 \nabla v | \nabla u \rangle_{L^2(\Omega)} - \langle \Delta u | c^2 v \rangle_{L^2(\Omega)} \\ &= -\langle v | u \rangle_{L^2(\Omega)} - \langle c^2 \nabla v | \nabla u \rangle_{L^2(\Omega)} + \langle \nabla u | \nabla(c^2 v) \rangle_{L^2(\Omega)} \\ &= -\langle v | u \rangle_{L^2(\Omega)} - \langle c^2 \nabla v | \nabla u \rangle_{L^2(\Omega)} + \langle \nabla u | c^2 \nabla v \rangle_{L^2(\Omega)} + 2 \langle \nabla u | cv \nabla c \rangle_{L^2(\Omega)} \\ &= -\langle v | u \rangle_{L^2(\Omega)} + 2i \operatorname{Im} \left(\langle \nabla u | c^2 \nabla v \rangle_{L^2(\Omega)} \right) + 2 \langle \nabla u | cv \nabla c \rangle_{L^2(\Omega)} \end{aligned}$$

Then, taking the real part and using Cauchy-Schwarz's inequality, it yields

$$\begin{aligned} \operatorname{Re} \left\langle A \begin{pmatrix} u \\ v \end{pmatrix} \middle| \begin{pmatrix} u \\ v \end{pmatrix} \right\rangle &\geq -\|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} - 2 \|c \nabla u\|_{L^2(\Omega)} \|v \nabla c\|_{L^2(\Omega)} \\ &\geq -\|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} - 2 \|c \nabla u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \|\nabla c\|_{\infty} \\ &\geq -(1 + 2 \|\nabla c\|_{\infty}) \|v\|_{L^2(\Omega)} \sqrt{\|u\|_{L^2(\Omega)}^2 + \|c \nabla u\|_{L^2(\Omega)}^2} \\ &\geq -\frac{1 + 2 \|\nabla c\|_{\infty}}{2} \left(\|u\|_{L^2(\Omega)}^2 + \|c \nabla u\|_{L^2(\Omega)}^2 + \|v\|_{L^2(\Omega)}^2 \right) \end{aligned}$$

where $\|\nabla c\|_{\infty}$ denotes the supremum over $\overline{\Omega}$ which is compact, since $c \in \mathcal{C}^1(\overline{\Omega})$.

Thus, $A + \|\nabla c\|_{\infty} + \frac{1}{2}$ is accretive.

Let $\lambda = 1 + 2 \|\nabla c\|_{\infty}$. It remains to show that $A + \lambda$ is surjective to prove that $A + \|\nabla c\|_{\infty} + \frac{1}{2}$ is maximal accretive. Let $(f, g) \in \mathcal{H}$. We search $(u, v) \in D(A)$ such that $(A + \lambda) \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}$.

We define the sesquilinear form

$$\begin{aligned} a : H^1(\Omega) \times H^1(\Omega) &\longrightarrow \mathbb{C} \\ (u, v) &\longmapsto \lambda^2 \langle u | v \rangle_{L^2(\Omega)} + \langle c^2 \nabla u | \nabla v \rangle_{L^2(\Omega)} + 2 \langle c \nabla u | v \nabla c \rangle_{L^2(\Omega)} \end{aligned}$$

We show that a is continuous using Cauchy-Schwarz's inequality and $c(x) \leq 1$. For all $u, v \in H^1(\Omega)$,

$$\begin{aligned} |a(u, v)| &\leq \lambda^2 \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + 2 \|\nabla c\|_{\infty} \|\nabla u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \\ &\leq (1 + \lambda^2 + 2 \|\nabla c\|_{\infty}) \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}. \end{aligned}$$

Then, we prove that a is coercive. For all $u \in H^1(\Omega)$,

$$\begin{aligned} a(u, u) &= \lambda^2 \|u\|_{L^2(\Omega)}^2 + \|c \nabla u\|_{L^2(\Omega)}^2 + 2 \langle c \nabla u | u \nabla c \rangle_{L^2(\Omega)} \\ &\geq \lambda^2 \|u\|_{L^2(\Omega)}^2 + \|c \nabla u\|_{L^2(\Omega)}^2 - 2 \|c \nabla u\|_{L^2(\Omega)} \|u \nabla c\|_{L^2(\Omega)}. \end{aligned}$$

For $\varepsilon > 0$, using $2ab \leq a^2 + b^2$,

$$\begin{aligned} a(u, u) &\geq \lambda^2 \|u\|_{L^2(\Omega)}^2 + \|c \nabla u\|_{L^2(\Omega)}^2 - \varepsilon^2 \|c \nabla u\|_{L^2(\Omega)}^2 - \frac{1}{\varepsilon^2} \|u \nabla c\|_{L^2(\Omega)}^2 \\ &\geq \left(\lambda^2 - \frac{\|\nabla c\|_\infty^2}{\varepsilon^2} \right) \|u\|_{L^2(\Omega)}^2 + (1 - \varepsilon^2) \|c \nabla u\|_{L^2(\Omega)}^2. \end{aligned}$$

Using the choice of λ and for $\varepsilon = \frac{1}{2}$, we obtain

$$a(u, u) \geq (1 + 4 \|\nabla c\|_\infty) \|u\|_{L^2(\Omega)}^2 + \frac{3c_0^2}{4} \|\nabla u\|_{L^2(\Omega)}^2.$$

Then a is coercive. On the other hand, we define the continuous semilinear form

$$\begin{aligned} l : H^1(\Omega) &\longrightarrow \mathbb{C} \\ v &\longmapsto \langle g + \lambda f \mid v \rangle_{L^2(\Omega)}. \end{aligned}$$

By Lax-Milgram's Theorem, there exists a unique $u \in H^1(\Omega)$ such that for every $v \in H^1(\Omega)$:

$$a(u, v) = l(v) \quad (13)$$

Then, for every $\varphi \in \mathcal{D}(\Omega)$

$$\begin{aligned} \lambda^2 \langle u, \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} + \langle c^2 \nabla u, \nabla \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} + 2 \langle c \nabla u, \varphi \nabla c \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} &= \langle g + \lambda f, \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} \\ \lambda^2 \langle u, \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} - \langle c^2 \Delta u, \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)} &= \langle g + \lambda f, \varphi \rangle_{\mathcal{D}'(\Omega) \times \mathcal{D}(\Omega)}. \end{aligned}$$

Thus, in $\mathcal{D}'(\Omega)$,

$$-c^2 \Delta u + \lambda^2 u = g + \lambda f. \quad (14)$$

Since, f, g and $u \in L^2(\Omega)$ and that c is bounded, then $\Delta u \in L^2(\Omega)$. Furthermore, Equation (13) and Equation (14) ensure that

$$\forall v \in H^1(\Omega), \quad \int_{\Omega} \Delta u v = - \int_{\Omega} \nabla u \cdot \nabla v.$$

Thus, $u \in D(\Delta_N)$. We define also $v = \lambda u - f \in L^2(\Omega)$. We easily check from these definitions that $(A + \lambda) \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}$. Thus, the proof is complete. $\square$

Through the existence of a semi-group, we can now show a theorem of existence and uniqueness of a weak solution to Problem (1) in the case of homogeneous Neumann boundary conditions (*ie.* $p \equiv 0$) and with an internal source $f(t, x)$.

Lemma 12. *Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$. Let $f \in \mathcal{C}(\mathbb{R}_+, L^2(\Omega))$.*

Then there exists a unique $u \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$ such that

$$\forall t > 0, \quad u(t) \in D(\Delta_N),$$

$$u(0) = u_0, \quad \frac{\partial u}{\partial t}(0) = v_0,$$

and such that in $\mathcal{D}'(\mathbb{R}_+^ \times \Omega)$*

$$a(x) \frac{\partial^2 u}{\partial t^2}(t, x) - \Delta u(t, x) = f(t, x).$$

Proof. First, we begin with the existence. We denote by $(S(t))_{t \geq 0}$ the semi-group on $\mathcal{H}$ whose infinitesimal generator is A . Then, we define using Duhamel's formula

$$\forall t \geq 0, \quad \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} = S(t) \begin{pmatrix} u_0 \\ v_0 \end{pmatrix} + \int_0^t S(t-\tau) \begin{pmatrix} 0 \\ c^2 f(\tau) \end{pmatrix} d\tau.$$

According to Theorem on the infinitesimal generators Duhamel's formula, we obtain that for all $\psi \in \mathcal{C}_c^\infty(\mathbb{R}_+^*)$

$$\int_0^{+\infty} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} \psi(t) dt \in D(A),$$

and

$$A \int_0^{+\infty} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} \psi(t) dt = \int_0^{+\infty} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} \psi'(t) dt + \int_0^{+\infty} \begin{pmatrix} 0 \\ c^2 f(t) \end{pmatrix} \psi(t) dt.$$

Then

$$\int_0^\infty u(t) \psi'(t) dt = - \int_0^\infty v(t) \psi(t) dt,$$

and

$$\int_0^\infty v(t) \psi'(t) dt + c^2(x) \int_0^\infty f(t) \psi(t) dt = -c^2(x) \int_0^\infty \Delta u(t) \psi(t) dt.$$

Then for every $\varphi \in \mathcal{C}_c^\infty(\Omega)$, it yields

$$\langle u, \psi' \varphi \rangle_{\mathcal{D}'(\mathbb{R}_+^* \times \Omega) \times \mathcal{D}(\mathbb{R}_+^* \times \Omega)} = - \langle v, \psi \varphi \rangle_{\mathcal{D}'(\mathbb{R}_+^* \times \Omega) \times \mathcal{D}(\mathbb{R}_+^* \times \Omega)},$$

$$\langle v, \psi' \varphi \rangle_{\mathcal{D}'(\mathbb{R}_+^* \times \Omega) \times \mathcal{D}(\mathbb{R}_+^* \times \Omega)} + \langle c^2 f, \psi \varphi \rangle_{\mathcal{D}'(\mathbb{R}_+^* \times \Omega) \times \mathcal{D}(\mathbb{R}_+^* \times \Omega)} = - \langle c^2 \Delta u, \psi \varphi \rangle_{\mathcal{D}'(\mathbb{R}_+^* \times \Omega) \times \mathcal{D}(\mathbb{R}_+^* \times \Omega)}.$$

By density of the span of the functions with separated variables, we can replace $\psi \varphi$ by any function $\Phi \in \mathcal{D}(\mathbb{R}_+^* \times \Omega)$. Thus, the first equation means that in $\mathcal{D}'(\mathbb{R}_+^* \times \Omega)$

$$\frac{\partial u}{\partial t} = v,$$

and the second equation means that

$$\frac{\partial v}{\partial t} = c^2 \Delta u + c^2 f.$$

Thus, since $u(0) = u_0$ and $v(0) = v_0$, u verifies the initial conditions. Furthermore, from the two equations, we obtain that u verifies the expected equation.

Finally, it remains to prove the domain for $u(t)$. We have shown that $\int_0^\infty u(t) \psi(t) dt \in D(\Delta_N)$.

Thus, as it is true for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R}_+^*)$, then for $t > 0$, $\Delta u(t) \in L^2(\Omega)$. Furthermore, the equality

$$\int_0^\infty \int_\Omega \Delta u(t)(x) v(x) dx \psi(t) dt = - \int_0^\infty \int_\Omega \nabla u(t)(x) \cdot \nabla v(x) dx \psi(t) dt,$$

holds for every $\psi \in \mathcal{C}_c^\infty(\mathbb{R}_+^*)$ and $v \in H^1(\Omega)$ and thus the equality holds for $t > 0$.

As a conclusion, $u(t) \in D(\Delta_N)$ for $t > 0$ and u is the solution.

Then, we prove the uniqueness. Let u_1 and u_2 be two solutions as in the lemma. We define $u := u_1 - u_2$. Then u verifies the same equations than u_1 and u_2 with $u_0 = 0$, $v_0 = 0$

and $f = 0$. By applying similar calculus than in the existence part backward, we obtain that for all $\psi \in \mathcal{C}_c^\infty(\mathbb{R}_+^*)$

$$\int_0^\infty u(t)\psi'(t) dt = - \int_0^\infty \frac{\partial u}{\partial t}(t)\psi(t) dt,$$

and

$$\int_0^\infty \frac{\partial u}{\partial t}(t)\psi'(t) dt = -c^2(x) \int_0^\infty \Delta u(t)\psi(t) dt.$$

We easily check that

$$\int_0^{+\infty} \begin{pmatrix} u(t) \\ \frac{\partial u}{\partial t}(t) \end{pmatrix} \psi(t) dt \in D(A).$$

Then, from uniqueness of the semi-group associated to an infinitesimal generator, then

$$\forall t \geq 0, \quad \begin{pmatrix} u(t) \\ \frac{\partial u}{\partial t}(t) \end{pmatrix} = S(t) \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0.$$

Then $u_1 = u_2$ and this ends the proof. $\square$

In Lemma 12, the fact that $u(t) \in D(\Delta_N)$ ensures that if the solution is smooth enough (ie. $u(t) \in H^2(\Omega)$), using Trace Theorem, u verifies the homogeneous Neumann boundary condition. Thus, this condition is the weak formulation of the boundary condition in the case that $p \equiv 0$.

Finally, using Assumptions 2 and 3, we can prove Theorem 1. The existence is stated here in a weak formulation of the problem considering weak derivatives.

Theorem 13 (Existence-Uniqueness of a weak solution to Problem (1)). *Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$.*

We consider that Assumptions 1, 2 and 3 hold and we denote by v the same function as in Assumption 3. Then there exists a unique $u \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$ such that

$$u(0) = u_0, \quad \frac{\partial u}{\partial t}(0) = v_0,$$

$$\forall t > 0, \quad u(t) - v(t) \in D(\Delta_N), \tag{**}$$

and such that in $\mathcal{D}'(\mathbb{R}_+^ \times \Omega)$*

$$a(x) \frac{\partial^2 u}{\partial t^2}(t, x) - \Delta u(t, x) = 0.$$

Theorem 13 is the same theorem as Theorem 1. Indeed, the only difference is Equation (**). This condition means that $\Delta u(t) - \Delta v(t) \in L^2(\Omega)$ and since $v(t) \in H^2(\Omega)$, then $\Delta u(t) \in L^2(\Omega)$. Furthermore Equation (**) means that

$$\forall t > 0, \forall w \in H^1(\Omega), \quad \int_\Omega \Delta u(t)w - \int_\Omega \Delta v(t)w = \int_\Omega \nabla v(t) \cdot \nabla w - \int_\Omega \nabla u(t) \cdot \nabla w.$$

Or, as $v(t) \in H^2(\Omega)$ and by Assumption 3, Green's Formula gives us that

$$\forall t > 0, \forall w \in H^1(\Omega), \quad \int_\Omega \Delta v(t)w = \int_{\partial_1 \Omega} p(t)w d\sigma - \int_\Omega \nabla v(t) \cdot \nabla w.$$

Thus, it yields

$$\forall t > 0, \forall w \in H^1(\Omega), \quad \int_{\Omega} \Delta u(x, t) w(x) \, dx = \int_{\partial_1 \Omega} p(t) w(x) \, d\sigma(x) - \int_{\Omega} \nabla u(x, t) \cdot \nabla w(x) \, dx,$$

which is exactly Equation (*) from Theorem 1.

Proof. The proof is based on Lemma 12. To apply it, we reason through an analysis-synthesis method.

First, we assume that there exists $u \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$ which verifies the results from the theorem.

We define $w := u - v$. By assumptions on v , $w \in \mathcal{C}^1(\mathbb{R}_+, L^2(\Omega)) \cap \mathcal{C}(\mathbb{R}_+, H^1(\Omega))$. Furthermore, we have

$$\forall t > 0, \quad w(t) \in D(\Delta_N),$$

and

$$w(0) = u_0 - v(0), \quad \frac{\partial w}{\partial t}(0) = v_0 - \frac{\partial v}{\partial t}(0).$$

Since $v \in \mathcal{C}^2(\mathbb{R}_+, L^2(\Omega))$, $w(0) \in H^1(\Omega)$ and $\frac{\partial w}{\partial t}(0) \in L^2(\Omega)$.

Finally, we can calculate that in $\mathcal{D}'(\mathbb{R}_+^* \times \Omega)$

$$a(x) \frac{\partial^2 w}{\partial t^2}(t, x) - \Delta w(t, x) = -a(x) \frac{\partial^2 v}{\partial t^2}(t, x) + \Delta v(t, x).$$

If we define $f := -a \frac{\partial^2 v}{\partial t^2} + \Delta v$, by Assumption 3 and since a is bounded, then $f \in \mathcal{C}(\mathbb{R}_+, L^2(\Omega))$.

Thus, from Lemma 12, w is the unique function which verifies these properties. Since v is chosen, it is clear that u is unique (the formal proof would have consisted in taking the difference between two solutions).

Secondly, we take w solution to the exposed problem in the analysis part. In other terms, we take w as in Lemma 12 for $u_0 - v(0)$, $v_0 - \frac{\partial v}{\partial t}(0)$ and $f(t, x) = -a(x) \frac{\partial^2 v}{\partial t^2}(t, x) + \Delta v(t, x)$.

We easily check that $u := v + w$ suits for the theorem. This concludes the proof. $\square$

B.2 Proof of the energy estimates

In this subsection, we propose a demonstration of Theorem 2 and Corollary 3 inspired from demonstrations of similar results from [5]. First, we prove Theorem 2.

Proof. Let $u \in H^2(\Omega \times]0, T[)$ be a weak solution to either Problem (1), (2) or (3) for $p \equiv 0$. We denote by $\Sigma_{abs} \subset \partial\Omega \times]0, T[$ the part on which the absorbing boundary condition is set in the considered problem. Potentially, $\Sigma_{abs} = \emptyset$.

Since $u \in H^2(\Omega \times]0, T[)$, the acoustic wave equation holds in $L^2(\Omega \times]0, T[)$. Then we do the inner product with $\frac{\partial u}{\partial t} \in L^2(\Omega \times]0, t[)$ for $t \in]0, T[$,

$$2 \int_0^t \int_{\Omega} a(x) \frac{\partial^2 u}{\partial t^2} \frac{\partial u}{\partial t} \, dx \, d\tau = 2 \int_0^t \int_{\Omega} \Delta u \frac{\partial u}{\partial t} \, dx \, d\tau.$$

Using Green's formula from Theorem 9, we obtain

$$\int_0^t \int_{\Omega} \frac{\partial}{\partial t} \left(a(x) \left| \frac{\partial u}{\partial t} \right|^2 \right) dx d\tau = 2 \int_0^t \int_{\partial\Omega} \frac{\partial u}{\partial n} \frac{\partial u}{\partial t} dx d\tau - 2 \int_0^t \int_{\Omega} \nabla u \cdot \frac{\partial \nabla u}{\partial t} dx d\tau.$$

Then

$$\int_0^t \int_{\Omega} \frac{\partial}{\partial t} \left(a(x) \left| \frac{\partial u}{\partial t} \right|^2 \right) dx d\tau + \int_0^t \int_{\Omega} \frac{\partial}{\partial t} (|\nabla u|^2) dx d\tau = 2 \int_0^t \int_{\partial\Omega} \frac{\partial u}{\partial n} \frac{\partial u}{\partial t} dx d\tau.$$

Thus, by the definition of the boundary conditions and since $p \equiv 0$, it yields

$$\begin{aligned} \int_{\Omega} a(x) \left| \frac{\partial u}{\partial t} \right|^2 (x, t) dx + \int_{\Omega} |\nabla u|^2 (x, t) dx &= \int_{\Omega} a(x) |v_0|^2 (x) dx + \int_{\Omega} |\nabla u_0|^2 (x) dx \\ &\quad - 2 \int_{\Sigma_{abs}} 1_{\tau \leq t} \left| \frac{\partial u}{\partial t} \right|^2 dx d\tau. \end{aligned}$$

Since the last integral from the right-hand side is negative, this concludes the proof. $\square$

Finally, we deduce Corollary 3.

Proof. Let u_1 and $u_2 \in H^2(\Omega \times]0, T[)$ be two weak solutions to either Problem (2) or (3). Then, we define $v := u_1 - u_2$.

It is obvious that $v \in H^2(\Omega \times]0, T[)$ and that v is a weak solution to the same problem as u_1 and u_2 but with $u_0 \equiv 0$, $v_0 \equiv 0$ and $p \equiv 0$. Then, we can apply Theorem 2:

$$\left\| \sqrt{a} \frac{\partial v}{\partial t}(t) \right\|_{L^2(\Omega)}^2 + \|\nabla v(t)\|_{L^2(\Omega)}^2 \leq \|\sqrt{a} v_0\|_{L^2(\Omega)}^2 + \|\nabla u_0\|_{L^2(\Omega)}^2.$$

Thus for every $t \in]0, T[$, $\left\| \sqrt{a} \frac{\partial v}{\partial t}(t) \right\|_{L^2(\Omega)} = 0 = \|\nabla v(t)\|_{L^2(\Omega)}$.

By Assumption 1, $a \geq 1$ and so $\nabla v = \frac{\partial v}{\partial t} = 0$.

Therefore, v is a constant function and $v(0) = 0$, so $v \equiv 0$.

This concludes that $u_1 \equiv u_2$. $\square$

C Numerical integration of the forward problems and simulations

In this section, we describe the implementation of the discretisation in time to solve the forward problems. This section is an extension of Subsection 1.3.

C.1 The Newmark numerical integration for Problem (1)

In Subsection 1.3, we stated the discretisation in space of Problem (1) as

$$\begin{cases} M_h \frac{d^2 U_h}{dt^2}(t) + K_h U_h(t) = B_h(t), \\ U_h(0) = \frac{dU_h}{dt}(0) = 0. \end{cases} \quad (6)$$

We use the discretisation in time $0 = t_0 < t_1 < \dots < t_N = T$ with $t_k = k \, dt = k \frac{T}{N}$. We apply the Newmark numerical integration to solve (6). In this scheme, we consider two parameters $\frac{1}{2} \leq \delta \leq 1$ and $0 \leq \theta \leq \frac{1}{2}$. We approximate U_h and its derivative by three sequences $(U_{h,k})_k$, $(\dot{U}_{h,k})_k$ and $(\ddot{U}_{h,k})_k$ defined by

$$\begin{cases} M_h \ddot{U}_{h,k+1} + K_h U_{h,k+1} = B_h(t_{k+1}), \\ \dot{U}_{h,k+1} = \dot{U}_{h,k} + dt \left(\delta \ddot{U}_{h,k+1} + (1 - \delta) \ddot{U}_{h,k} \right), \\ U_{h,k+1} = U_{h,k} + dt \dot{U}_{h,k} + \frac{dt^2}{2} \left(2\theta \ddot{U}_{h,k+1} + (1 - 2\theta) \ddot{U}_{h,k} \right). \end{cases} \quad (15)$$

System (15) can be rewritten almost as a θ -scheme using only the sequence $(U_{h,k})_k$

$$\begin{aligned} M_h \frac{U_{h,k+1} - 2U_{h,k} + U_{h,k-1}}{dt^2} + K_h \left(\theta U_{h,k+1} + \left(\frac{1}{2} + \delta - 2\theta \right) U_{h,k} + \left(\frac{1}{2} - \delta + \theta \right) U_{h,k-1} \right) \\ = \theta B_h(t_{k+1}) + \left(\frac{1}{2} + \delta - 2\theta \right) B_h(t_k) + \left(\frac{1}{2} - \delta + \theta \right) B_h(t_{k-1}). \end{aligned}$$

Proof. From the third equation, we obtain

$$\begin{aligned} M_h \frac{U_{h,k+1} - 2U_{h,k} + U_{h,k-1}}{dt^2} &= M_h \frac{U_{h,k+1} - U_{h,k}}{dt^2} - M_h \frac{U_{h,k} - U_{h,k-1}}{dt^2} \\ &= M_h \frac{\dot{U}_{h,k} - \dot{U}_{h,k-1}}{dt} + \frac{M_h}{2} \left(2\theta \ddot{U}_{h,k+1} + (1 - 4\theta) \ddot{U}_{h,k} - (1 - 2\theta) \ddot{U}_{h,k-1} \right) \end{aligned}$$

By the second equation, it yields

$$M_h \frac{U_{h,k+1} - 2U_{h,k} + U_{h,k-1}}{dt^2} = \theta M_h \ddot{U}_{h,k+1} + \left(\frac{1}{2} + \delta - 2\theta \right) M_h \ddot{U}_{h,k} + \left(\frac{1}{2} - \delta + \theta \right) M_h \ddot{U}_{h,k-1}.$$

The first equation concludes. $\square$

To solve this recurrent sequence of order 2, we have to set the initial values according to the initial conditions. Thus $U_{h,0} = (u_0(x_j))_j$ and $\frac{U_{h,1} - U_{h,0}}{dt} = (v_0(x_j))_j$. In simulations, we set

$$U_{h,0} = U_{h,1} = 0.$$

A simple calculation shows that $U_{h,k+1}$ can be calculated by solving a linear system of the form

$$Ax = b,$$

where b contains the terms that do not depend on $U_{h,k+1}$ and $A = M_h + \theta dt^2 K_h$.

C.2 Practical implementation of the forward solver

To numerically solve this forward problem, we need to generate the matrices K_h , M_h and $B_h(t)$. To assemble these matrices, we use the easy-to-use library, *Gypsilab* [4] made by Matthieu Aussal. We generate a uniform triangular mesh on $\overline{\Omega}$ with the function `mshSquare2` (personal version of `mshSquare`). Next, we generate a quadrature for the calculation of the integrals with the function `dom`. Finally, we create Lagrange elements using `fem(mesh, 'P1')`. Matrices are finally computed using `integral` in *Gypsilab*. For more precision on these functions, we refer to the documentation made by Matthieu Aussal and François Alouges [2].

All these steps are done in `forward_solver_method2_1`. After that, we use the function `undamped_newmark` to apply the Newmark numerical integration for Problem (1). To solve multiple times $Ax = b$, we use the function `decomposition` in Matlab which is 5 times faster than `A\b`.

Here we present some results given by the function `forward_solver_method2_1` in Figures 19 and 20.

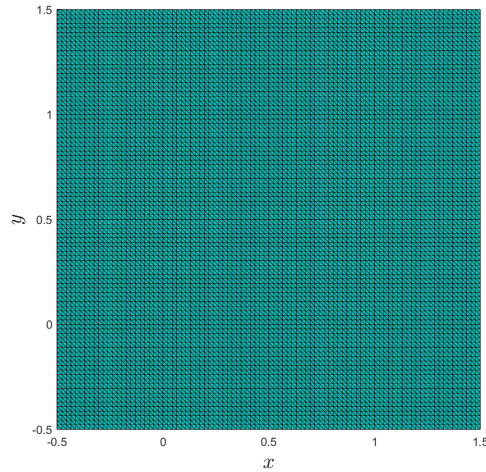


Figure 18: Mesh used to solve Problems (1), (2) and (3).

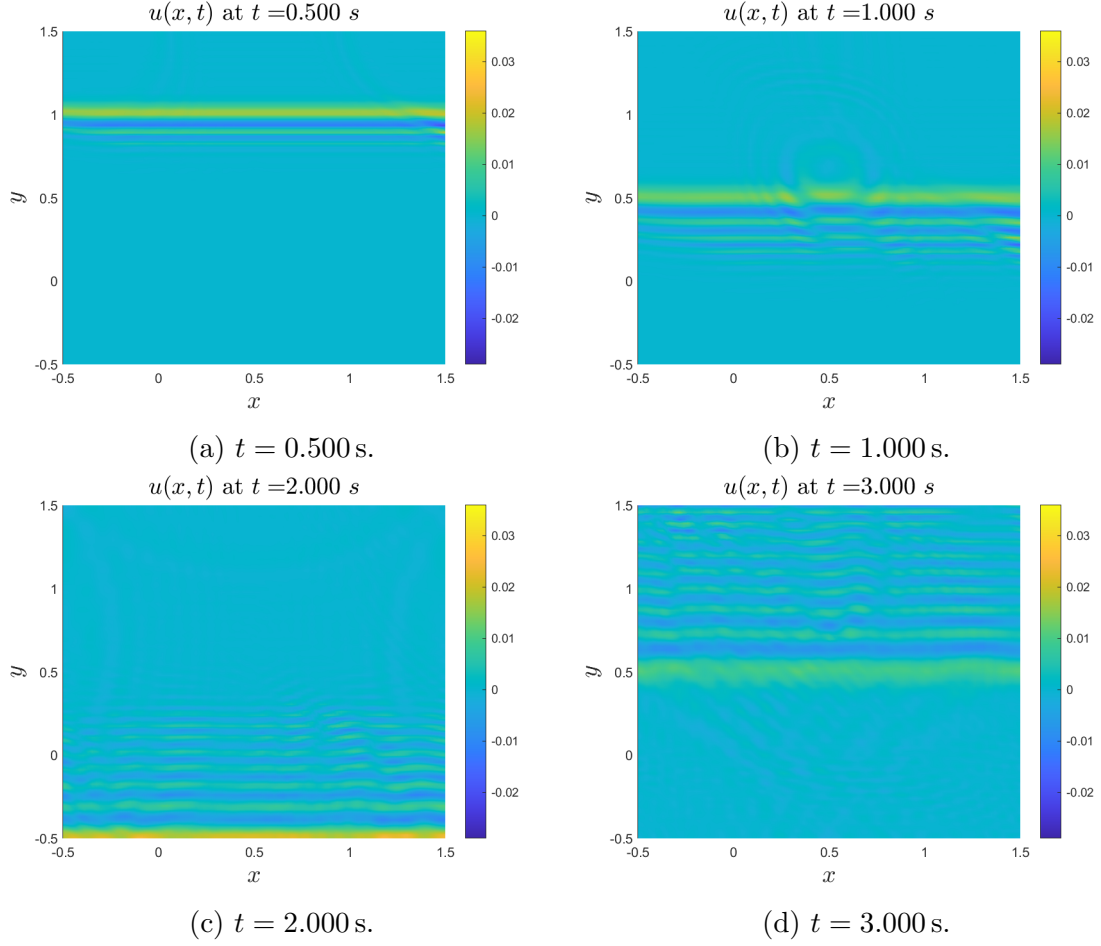


Figure 19: Numerical integration of Problem (1) for Test 1 at various time. See simulation at <https://www.youtube.com/watch?v=huPQ5o9bEQY>.

C.3 Numerical resolution of Problems (2) and (3)

Problems (2) and (3) are solved using a similar method. The only difference is that, because of the boundary conditions, the weak formulation (4) becomes

$$\begin{aligned}
 & \forall t \geq 0, \forall v \in H^1(\Omega), \\
 & \frac{d^2}{dt^2} \left(\int_{\Omega} a(x) u(x, t) v(x) dx \right) + \frac{d}{dt} \left(\int_{\Sigma} u(x, t) v(x) dx \right) + \\
 & \mathbf{1}_{t > t_1} \frac{d}{dt} \left(\int_{\partial_1 \Omega} u(x, t) v(x) dx \right) + \int_{\Omega} \nabla u(x, t) \cdot \nabla v(x) dx = p(t) \int_{\partial_1 \Omega} v(x) dx \quad (16)
 \end{aligned}$$

where $\Sigma = \partial_3 \Omega$ for (2) and $\Sigma = \partial \Omega \setminus \partial_1 \Omega$ for (3).

Thus, applying the same Finite Element Method, we obtain the linear system

$$\begin{cases} M_h \frac{d^2 U_h}{dt^2}(t) + C_h(t) \frac{d U_h}{dt}(t) + K_h U_h(t) = B_h(t), \\ U_h(0) = \frac{d U_h}{dt}(0) = 0, \end{cases} \quad (17)$$

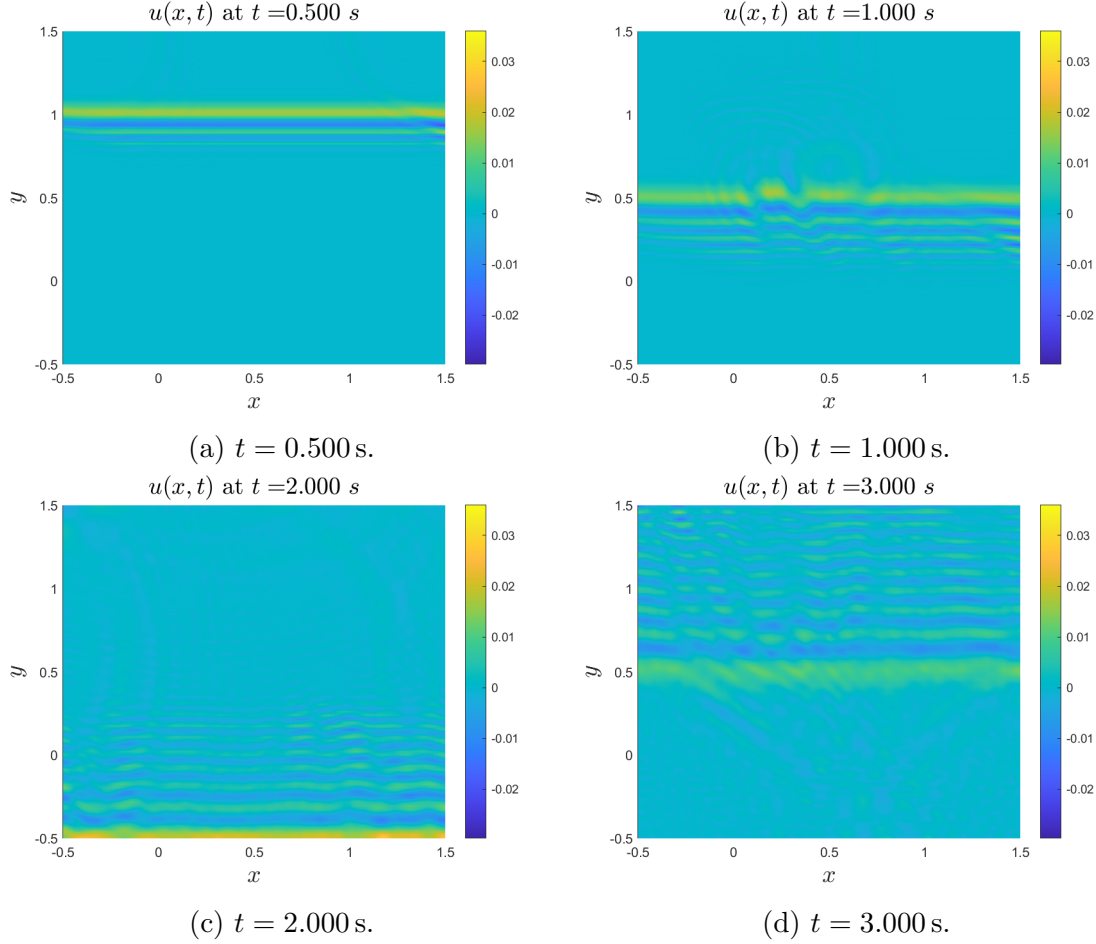


Figure 20: Numerical integration of Problem (1) for Test 2 at various time. See simulation at <https://www.youtube.com/watch?v=kaF8wWdIH5k>.

where we have introduced the damping matrix

$$C_h(t) = \left(\int_{\Sigma} \varphi_k(x) \varphi_j(x) dx + \mathbf{1}_{t > t_1} \int_{\partial_1 \Omega} \varphi_k(x) \varphi_j(x) dx \right)_{0 \leq j, k \leq n}.$$

We use the Newmark numerical integration to discretise and solve this ODE

$$\begin{cases} M_h \ddot{U}_{h,k+1} + C_h(t_{k+1}) \dot{U}_{h,k+1} + K_h U_{h,k+1} = B_h(t_{k+1}), \\ \dot{U}_{h,k+1} = \dot{U}_{h,k} + dt \left(\delta \ddot{U}_{h,k+1} + (1 - \delta) \ddot{U}_{h,k} \right), \\ U_{h,k+1} = U_{h,k} + dt \dot{U}_{h,k} + \frac{dt^2}{2} \left(2\theta \ddot{U}_{h,k+1} + (1 - 2\theta) \ddot{U}_{h,k} \right). \end{cases} \quad (18)$$

This scheme can not be rewritten in a θ -scheme as for Problem (1) but it can be solved by substituting elements at each line. As previously, it requires to solve a linear system $Ax = b$ where b only depends on the terms at rank k and $k - 1$ and $A = M_h + \delta dt C_h(t_{k+1}) + \theta dt^2 K_h$.

The numerical resolution of this Newmark numerical integration is effectively implemented in the function `damped_newmark`. Every step of the numerical implementation of this method for Problems (2) and (3) can be respectively found in `forward_solver_method2_2` and `forward_solver_method2_3`. In Figures 21, 22, 23 and 24, we present numerical results for solutions to these problems.

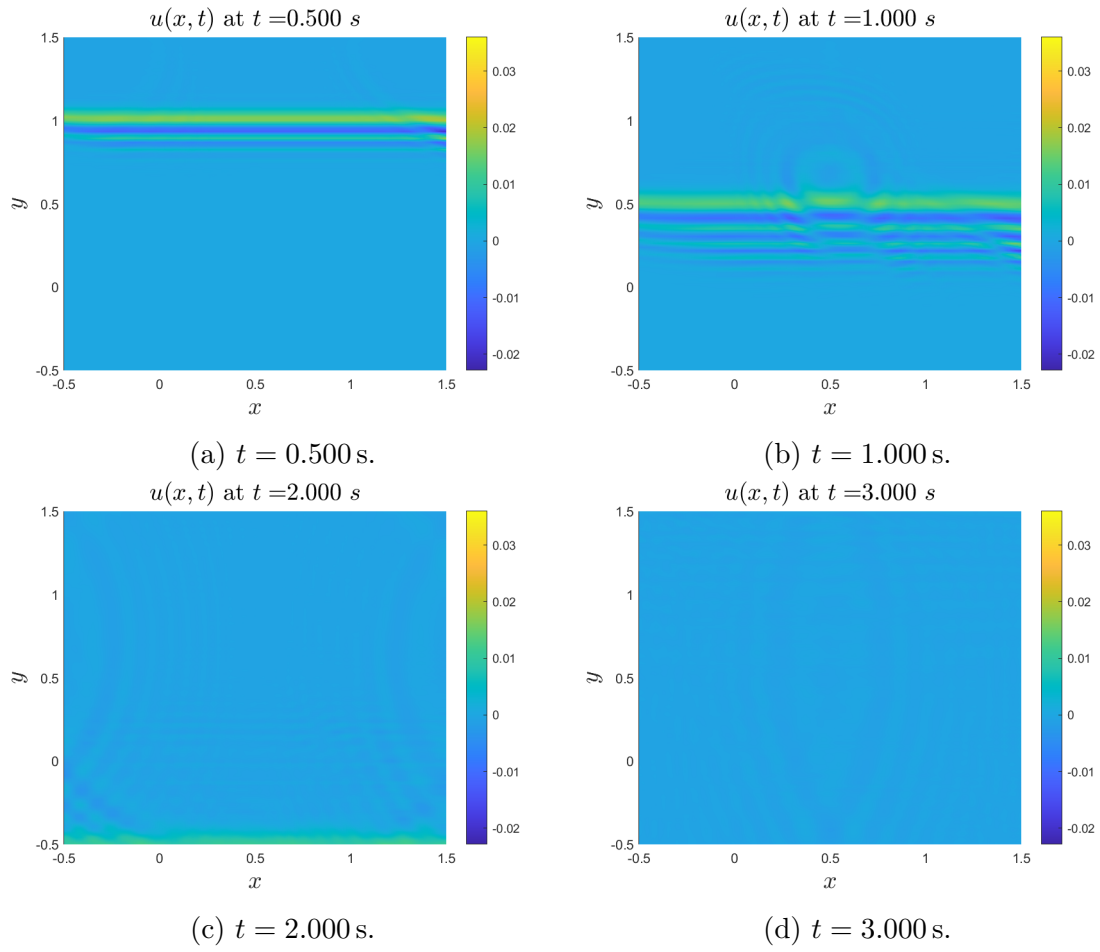


Figure 21: Numerical integration of Problem (2) for Test 1 at various time. See simulation at <https://www.youtube.com/watch?v=9fxZfK1tMt0>.

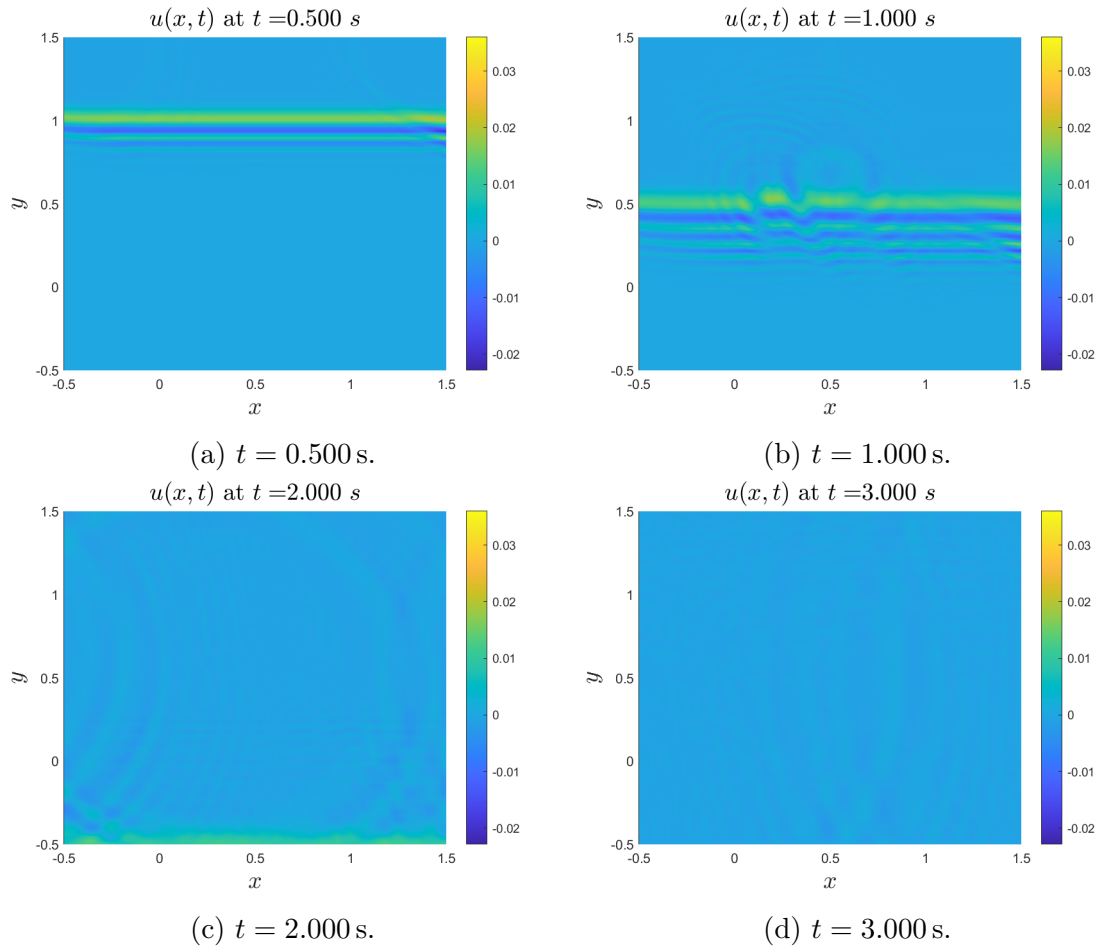


Figure 22: Numerical integration of Problem (2) for Test 2 at various time. See simulation at https://www.youtube.com/watch?v=VJ6r_edPCrA.

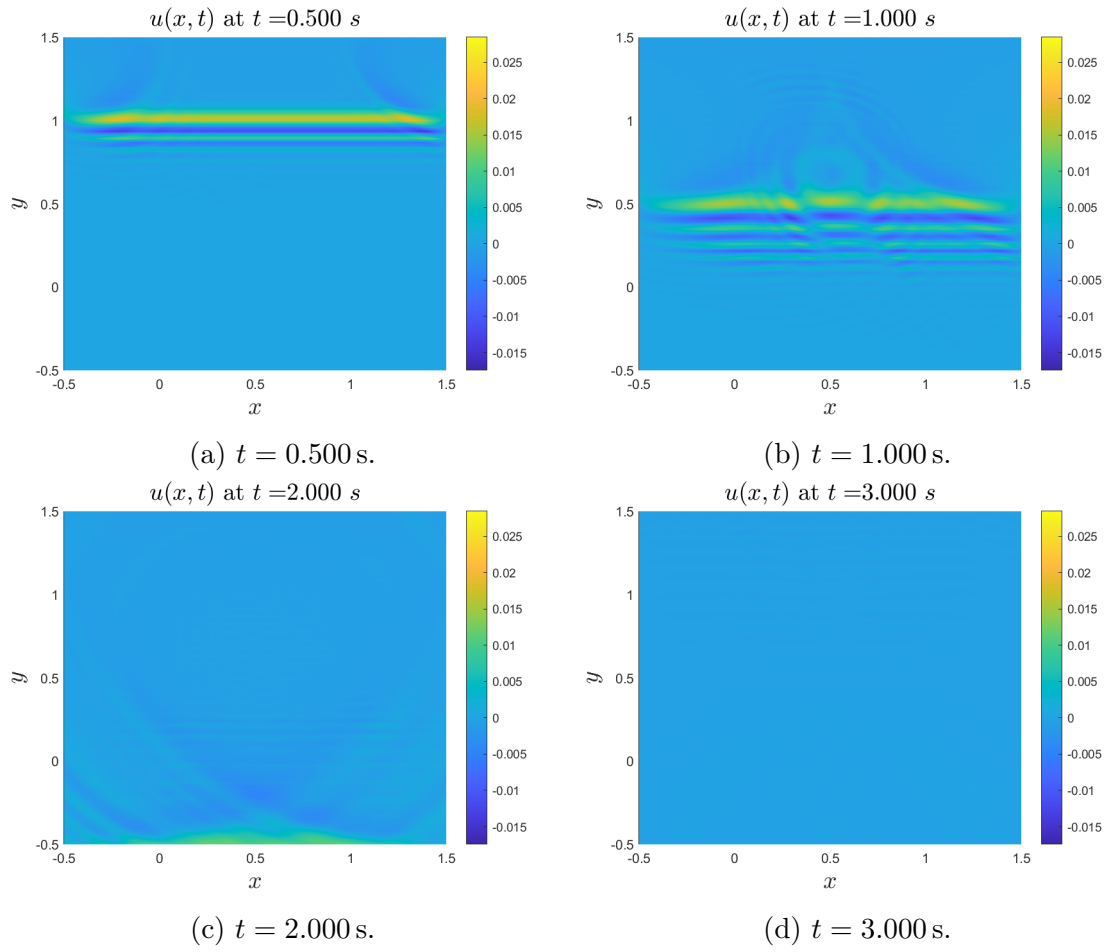


Figure 23: Numerical integration of Problem (3) for Test 1 at various time. See simulation at <https://www.youtube.com/watch?v=-Ji20zW5tC0>.

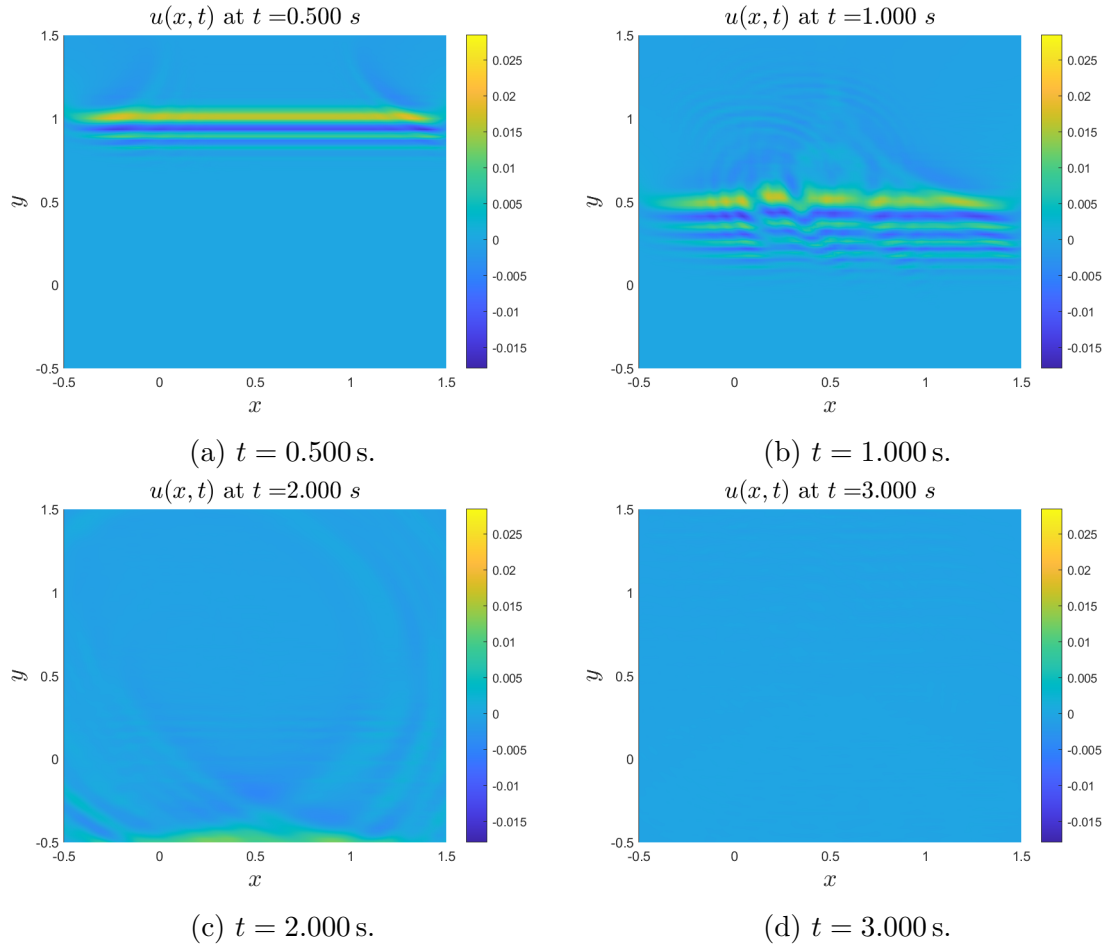


Figure 24: Numerical integration of Problem (3) for Test 2 at various time. See simulation at <https://www.youtube.com/watch?v=Ppcfzaz-wRk>.

D Numerical analysis without dependency on time

In this section, we present an initial idea of generation of the data called in the algorithms *Method 1*. We also present some initial studies for the computation of the normal derivative $\frac{\partial w}{\partial n}$. We conclude this part by the study of a simple idea that was not effective.

D.1 Explanation of Method 1

A first idea to generate the data for Inverse problem 1 has been to generate only $w(x, s)$ for a certain s and not $u(x, t)$ and then calculate the discrete Laplace transform. This idea has been implemented in the case of Model 1. Thus, applying the Laplace transform as in Subsection 2.1 for Problem (1), it yields

$$\forall s > 0, \quad \begin{cases} -\Delta w(x, s) + s^2 a(x) w(x, s) = 0 & \text{in } \Omega, \\ \frac{\partial w}{\partial n} = \mathcal{L}(p)(s) & \text{on } \partial_1 \Omega, \\ \frac{\partial w}{\partial n} = 0 & \text{on } \partial \Omega \setminus \partial_1 \Omega. \end{cases} \quad (19)$$

Using the same approach as in Subsection 2.1, Equation (19) can be solved using Finite Element Method. Thus, we can generate $w(x, s)$ for a certain s . This is done in the file `Simulation_Method_1`. Figure 25 presents some reconstructions by applying the inverse solver on data generated using this method.

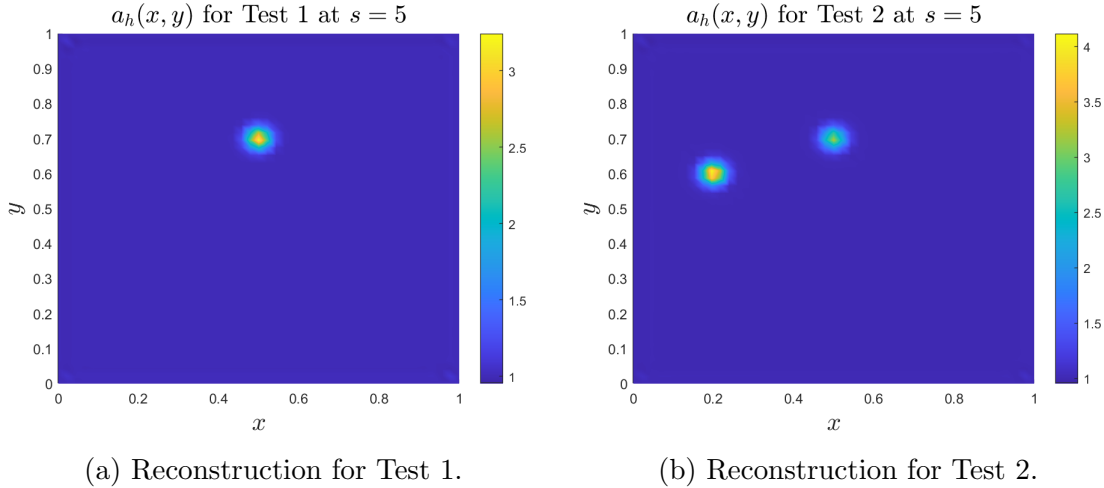


Figure 25: Reconstruction of the test functions using generated data from Method 1. (a) Reconstruction of a for Test 1. $e_{quad} = 0.0105$ and $e_{max} = 0.281$. (b) Reconstruction of a for Test 2. $e_{quad} = 0.0172$ and $e_{max} = 0.385$.

In this method, we should calculate the Laplace transform of p . Even if this could be done using a discrete Laplace transform, we can analytically calculate it in the case of $p(t) = \sin(\omega t)$:

$$\mathcal{L}(p)(s) = \omega \frac{1 - e^{-\frac{2s\pi}{\omega}}}{\omega^2 + s^2}.$$

To obtain Equation (19), we assumed that either $T = +\infty$ and the Laplace transform of $u(x, \cdot)$ exists which requires s to be greater than the abscissa of absolute convergence of the integral, or that we can disregard the term at $t = T$ in the integration by parts. These assumptions could be discussed. However, we can notice through the quantitative errors in Figure 25 that the reconstruction is effective. Furthermore, the relative quadratic error is lower when we use data from Method 1 as in Figure 25 in comparison with results from Figure 7 whereas the maximal error is slightly greater. Thus, a further analysis on the efficacy of the reconstruction using data generated from Equation (19) should be develop and is available in Appendix E.

D.2 Computation of the normal derivative

In this subsection, we present the method of calculation of $\frac{\partial w}{\partial n}$ and we numerically analyse the efficacy of these methods on test functions using the generation of data from Method 1 as explained in the previous subsection. The computation of the normal derivative might be one of the most important source for computational errors. In the inverse solver presented in Subsection 2.1, we use scattered data of w to compute $\frac{\partial w}{\partial n}$ on scattered points and we use these values to compute the integrals

$$\left(\int_{\partial\Omega_1} f(x) \varphi_j(x) dx \right)_j.$$

These integrals are computed using a quadrature method on each side of $\partial\Omega_1$. Thus, we need to generate a function f that approximate $\frac{\partial w}{\partial n}$. On each side, we compute scattered values of $\frac{\partial w}{\partial n}$ using the slope in the correct direction. For instance, on the top side:

$$\frac{\partial w}{\partial n}(x, y) \simeq \frac{w(x, y + dy) - w(x, y - dy)}{2 dy}$$

We use similar method to compute $\frac{\partial w}{\partial n}$ on each side. However, $\frac{\partial w}{\partial n}$ is not analytically defined on the vertices. As we need values for the computation of f we need to set values at the vertices. After putting a value at each point (x_j, y_j) , we compute $f(x, y)$ as the value of the nearest point on $\partial\Omega_1$.

For the values at the vertices, three proposals have been studied and quantitatively compared to select the one with the better results. This study has been done in the folder **Simulation_1**. We have generated various w for various pseudo-frequencies and mesh sizes and we have applied the inverse solver for each proposal to reconstruct A_h . Then, we have stocked A_h , e_{quad} and e_{max} for each combination. Finally, we present some numerical analyses of the results.

We present the three proposals for the computation of f at the vertices and the way to calculate f at the top right vertex (x_*, y_*) .

Proposal 1: Compute the mean value of the two directional derivatives

$$f(x_*, y_*) \simeq \frac{1}{2} \frac{w(x_*, y_* + dy) - w(x_*, y_* - dy)}{2 dy} + \frac{1}{2} \frac{w(x_* + dx, y_*) - w(x_* - dx, y_*)}{2 dx}.$$

Proposal 2: Compute the diagonal derivative

$$f(x_*, y_*) \simeq \frac{w(x_* + dx, y_* + dy) - w(x_* - dx, y_* - dy)}{2\sqrt{dx^2 + dy^2}}.$$

Proposal 3: For the calculation of the integral on each side, we take the nearest non-vertex point to compute f

$$f(x, y) = \begin{cases} f(x_*, y_j) & \text{if } x = x_* \text{ and } y_j \text{ is the nearest point with } y_j \neq y_*, \\ f(x_j, y_*) & \text{if } y = y_* \text{ and } x_j \text{ is the nearest point with } x_j \neq x_*. \end{cases}$$

Figure 26 shows the comparison between these three proposals. For each combination of parameters h and s , we select the index of the proposal with the lowest error according to e_{quad} and e_{max} . We compute this result in the case in which a post-processing on the boundary has been applied or not. We can notice that Proposal 3 clearly provides better result on most of the simulations. Thus, we decide to keep this method for the computation of f .

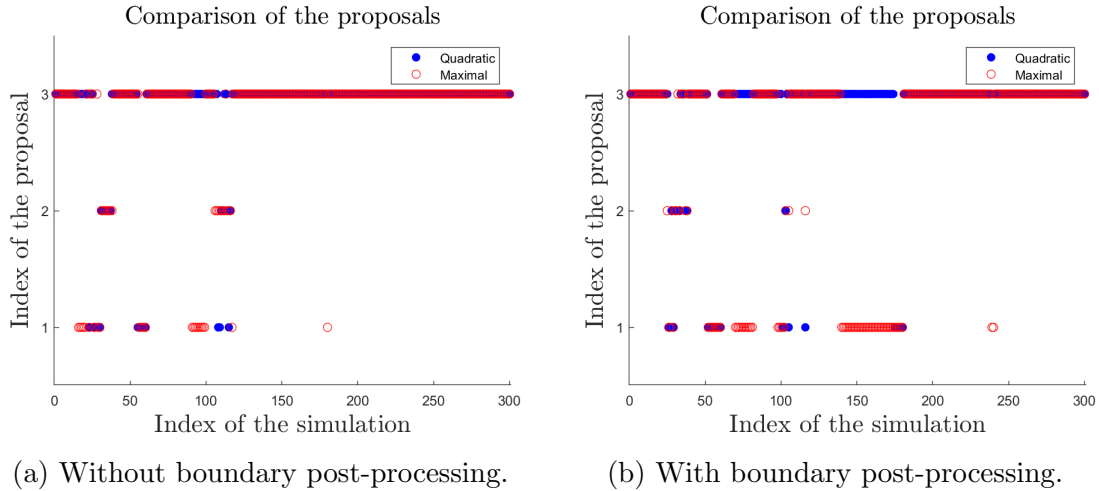


Figure 26: Comparison of the proposals for computation of f . (a) Selection of the proposal with the smallest e_{quad} and e_{max} for various parameters h and s without boundary post-processing. (b) Analogous results with boundary post-processing.

In Figure 27, we present the results of the reconstruction with or without a boundary post-processing. The qualitative result and the quantitative errors prove the efficacy of the boundary post-processing. Thus, in the case of the reconstruction using internal measurements, we always apply a boundary post-processing. This post-processing is an essential corrector of the computational errors linked to the approximation of f on the vertices.

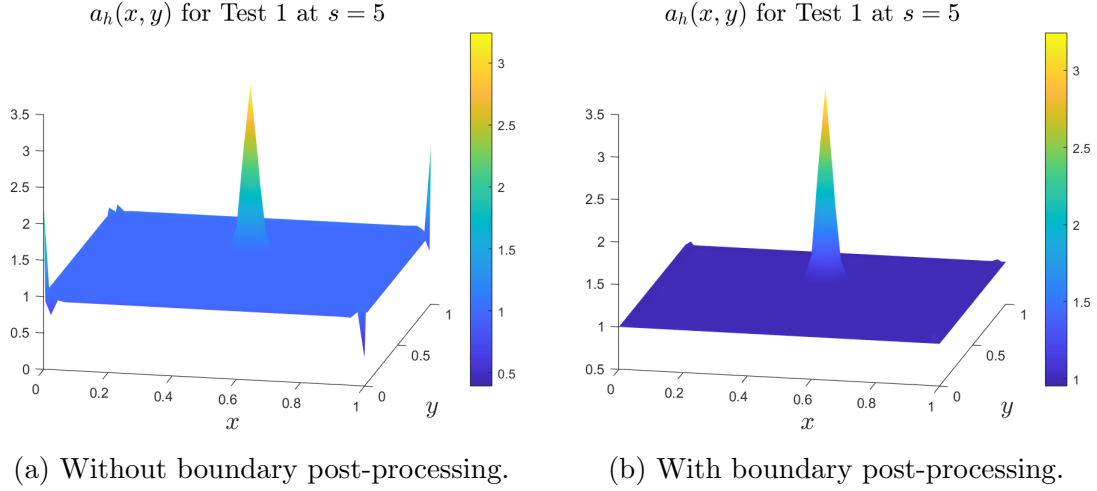


Figure 27: Effect of the boundary post-processing. (a) Reconstruction of a for Test 1 without a boundary post-processing. $e_{quad} = 0.0438$ and $e_{max} = 1.21$. (b) Application of the boundary post-processing. $e_{quad} = 0.0105$ and $e_{max} = 0.281$.

D.3 A reconstruction method in Ω

An initial idea has been to solve the problem not only in Ω_1 but in Ω . The idea is legitimate since the principle that has been developed in Subsection 2.1 can be done in Ω instead of Ω_1 . However, one could argue that since a is known outside Ω_1 , there is no need to solve in whole Ω . Nonetheless, we can see in the method below that the results are better.

As described for the resolution of the forward solver using the Laplace transform in Subsection D.1, the formulation of Problem (1) can be written in term of Laplace transform as:

$$\forall s > 0, \quad \begin{cases} -\Delta w(x, s) + s^2 a(x) w(x, s) = 0 & \text{in } \Omega, \\ \frac{\partial w}{\partial n} = \mathcal{L}(p)(s) & \text{on } \partial_1 \Omega, \\ \frac{\partial w}{\partial n} = 0 & \text{on } \partial \Omega \setminus \partial_1 \Omega. \end{cases} \quad (19)$$

Thus, by applying Finite Element Method, it yields to the discretisation

$$M_h A_h = \frac{1}{s^2} (\mathcal{L}(p)(s) B_h - K_h W_h) \quad (20)$$

where

$$\begin{aligned} W_h &= (w(x_j))_{0 \leq j \leq n} \text{ are the known values of } w(x, s) \text{ over the mesh,} \\ K_h &= \left(\int_{\Omega} \nabla \varphi_k(x) \cdot \nabla \varphi_j(x) dx \right)_{0 \leq j, k \leq n} \text{ is the stiffness matrix,} \\ M_h &= \left(w(x_k) \int_{\Omega} \varphi_k(x) \varphi_j(x) dx \right)_{0 \leq j, k \leq n} \text{ is the weighted mass matrix,} \\ B_h &= \left(\int_{\partial_1 \Omega} \varphi_j(x) dx \right)_{0 \leq j \leq n} \text{ is the boundary vector,} \\ A_h &= (a_{h,j})_{0 \leq j \leq n} \text{ is the unknown vector.} \end{aligned}$$

We notice that in this discrete equation, we must not compute the normal derivative of w . Indeed, the boundary function f is here known by the knowledge of the model. Thus, we could expect better results from this solver than from the previously described method.

This method has been implemented in the function `inverse_solver_2`. We also have chosen to extract only the values of the reconstructed a_h inside Ω_1 to obtain results that could be compared to the results obtained from the previous method. To compare the results obtained by this method, we apply the same parameters as in Figure 26 and we again select the best method between the resolution in Ω_1 (index 1) and the resolution in Ω (index 2). The results could be seen in Figure 28.

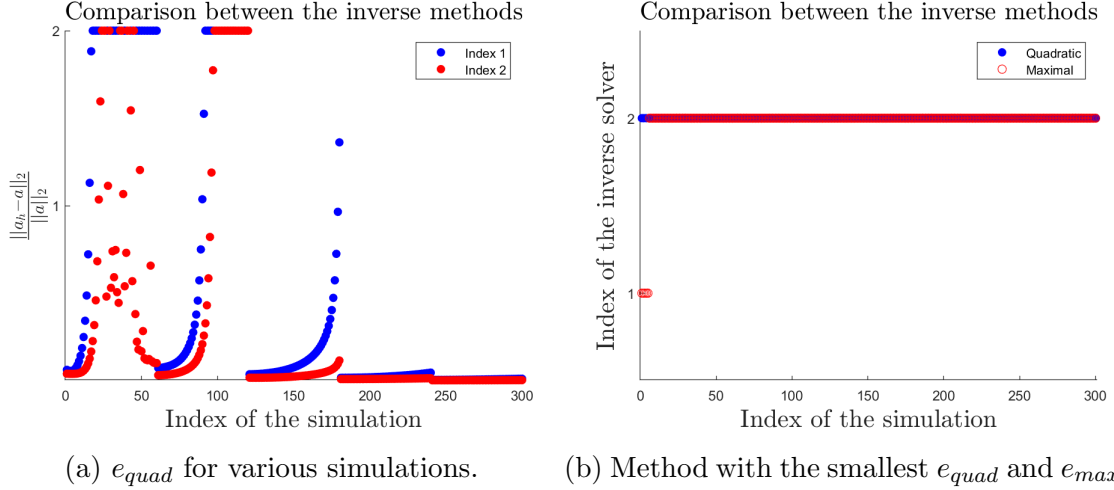


Figure 28: Comparison of the methods for the resolution of the inverse problem. (a) Values of e_{quad} for various mesh sizes and pseudo-frequencies and for the two methods. Values greater than 2 are set to 2. (b) Selection of the inverse solver with the smallest e_{quad} and e_{max} for various parameters h and s with boundary post-processing.

Figure 28 illustrates the large efficacy of this method of resolution in Ω . Thus, this method is a great improvement to the original method of resolution. However, this solver requires the knowledge of p to compute the matrices. Thus, contrary to the previous method in which only u should be known (which implies the knowledge of the initial conditions and the possibility to compute w and the normal derivative), the new inverse solver requires a control of the external source. Furthermore, the inverse method depends on the choice of the model as the boundary conditions change. For those reasons, we have chosen not to pursue this idea and to only analyse the previous method as described in [8].

E Further analysis of the effect of the numerical parameters

In this appendix, we present some complementary analyses of the described method from Subsection 2.2. In this part, we describe the effect of some numerical methods or parameters whose effect is less significant on the method of reconstruction. In Subsection 2.2, we have described the effect of the pseudo-frequency and the parameters linked to the integration time. Here we analyse two additional parameters, $\theta_{Newmark}$ and the quadrature rule.

First, we analyse the efficacy of reconstruction according to $\theta_{Newmark}$. Even if this parameter does not appear in the reconstruction method, it is an important parameter for the resolution of the forward problem. As described in Appendix C, the Newmark numerical integration relies on two parameters $\frac{1}{2} \leq \delta_{Newmark} \leq 1$ and $0 \leq \theta_{Newmark} \leq \frac{1}{2}$. These parameters control if the numerical scheme is explicit or implicit and conditionally or unconditionally stable.

In this work, we have set $\delta_{Newmark} = \frac{1}{2}$ as it is usually done. Then, four usual values of $\theta_{Newmark}$ exists. $\theta_{Newmark} = 0$ characterises an explicit scheme which is conditionally stable. $\theta_{Newmark} = \frac{1}{12}$ corresponds to the Fox Goodwin scheme which is conditionally stable. $\theta_{Newmark} = \frac{1}{4}$ corresponds to an average constant acceleration scheme which is unconditionally stable. Finally, $\theta_{Newmark} = \frac{1}{2}$ means a fully implicit scheme which is unconditionally stable.

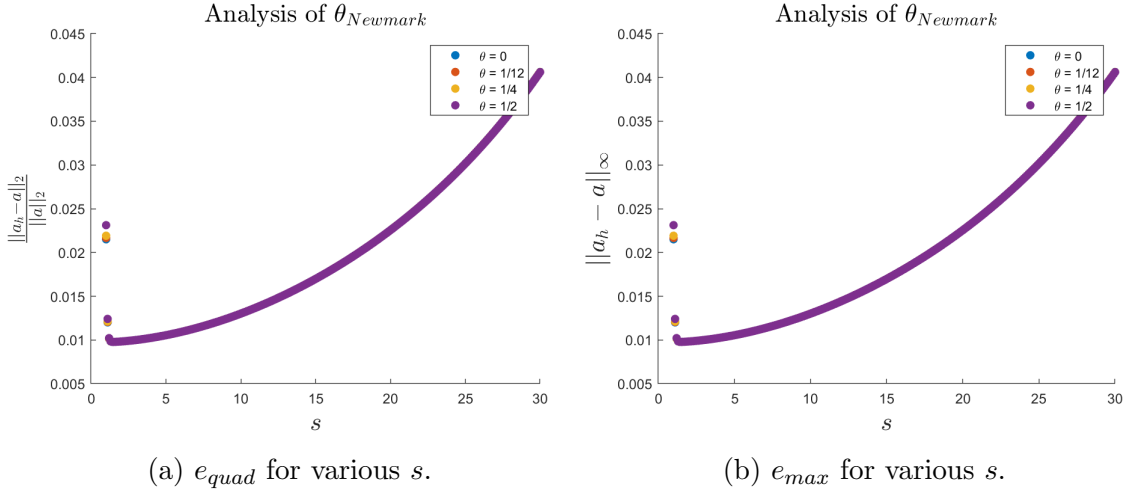


Figure 29: Comparison of the Newmark numerical integrations for various $\theta_{Newmark}$ for Model 1. (a) Values of e_{quad} for each $\theta_{Newmark}$ for various pseudo-frequencies. (b) Values of e_{max} for each $\theta_{Newmark}$ for various pseudo-frequencies.

Figure 29 shows the results of the reconstruction of the test function for Test 1 for those values of $\theta_{Newmark}$ for Model 1. We note that the errors are almost equal for every pseudo-frequency and for each $\theta_{Newmark}$. Thus, the choice of $\theta_{Newmark}$ is not very significant. Furthermore, similar results have been observed for the other models. In Figure 30, we have selected the $\theta_{Newmark}$ with the minimal error for each pseudo-frequency and for every model. We can notice that for the smallest pseudo-frequencies the explicit scheme can return better results. However, for the range of the selected frequencies from Subsection 2.2 (*ie.* $s \in [2, 7]$),

the fully implicit scheme seems to give better results. Thus, we choose to set $\theta_{Newmark} = \frac{1}{2}$ for the simulations.

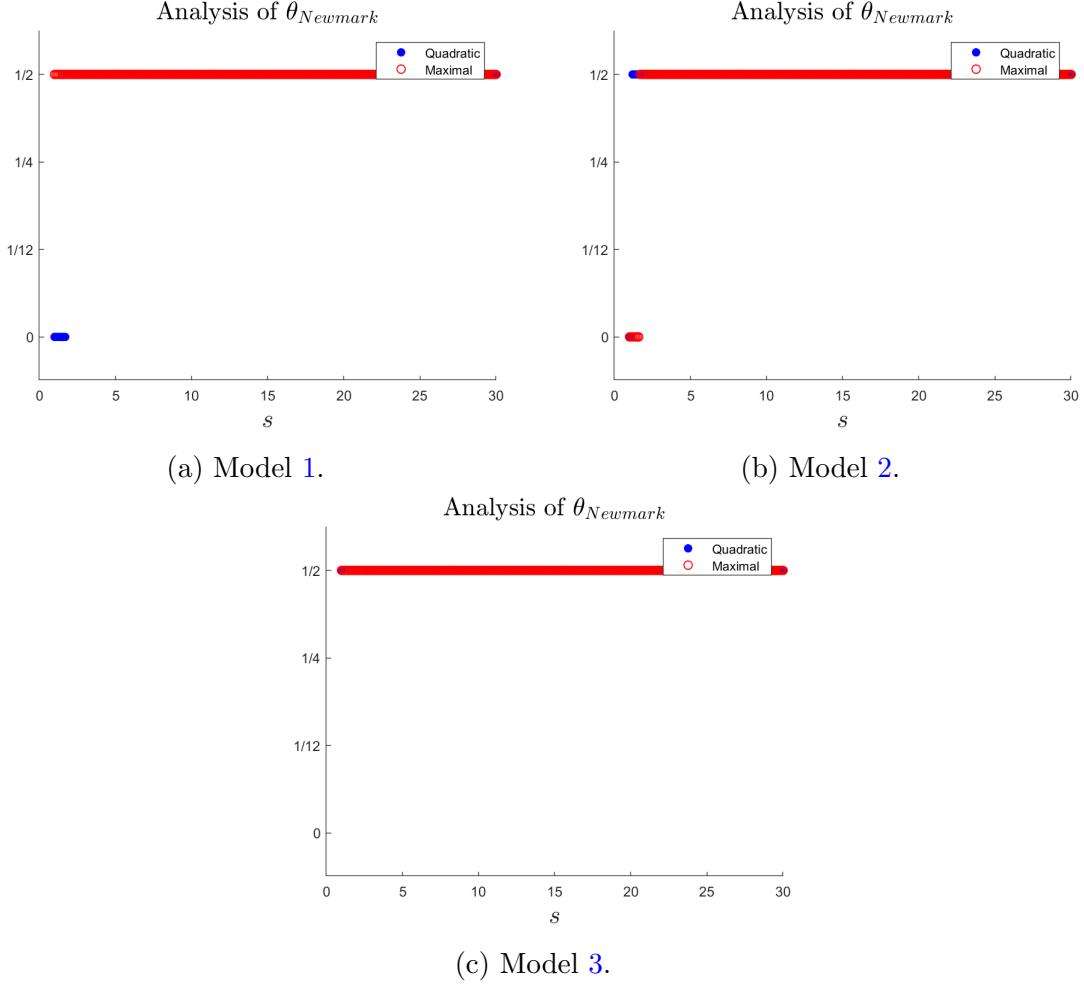


Figure 30: Selection of the $\theta_{Newmark}$ with the smallest e_{quad} and e_{max} for each pseudo-frequency for each model. (a) Case of Model 1 for the forward solver. (b) Case of Model 2 for the forward solver. (c) Case of Model 3 for the forward solver.

Besides the selection of $\theta_{Newmark}$, another numerical method appears in the generation of the measurements for the inverse solver. Indeed, to apply the reconstruction method, it requires to compute the discrete Laplace transform of u at a certain pseudo-frequency. The numerical computation of the integral is done using a quadrature rule. Three simple quadrature rules have been tested, the rectangle rules (left and right) and the trapezoidal rule. Similarly as before, we have tested the efficacy of each quadrature rule.

In Figure 31, we have presented the results of the tests on the quadrature rule. We can note that there is no method that seems clearly better than the others over the various simulations even in the range of $s \in [2, 7]$. However, the numerical results show that the difference between the quadrature rules is insignificant. As a consequence, each quadrature rule seems to be satisfying for the computation of the discrete Laplace transform. Thus, we have selected the trapezoidal rule since the convergence order is 1 whereas it is 0 for the rectangle rules.

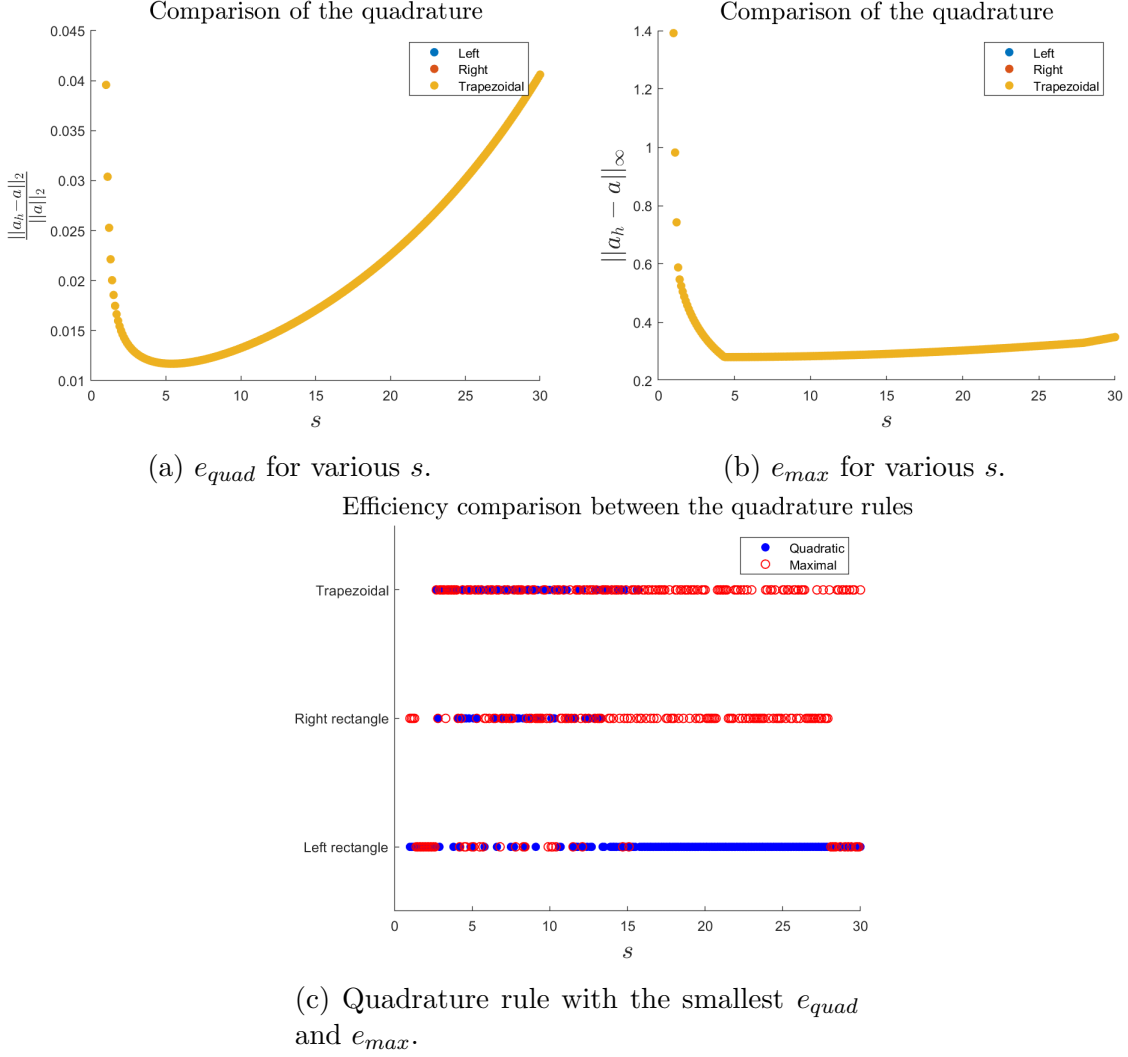


Figure 31: Comparison of the numerical quadrature rules for the discrete Laplace transform for Model 3. (a) Values of e_{quad} for each quadrature rule for various pseudo-frequencies. (b) Values of e_{max} for each quadrature rule for various pseudo-frequencies. (c) Selection of the quadrature rule with the smallest e_{quad} and e_{max} for each pseudo-frequency.

Finally, we have presented another method of resolution of Problem (1) in Appendix D and more precisely in Subsection D.1, named Method 1. This method consists in solving only the Laplace transform of Problem (1) at the correct pseudo-frequency to directly get the Laplace transform of u . Thus, we have compared the effectiveness of the reconstruction using either $\mathcal{L}(u)(s)$ coming from Method 1 or from the resolution in time and then a discrete Laplace transform. Figure 32 presents the results of this comparison. Despite a better reconstruction using data from Method 1 for $s < 2$, it seems that the reconstruction using data from the resolution in time is clearly better. Even if one could believe that Method 1 generates less significant errors than a discrete Laplace transform, it is important to remind that Method 1 relies on the assumption that the terms at $t = T$ are insignificant. Consequently, it appears that this assumption might not hold in practical situation. Thus, this justifies that we prefer the forward solver in time than Method 1.

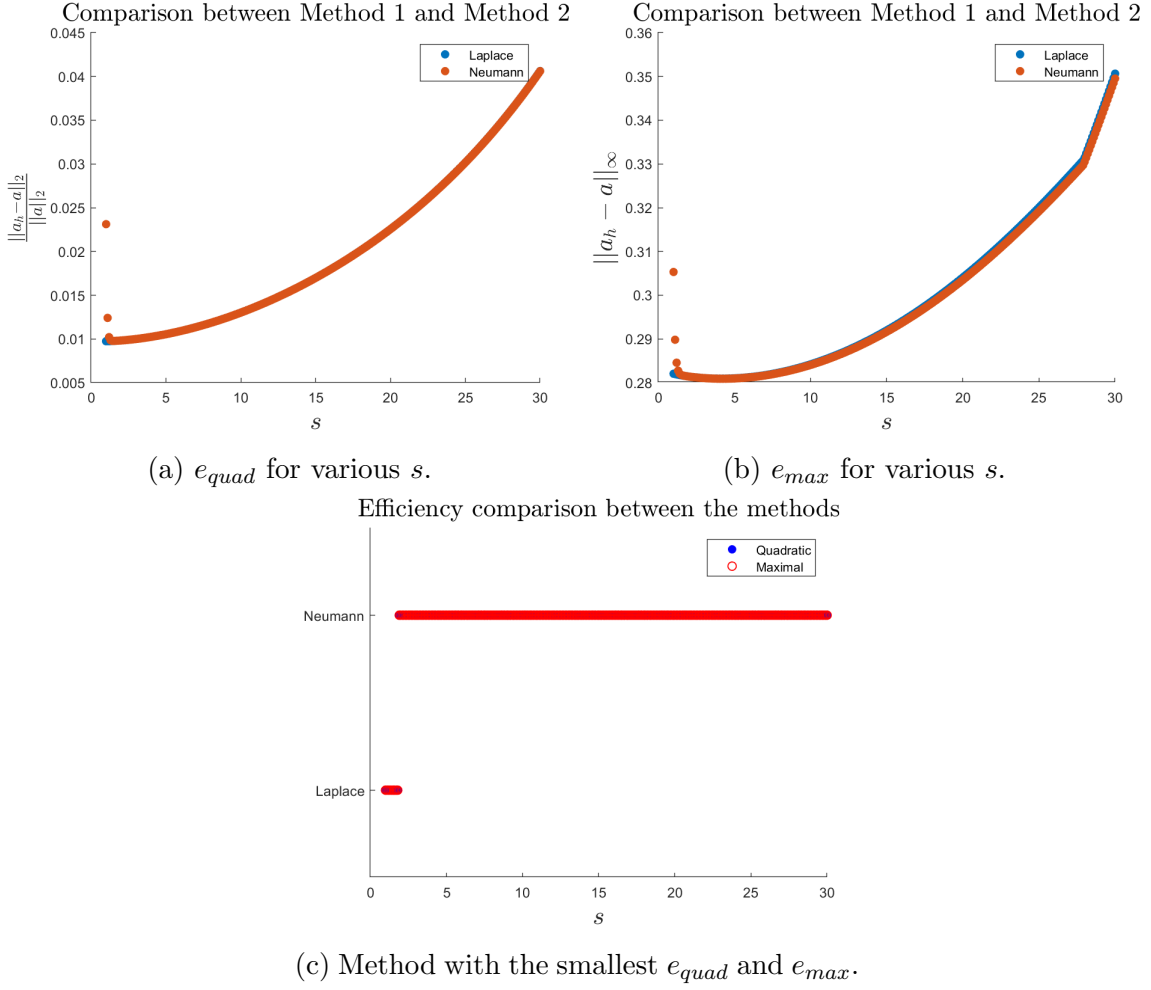


Figure 32: Comparison of the methods of resolution of Problem (1) for Test 1. (a) Values of e_{quad} for each method for various pseudo-frequencies. (b) Values of e_{max} for each method for various pseudo-frequencies. (c) Selection of the method with the smallest e_{quad} and e_{max} for each pseudo-frequency. Laplace refers to the resolution of Problem (1) using Method 1 and Neumann to the resolution in time.

To conclude this section, we present some additional simulations. In Subsection 2.2, we have illustrated the reconstruction using data generated using Model 3. In Figures 33 and 34, we present analogous results with data respectively generated using Model 1 and Model 2. We note that the results are strongly similar to those obtained in Figure 7. More precisely, the reconstructions using data from Models 1 and 2 are mainly similar even for the quantitative errors. On the one hand, these reconstructions are more precise according to the relative quadratic error than the reconstruction using data from Model 3. On the other hand, they are less precise according to the maximal error. Thus, the three models are interesting for our study.

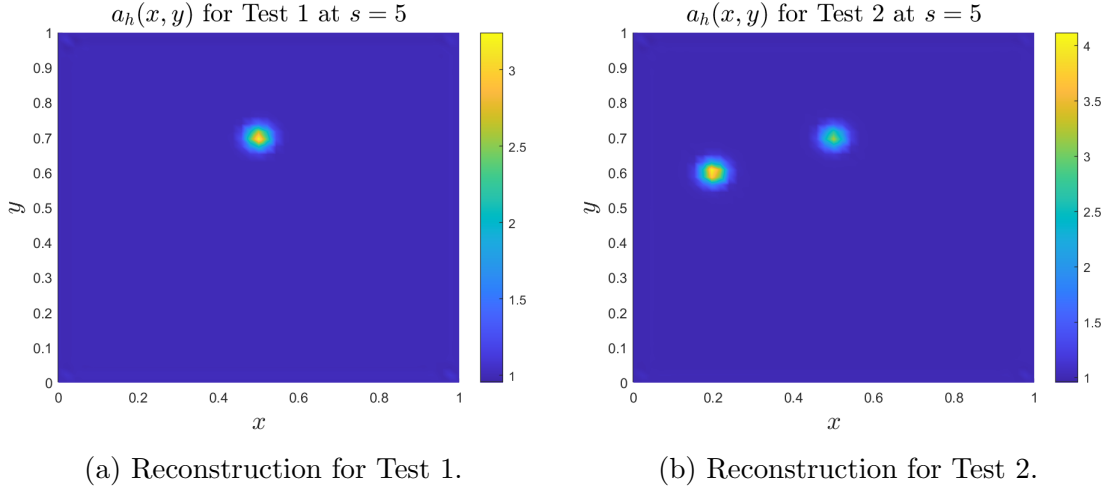


Figure 33: Reconstruction of the test functions using Model 1. (a) Reconstruction of a for Test 1. $e_{quad} = 0.0105$ and $e_{max} = 0.281$. (b) Reconstruction of a for Test 2. $e_{quad} = 0.0172$ and $e_{max} = 0.385$.

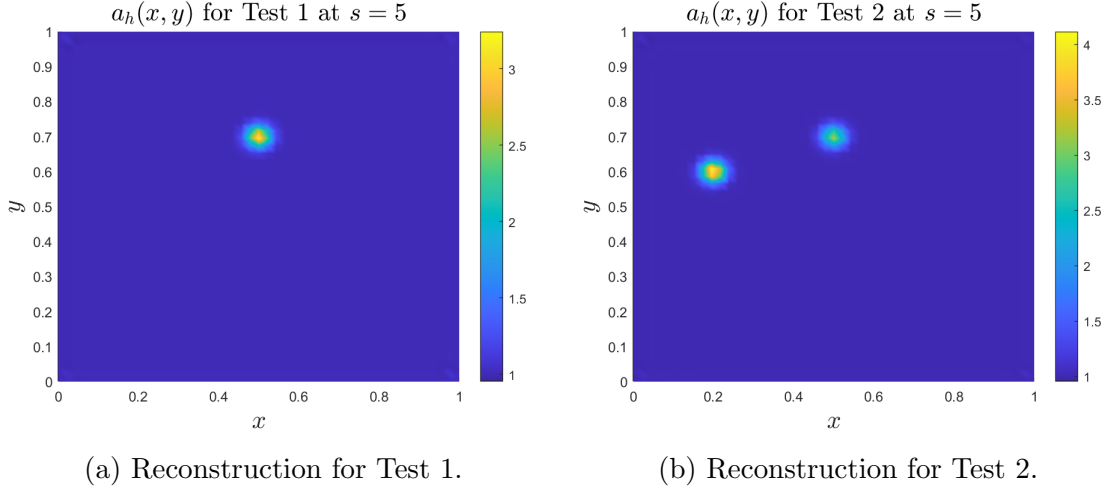
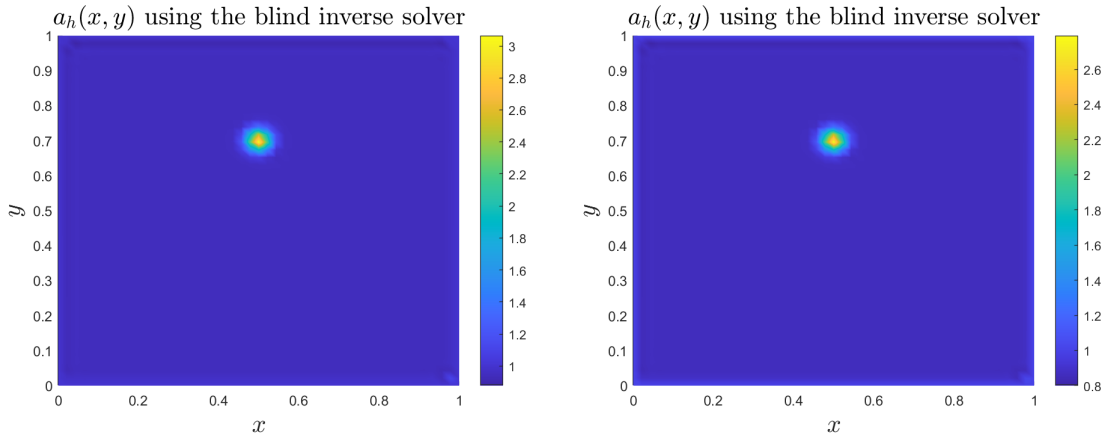


Figure 34: Reconstruction of the test functions using Model 2. (a) Reconstruction of a for Test 1. $e_{quad} = 0.0105$ and $e_{max} = 0.281$. (b) Reconstruction of a for Test 2. $e_{quad} = 0.0172$ and $e_{max} = 0.385$.

F Complements on the reconstruction for an unknown pseudo-frequency

In this section, we present some additional analyses of the method of blind reconstruction, described in Subsection 2.3. Previously, we have raised the main issue for the blind solver that is the requirement of knowledge of the Model and of the external source. Even if this requirement seems to be natural since the pseudo-frequency is unknown, we might try some further simulations.

First, we try to apply the method of blind reconstruction using the incorrect model. Thus, we test the robustness of the blind inverse solver to the model. In Figure 35a, we have applied `blind_inverse_solver_time` to some measurements $(w(x_k, s_*))_k$ generated using Model 3. However, the function has been applied with the forward solver associated to Model 2. In other terms, for each tested pseudo-frequency s_k , we have reconstructed the acoustic coefficient a_h using the inverse solver at s_k . Then we have used this coefficient to generate w using the forward solver for Model 2 and the discrete Laplace transform at s_k . The similar approach has been done in Figure 35b using the forward solver from Method 1 (see Appendix D.1 for the explanation of Method 1).



(a) Blind reconstruction with Model 2. (b) Blind reconstruction with Method 1.

Figure 35: Reconstruction for Test 1 using data from Model 3 and the blind inverse solver with other models. (a) Reconstruction of a using the forward solver associated to Model 2 in the blind inverse solver. $s_{reconstructed} = 10.5$, $e_{quad} = 0.0550$ and $e_{max} = 0.190$. (b) Reconstruction of a using the forward solver associated to Model 1 as in Method 1 in the blind inverse solver. $s_{reconstructed} = 11$, $e_{quad} = 0.134$ and $e_{max} = 0.318$.

Figure 35 illustrates that the results of a blind reconstruction with an incorrect forward solver are still satisfying. Indeed, on the one hand, we can note that the inclusion is qualitatively reconstructed and the height of the peak is almost the expected one. Furthermore, the quantitative aspect is also quite satisfying. Despite the high e_{quad} in Figure 35b, this error in Figure 35a is similar to the one obtained in Figure 10a. Moreover, the maximal error is even better for Figure 35 than for Figure 10a. Thus, an incorrect model for the blind solver is not a great issue for the method.

Nonetheless, this method still requires the knowledge of the external source $p(t)$. In the case of a measurement of $u(x, t)$, $p(t)$ could have been compute by calculating the discrete

normal derivative on $\partial_1\Omega$ at each time. However, $p(t)$ could not be calculated from w . In consequence, we have developed a method of blind reconstruction which does not require the knowledge of $p(t)$ but only its Laplace transform. The principle of this method is used the function `blind_inverse_solver`.

To understand this method, we should assume that we have the solution to Problem (1). Then, as explained in Method 1 (Appendix D.1), if we apply the Laplace transform to Problem (1), we obtain Problem (19). In this problem, we only require $\mathcal{L}(p)(s)$ to compute $w(x, s)$. Thus, this method could be used for the computation of w at each s_k after reconstruction of a_h using $(w(x_k, s_*))_k$. One could argue that Problem (19) still requires $p(t)$ to compute the Laplace transform and would only work if we compute $\mathcal{L}(p)(s_*)$. However, $\mathcal{L}(p)(s_*)$ could be approximated from $(w(x_k, s_*))_k$ by calculating the normal derivative on $\partial_1\Omega$ and taking the average value. Thus, we could assume to have access to $\mathcal{L}(p)(s_*)$, eventually with an additive noise.

A result from this approach have been computed in Figure 35b and is therefore effective. Furthermore, this approach is largely faster to reconstruct s_* and the acoustic coefficient. To conclude this section, we propose an analysis of the effect of an error on $\mathcal{L}(p)(s_*)$ in this new blind reconstruction method.

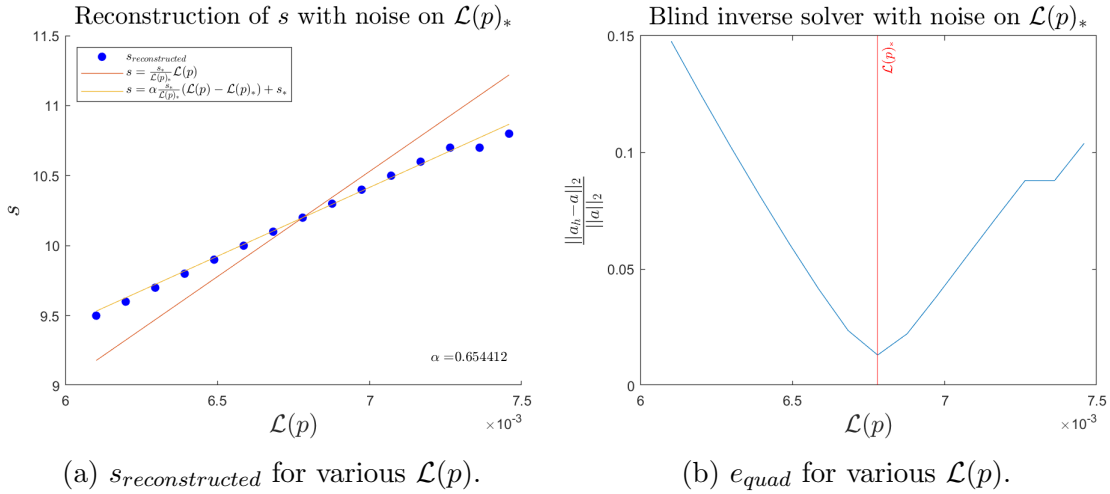


Figure 36: Analysis of the effect of an error on the Laplace transform of p in the blind reconstruction. (a) Reconstruction of s_* using the blind solver for various values of $\mathcal{L}(p)$. The line in red represents the identity between the relative error on s and $\mathcal{L}(p)$, $\frac{s}{s_*} = \frac{\mathcal{L}(p)}{\mathcal{L}(p)(s_*)}$. The line in orange represents the simple linear regression between the relative error on s and $\mathcal{L}(p)$ passing through $(\mathcal{L}(p)(s_*), s_*)$, $\frac{s}{s_*} = \alpha \frac{\mathcal{L}(p) - \mathcal{L}(p)(s_*)}{\mathcal{L}(p)(s_*)} + 1$. Numerically $\alpha = 0.65$. (b) Values of e_{quad} for various values of $\mathcal{L}(p)$.

Figure 36 presents the results of the application of `blind_inverse_solver` to data $(w(x_k, s_*))_k$ with an incorrect value of $\mathcal{L}(p)(s_*)$. We have tested several values of $\mathcal{L}(p)$ in $[\mathcal{L}(p)(s_*) - 10\%, \mathcal{L}(p)(s_*) + 10\%]$. Figure 36a illustrates the reconstructed values of the pseudo-frequency $s_{reconstructed}$. On the simple linear regression line, we can notice that the relative error done on $s_{reconstructed}$ is smaller than the one set on $\mathcal{L}(p)(s_*)$ with a factor $\alpha \simeq 0.65$. Thus, even a significant error on $\mathcal{L}(p)(s_*)$ is less significant on $s_{reconstructed}$. Furthermore, we can observe that for the various values of $\mathcal{L}(p)$, e_{quad} is less significant than it

was in Figure 8. Thus, the method gives better result than a reconstruction without applying the blind solver.

Finally, a last idea that has been studied is the efficacy of the blind inverse solver to correct an error on s_* . Indeed, if we assume that $p(t)$ is known (or its Laplace transform at s can be calculated) and we have a guess of s_* denoted by s , then we can compute $\mathcal{L}(p)(s)$ and apply `blind_inverse_solver` to obtain $s_{reconstructed}$ and the acoustic coefficient. In Figure 37, we have tried several values of $s \in [s_* - 10\%, s_* + 10\%]$. As we can see, $s_{reconstructed}$ and e_{quad} are optimal for many values around s_* . Furthermore, the coefficient of the linear regression in $s_{reconstructed}$ is about $\alpha \simeq 0.46$. Thus, it might be really efficient to apply `blind_inverse_solver` to improve an approximate value of s_* .

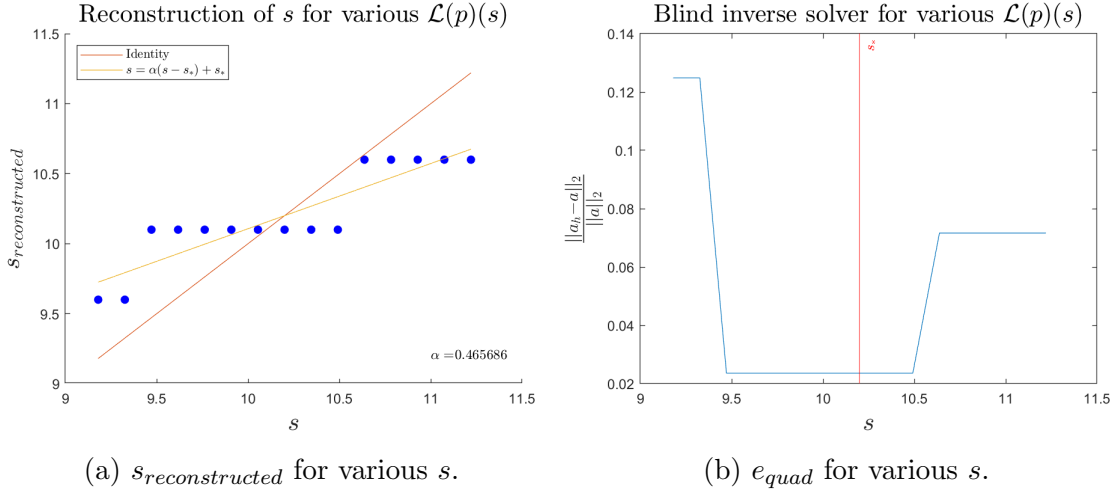


Figure 37: Analysis of the efficacy of the blind inverse solver for s_* known with an error. (a) Reconstruction of s_* using the blind solver for various values of s to compute $\mathcal{L}(p)(s)$. The line in red represents the identity. The line in orange represents the simple linear regression passing through (s_*, s_*) , $s = \alpha(s - s_*) + s_*$. Numerically $\alpha = 0.47$. (b) Values of e_{quad} for various values of s to compute $\mathcal{L}(p)(s)$.

G Additional simulations of reconstruction with noised data

In this section, we present some additional results concerning the reconstruction using noised measurements. First, we present an ineffective trial to smooth the noised data at each time step. The idea is analogous to the described method in Subsection 2.4. We apply a 2D-Gaussian-smoothing on the data at each time step t_k before the computation of the discrete Laplace transform. Then we use this smooth $u(x_j, t_k)$ to reconstruct a . Figure 38 shows the reconstructed a_h and the same result after another smoothing. It can be clearly seen that this method is ineffective to suppress the noise. Thus, it is more satisfying to apply the smoothing technique after the computation of the discrete Laplace transform. This phenomenon can be explained by the fact that the smoothing before the transformation consists in a smoothing in space but not in time. However, independent random variables are also applied at each time step. Thus, the correct smoothing technique would be to create a 3D-smoothing to smooth the data in (x, y, t) to obtain a better deblurring technique.

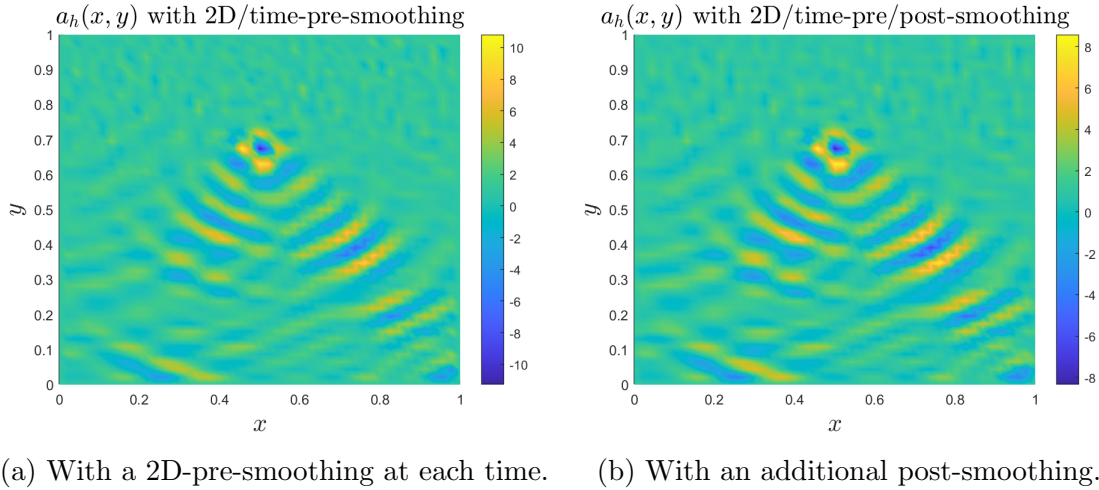


Figure 38: Reconstruction of the test function for Test 1 using noised data and a smoothing at each time step. (a) Reconstruction of a with noised data and with a smoothing at each time step before inversion. $e_{quad} = 1.58$. (b) Reconstruction of a with noised data and with a smoothing at each time step before inversion and a smoothing after inversion. $e_{quad} = 1.44$.

Figure 39 presents some results for the reconstruction for Test 2 obtained as in Figure 11. The results are similar to those obtained in Figure 11. Furthermore, we can observe that we obtain a qualitatively acceptable reconstruction of the two inclusions with the correct location. In addition, we can check that the ratio of the heights of the peaks is near to the ratio obtained for the test function.

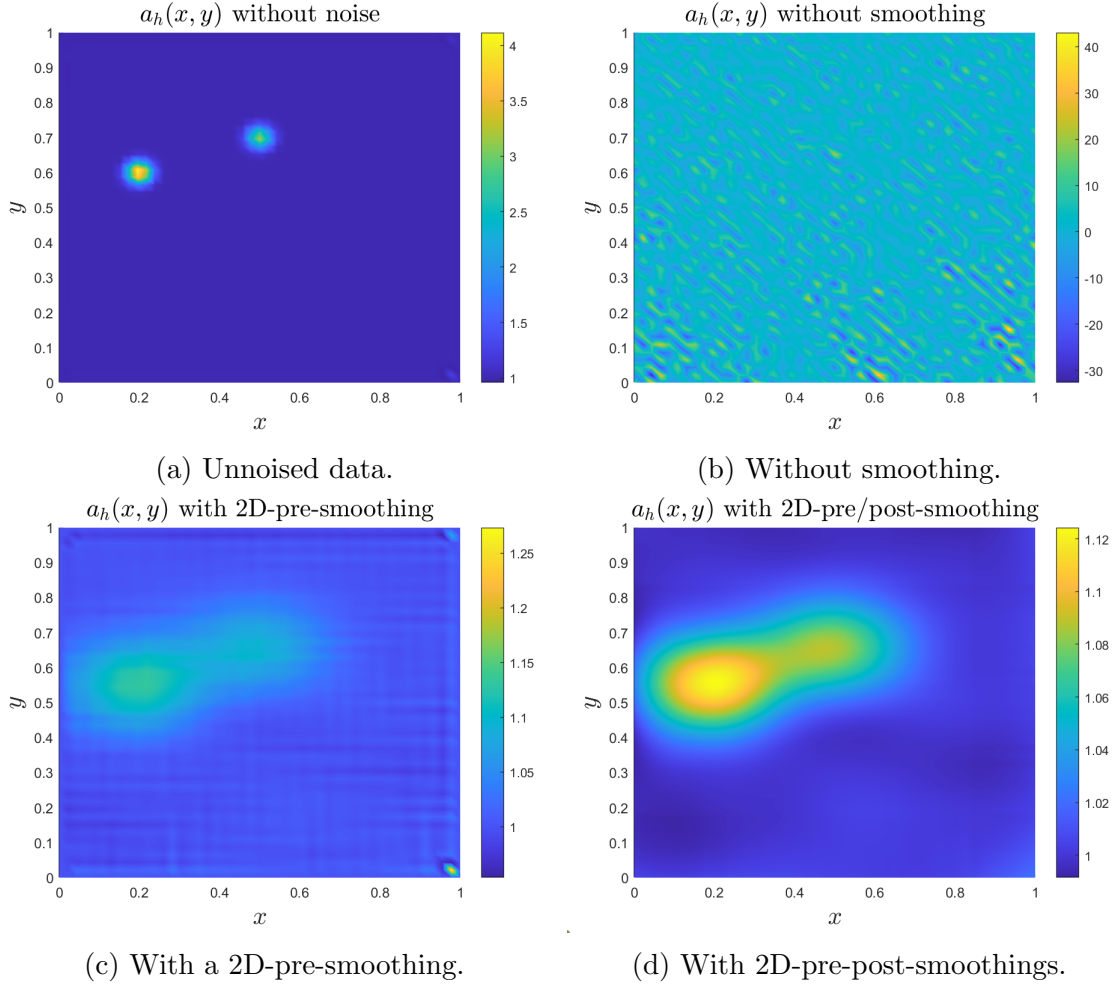


Figure 39: Reconstruction of the test function for Test 2 using noised data. (a) Reconstruction of a with unnoised data. $e_{quad} = 0.0179$. (b) Reconstruction of a with noised data and without smoothing methods. $e_{quad} = 7.31$. (c) Reconstruction of a with noised data and with a smoothing before inversion. $e_{quad} = 0.129$. (d) Reconstruction of a with noised data and with smoothings before and after inversion. $e_{quad} = 0.129$.

H Proof of the uniqueness theorem for the inverse problem using boundary measurements

In this appendix, we present the proof of Theorem 4. We refer to the notations introduced in Subsection 3.1. First, we introduce some additional notations used in the proof. We also state a Carleman estimate for solutions to the acoustic wave equation whose proof is done in [10] and is admitted here. We simply prove that the assumptions required for the theorem hold. Then, we develop several lemmas which are useful to finally prove the theorem. The proof is an adaptation of the one proposed in [10].

H.1 Notations and Carleman estimate for solutions to the wave equation

In this paragraph, we introduce the notations that are presented in [10] and the Carleman estimate that has been proven by M. Bellassoued. We invite the reader to read [10] for the proof and more explanation of the assumptions. A simplification of the notations from [10] can be done in our case. Indeed, M. Bellassoued has considered a general hyperbolic second-order operator $A(x, D_x)$ whereas we only consider the Laplacian operator in dimension 2. Thus, the Riemannian metric is simply

$$g(X, Y) = \frac{1}{c(x)} \langle X | Y \rangle_2,$$

where $\langle \cdot | \cdot \rangle_2$ denotes the Euclidean inner product on $\mathbb{R}^2$.

Then, two assumptions are introduced denoted by Assumptions H.1 and H.2. As explained in Remark 1, these assumptions hold in the case of the Laplacian by taking the following function

$$\theta(x) = \|x - x_0\|_2^2,$$

where $x_0 \notin \overline{\Omega}$ is the selected point as in Subsection 3.1. Then we prove the following lemma to ensure Assumptions H.1 and H.2 hold.

Proposition 14. *Under the notations from Subsection 3.1, then for every $c \in \mathcal{C}$ the following inequality holds for every $X \in \mathbb{R}^2$ and $x \in \overline{\Omega}$*

$$\mathbb{D}^2\theta(X, X)(x) \geq 2c_0 \left(1 - \frac{3}{2}r_0\right) |X|_g^2,$$

where $\mathbb{D}^2$ denotes the Hessian with respect to the Riemannian metric g and $|X|_g^2 = g(X, X)$.

Furthermore, θ has no critical point on $\overline{\Omega}$.

Proof. First, we prove that θ has no critical point on $\overline{\Omega}$. It is sufficient to prove that

$$\inf_{x \in \Omega} |\nabla_g \theta(x)|^2 > 0.$$

We calculate that

$$\begin{aligned} \forall x \in \Omega, \quad \nabla_g \theta(x) &= c(x) \nabla \theta(x) \\ &= 2c(x) \begin{pmatrix} x_1 - x_{0,1} \\ x_2 - x_{0,2} \end{pmatrix}. \end{aligned}$$

Then

$$\forall x \in \Omega, \quad |\nabla_g \theta(x)|^2 = 4c^2(x) \|x - x_0\|_2^2.$$

Since $c(x) \geq c_0$ and $x_0 \notin \bar{\Omega}$,

$$\forall x \in \Omega, \quad |\nabla_g \theta(x)|^2 \geq 4c_0^2 \inf_{y \in \Omega} \|y - x_0\|_2^2 > 0.$$

Thus, θ has no critical point.

Secondly, we prove the inequality. We make the Hessian explicit for $X = \alpha \frac{\partial}{\partial x_1} + \beta \frac{\partial}{\partial x_2}$

$$\begin{aligned} \mathbb{D}^2 \theta(X, X)(x) &= \alpha^2 \left(\frac{1}{c(x)} \frac{\partial (2c(x)(x_1 - x_{0,1}))}{\partial x_1} - \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_1} - \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_2} \right) \\ &\quad + \beta^2 \left(\frac{1}{c(x)} \frac{\partial (2c(x)(x_2 - x_{0,2}))}{\partial x_2} - \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_1} - \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_2} \right) \\ &\quad + \alpha\beta \left(\frac{1}{c(x)} \frac{\partial (2c(x)(x_2 - x_{0,2}))}{\partial x_1} + \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_2} - \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_1} \right) \\ &\quad + \alpha\beta \left(\frac{1}{c(x)} \frac{\partial (2c(x)(x_1 - x_{0,1}))}{\partial x_2} - \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_2} + \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_1} \right) \\ &= \alpha^2 \left(2 + \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_1} - \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_2} \right) + \beta^2 \left(2 + \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_2} - \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_1} \right) \\ &\quad + 2\alpha\beta \left(\frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_1} + \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_2} \right) \\ &= 2 \|X\|_2^2 - \|X\|_2^2 \frac{\nabla c \cdot (x - x_0)}{c(x)} + 2\alpha\beta \left(\frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_1} + \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_2} \right) \\ &\quad + 2\alpha^2 \frac{x_1 - x_{0,1}}{c(x)} \frac{\partial c}{\partial x_1} + 2\beta^2 \frac{x_2 - x_{0,2}}{c(x)} \frac{\partial c}{\partial x_2} \\ &= 2 \|X\|_2^2 - \|X\|_2^2 \frac{\nabla c \cdot (x - x_0)}{c(x)} + \frac{2}{c(x)} (\nabla c \cdot X) (X \cdot (x - x_0)). \end{aligned}$$

Then, we obtain

$$\mathbb{D}^2 \theta(X, X)(x) \geq \|X\|_2^2 \left(2 - \left| \frac{\nabla c \cdot (x - x_0)}{c(x)} \right| \right) - \frac{2}{c(x)} |(\nabla c \cdot X) (X \cdot (x - x_0))|.$$

Applying Cauchy-Schwarz's inequality, it yields

$$\mathbb{D}^2 \theta(X, X)(x) \geq \|X\|_2^2 \left(2 - 3 \frac{|\nabla c| \|x - x_0\|_2}{c(x)} \right).$$

Since $c \in \mathcal{C}$, the inequality becomes

$$\mathbb{D}^2 \theta(X, X)(x) \geq 2 \|X\|_2^2 \left(1 - \frac{3}{2} r_0 \right).$$

By definition of the Riemannian metric, it yields

$$\mathbb{D}^2 \theta(X, X)(x) \geq 2c(x) |X|_g^2 \left(1 - \frac{3}{2} r_0 \right) \geq 2c_0 \left(1 - \frac{3}{2} r_0 \right) |X|_g^2.$$

□

As a consequence, the definition of T_0 is the half of the value of T_0 that is introduced in [10]. Then we introduce some notations that slightly differ from those introduced by M. Bellassoued. Let $T > T_0$ as in Theorem 4. We take $\delta > 0$ such that

$$c_0 \left(1 - \frac{3r_0}{2}\right) T^2 > \sup_{x \in \bar{\Omega}} \theta(x) + \delta.$$

And let $0 < \gamma < c_0 \left(1 - \frac{3r_0}{2}\right)$ such that

$$\gamma T^2 > \sup_{x \in \bar{\Omega}} \theta(x) + \delta.$$

We define for $x \in \Omega$ and $t \in [0, 2T]$

$$\psi(x, t) = \theta(x) - \gamma (t - T)^2.$$

Let $\beta > 0$. We also define the Carleman weight function

$$\varphi(x, t) = e^{\beta \psi(x, t)}.$$

Through these notations, we can state the Carleman estimate from [10]. We admit the proof in this report.

Theorem 15 (Carleman estimate). *Let $\alpha \in [0, T]$ and $c \in \mathcal{C}$.*

Under the introduced notations, there exists $\lambda_ > 0$ and $C > 0$ which only depends on Ω , T , α , β and c such that for all $\lambda \geq \lambda_*$ and for all $u \in H^1(\Omega \times [0, 2T])$*

$$\begin{aligned} \lambda \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla u|^2 + \left| \frac{\partial u}{\partial t} \right|^2 + \lambda^2 |u|^2 \right) dx dt &\leq C \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| \frac{\partial^2 u}{\partial t^2} - c^2 \Delta u \right|^2 dx dt \\ &+ \lambda C \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(|\nabla u|^2 + \left| \frac{\partial u}{\partial t} \right|^2 + \lambda^2 |u|^2 \right) d\sigma(x) dt \\ &+ \lambda \int_{[0, 2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla u|^2 + \left| \frac{\partial u}{\partial t} \right|^2 + \lambda^2 |u|^2 \right) dx dt. \end{aligned}$$

H.2 Linearisation of the problem

In this subsection, we present a way of linearisation of the problem which is either Problem (1), (2) or (3). For everything coming after, we make the assumptions from Theorem 4. Let $u_0 \in H^1(\Omega)$ and $v_0 \in L^2(\Omega)$ such that

$$|\Delta u_0(x)| \geq \varepsilon_0, \quad \text{for almost every } x \in \bar{\Omega}.$$

Let $c, c_* \in \mathcal{C}$ and u, u_* the associated solutions to one of the problems on $\Omega \times]0, T[$. We assume that $u, u_* \in H^2(\Omega \times]0, T[) \cap W^{3,\infty}(\Omega \times]0, T[)$. We linearise the problems using the following way. First we define the symmetrised functions

$$\forall x \in \Omega, \forall t \in [-T, T], \quad U(x, t) = \begin{cases} u(x, t) & \text{if } t \geq 0, \\ u(x, -t) & \text{else.} \end{cases}$$

$$\forall x \in \Omega, \forall t \in [-T, T], \quad U_*(x, t) = \begin{cases} u_*(x, t) & \text{if } t \geq 0, \\ u_*(x, -t) & \text{else.} \end{cases}$$

Then we define $V = U - U_*$ and we make a translation of the time by defining

$$\forall x \in \Omega, \forall t \in [0, 2T], \quad v(x, t) = V(x, t - T).$$

One can easily show that $v \in H^2(\Omega \times]0, 2T[) \cap W^{3,\infty}(\Omega \times]0, 2T[)$ by testing on test functions and that the derivatives are related to the derivatives of u and u_* . We only prove here that v is solution to some problem.

Proof. First we prove that v is still solution to the wave equation in $\mathcal{D}'(\Omega \times]0, 2T[)$.

Let $\varphi \in \mathcal{D}(\Omega \times]0, 2T[)$, we have that

$$\begin{aligned} \int_0^{2T} \int_{\Omega} \frac{\partial^2 v}{\partial t^2}(x, t) \varphi(x, t) \, dx \, dt &= \int_0^{2T} \int_{\Omega} v(x, t) \frac{\partial^2 \varphi}{\partial t^2}(x, t) \, dx \, dt \\ &= \int_0^T \int_{\Omega} (u - u_*)(x, t) \frac{\partial^2 \varphi}{\partial t^2}(x, t + T) \, dx \, dt \\ &\quad + \int_{-T}^0 \int_{\Omega} (u - u_*)(x, -t) \frac{\partial^2 \varphi}{\partial t^2}(x, t + T) \, dx \, dt \\ &= \int_0^T \int_{\Omega} (u - u_*)(x, t) \frac{\partial^2 \varphi}{\partial t^2}(x, t + T) \, dx \, dt \\ &\quad + \int_0^T \int_{\Omega} (u - u_*)(x, t) \frac{\partial^2 \varphi}{\partial t^2}(x, T - t) \, dx \, dt \\ &= \int_0^T \int_{\Omega} \left(\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u_*}{\partial t^2} \right) (x, t) (\varphi(x, t + T) + \varphi(x, T - t)) \, dx \, dt \\ &= \int_0^T \int_{\Omega} (c^2(x) \Delta u - c_*^2(x) \Delta u_*)(x, t) (\varphi(x, t + T) + \varphi(x, T - t)) \, dx \, dt \\ &= \int_0^T \int_{\Omega} c^2(x) (\Delta u - \Delta u_*)(x, t) (\varphi(x, t + T) + \varphi(x, T - t)) \, dx \, dt \\ &\quad + \int_0^T \int_{\Omega} (c^2(x) - c_*^2(x)) \Delta u_*(x, t) (\varphi(x, t + T) + \varphi(x, T - t)) \, dx \, dt \\ &= \int_0^T \int_{\Omega} (u - u_*)(x, t) (\Delta (c^2 \varphi)(x, t + T) + \Delta (c^2 \varphi)(x, T - t)) \, dx \, dt \\ &\quad + \int_0^T \int_{\Omega} u_*(x, t) (\Delta ((c^2 - c_*^2) \varphi)(x, t + T) + \Delta ((c^2 - c_*^2) \varphi)(x, T - t)) \, dx \, dt \end{aligned}$$

Thus

$$\begin{aligned}
\int_0^{2T} \int_{\Omega} \frac{\partial^2 v}{\partial t^2}(x, t) \varphi(x, t) \, dx \, dt &= \int_0^T \int_{\Omega} (u - u_*)(x, t) \Delta (c^2 \varphi)(x, t + T) \, dx \, dt \\
&+ \int_{-T}^0 \int_{\Omega} (u - u_*)(x, -t) \Delta (c^2 \varphi)(x, t + T) \, dx \, dt \\
&+ \int_0^T \int_{\Omega} u_*(x, t) \Delta ((c^2 - c_*^2) \varphi)(x, t + T) \, dx \, dt \\
&+ \int_{-T}^0 \int_{\Omega} u_*(x, -t) \Delta ((c^2 - c_*^2) \varphi)(x, t + T) \, dx \, dt \\
&= \int_0^{2T} \int_{\Omega} c^2(x) \Delta v(x, t) \varphi(x, t) \, dx \, dt \\
&+ \int_0^{2T} \int_{\Omega} (c^2(x) - c_*^2(x)) \Delta U_*(x, t - T) \varphi(x, t) \, dx \, dt
\end{aligned}$$

Therefore in $\mathcal{D}'(\Omega \times]0, 2T[)$

$$\frac{\partial^2 v}{\partial t^2} - c^2(x) \Delta v = f(x)g(x, t),$$

where $f(x) = c^2(x) - c_*^2(x)$ and $g(x, t) = \Delta U_*(x, t - T)$.

We easily check that the initial conditions give

$$v(x, T) = 0, \quad \frac{\partial v}{\partial t}(x, T) = 0.$$

It remains to obtain the boundary conditions. Let $w \in H^1(\Omega)$, we calculate for $t \in [T, 2T]$

$$\begin{aligned}
\int_{\Omega} \Delta v(x, t) w(x) \, dx &= \int_{\Omega} (\Delta u(x, t - T) - \Delta u_*(x, t - T)) w(x) \, dx \\
&= \int_{\partial\Omega} \left(\frac{\partial u}{\partial n}(x, t - T) - \frac{\partial u_*}{\partial n}(x, t - T) \right) w(x) \, d\sigma(x) \\
&- \int_{\Omega} (\nabla u(x, t - T) - \nabla u_*(x, t - T)) \cdot \nabla w(x) \, dx \\
&= \int_{\partial\Omega} \left(\frac{\partial u}{\partial n}(x, t - T) - \frac{\partial u_*}{\partial n}(x, t - T) \right) w(x) \, d\sigma(x) \\
&- \int_{\Omega} \nabla v(x, t) \cdot \nabla w(x) \, dx
\end{aligned}$$

If we have an absorbing condition on $\Gamma \subset \partial\Omega$, then

$$\int_{\Omega} \Delta v(x, t) w(x) \, dx = - \int_{\Gamma} \frac{\partial v}{\partial t}(x, t) w(x) \, d\sigma(x) - \int_{\Omega} \nabla v(x, t) \cdot \nabla w(x) \, dx.$$

We do an analogous reasoning for $t \in [0, T]$ and we obtain

$$\int_{\Omega} \Delta v(x, t) w(x) \, dx = \int_{\Gamma} \frac{\partial v}{\partial t}(x, t) w(x) \, d\sigma(x) - \int_{\Omega} \nabla v(x, t) \cdot \nabla w(x) \, dx.$$

□

As a consequence, v is solution to the linearised problem

$$\begin{cases} \frac{\partial^2 v}{\partial t^2}(x, t) - c^2(x)\Delta v(x, t) = f(x)g(x, t) & \text{in } \Omega \times]0, 2T[, \\ v(x, T) = 0, \quad \frac{\partial v}{\partial t}(x, T) = 0 & \text{in } \Omega, \\ \frac{\partial v}{\partial n}(x, t) = -\frac{\partial v}{\partial t}(x, t) & \text{on } \Sigma_{abs,1}, \\ \frac{\partial v}{\partial n}(x, t) = \frac{\partial v}{\partial t}(x, t) & \text{on } \Sigma_{abs,2}, \\ \frac{\partial v}{\partial n}(x, t) = 0 & \text{on } \partial\Omega \times]0, 2T[\setminus (\Sigma_{abs,1} \cup \Sigma_{abs,2}). \end{cases} \quad (21)$$

where $f(x) = c^2(x) - c_*^2(x)$ and $g(x, t) = \Delta U_*(x, t - T)$ and:

- For Problem (1):

$$\Sigma_{abs,1} = \Sigma_{abs,2} = \emptyset$$

- For Problem (2):

$$\begin{aligned} \Sigma_{abs,1} &= \partial_1\Omega \times]T + t_1, 2T[\cup \partial_3\Omega \times]T, 2T[\\ \Sigma_{abs,2} &= \partial_1\Omega \times]0, T - t_1[\cup \partial_3\Omega \times]0, T[\end{aligned}$$

- For Problem (3):

$$\begin{aligned} \Sigma_{abs,1} &= \partial_1\Omega \times]T + t_1, 2T[\cup (\partial\Omega \setminus \partial_1\Omega) \times]T, 2T[\\ \Sigma_{abs,2} &= \partial_1\Omega \times]0, T - t_1[\cup (\partial\Omega \setminus \partial_1\Omega) \times]0, T[\end{aligned}$$

We have introduced those notations to unify the proof for the three models into a single proof.

Additionally, we can show similarly that $w = \frac{\partial v}{\partial t}$ is solution to a similar problem

$$\begin{cases} \frac{\partial^2 w}{\partial t^2}(x, t) - c^2(x)\Delta w(x, t) = f(x)\frac{\partial g}{\partial t}(x, t) & \text{in } \Omega \times]0, 2T[, \\ w(x, T) = 0, \quad \frac{\partial w}{\partial t}(x, T) = f(x)g(x, T) & \text{in } \Omega, \\ \frac{\partial w}{\partial n}(x, t) = -\frac{\partial w}{\partial t}(x, t) & \text{on } \Sigma_{abs,1}, \\ \frac{\partial w}{\partial n}(x, t) = \frac{\partial w}{\partial t}(x, t) & \text{on } \Sigma_{abs,2}, \\ \frac{\partial w}{\partial n}(x, t) = 0 & \text{on } \partial\Omega \times]0, 2T[\setminus (\Sigma_{abs,1} \cup \Sigma_{abs,2}). \end{cases} \quad (22)$$

where the second initial condition comes from the wave equation on v and the fact that $\Delta v(x, T) = 0$ from initial condition on v .

H.3 Proof of Theorem 4

In this subsection, we prove Theorem 4. We use in this part all the notations introduced until now in Subsection 3.1 and in the two previous subsections. Before beginning the proof of the theorem, we need to state two general lemmas.

The first lemma is called the energy identity for the acoustic wave equation. The proof of this lemma is simply an application of Green's formula.

Lemma 16 (Energy identity for the acoustic wave equation). *Let t_1 and $t_2 \in]0, 2T[$.*

Let $\Phi \in H^2(\Omega \times]0, 2T[)$. We define

$$\forall x \in \Omega, t \in]0, 2T[, \quad F(x, t) = \frac{\partial^2 \Phi}{\partial t^2}(x, t) - c^2(x) \Delta \Phi(x, t).$$

Then the following identity holds

$$\begin{aligned} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t_2) + |\nabla \Phi(x, t_2)|^2 dx &= \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t_1) + |\nabla \Phi(x, t_1)|^2 dx \\ &+ 2 \int_{t_1}^{t_2} \int_{\Omega} \frac{1}{c^2(x)} \frac{\partial \Phi}{\partial t}(x, t) F(x, t) dx dt + 2 \int_{t_1}^{t_2} \int_{\partial \Omega} \frac{\partial \Phi}{\partial t}(x, t) \frac{\partial \Phi}{\partial n}(x, t) d\sigma(x) dt. \end{aligned}$$

Proof. Since $\Phi \in H^2(\Omega \times]0, 2T[)$, we can apply Green's formula from Theorem 9:

$$\begin{aligned} \int_{t_1}^{t_2} \int_{\Omega} \frac{\partial \Phi}{\partial t}(x, t) \Delta \Phi(x, t) dx dt &= \int_{t_1}^{t_2} \int_{\partial \Omega} \frac{\partial \Phi}{\partial t}(x, t) \frac{\partial \Phi}{\partial n}(x, t) d\sigma(x) dt \\ &- \int_{t_1}^{t_2} \int_{\Omega} \frac{\partial \nabla \Phi}{\partial t}(x, t) \cdot \nabla \Phi(x, t) dx dt \\ &= \int_{t_1}^{t_2} \int_{\partial \Omega} \frac{\partial \Phi}{\partial t}(x, t) \frac{\partial \Phi}{\partial n}(x, t) d\sigma(x) dt \\ &- \frac{1}{2} \int_{t_1}^{t_2} \int_{\Omega} \frac{\partial |\nabla \Phi|^2}{\partial t}(x, t) dx dt \\ &= \int_{t_1}^{t_2} \int_{\partial \Omega} \frac{\partial \Phi}{\partial t}(x, t) \frac{\partial \Phi}{\partial n}(x, t) d\sigma(x) dt \\ &- \frac{1}{2} \int_{\Omega} |\nabla \Phi(x, t_2)|^2 dx + \frac{1}{2} \int_{\Omega} |\nabla \Phi(x, t_1)|^2 dx \end{aligned}$$

And:

$$\begin{aligned} \int_{t_1}^{t_2} \int_{\Omega} \frac{1}{c^2(x)} \frac{\partial \Phi}{\partial t}(x, t) \frac{\partial^2 \Phi}{\partial t^2}(x, t) dx dt &= \frac{1}{2} \int_{t_1}^{t_2} \int_{\Omega} \frac{\partial}{\partial t} \left(\frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t) \right) dx dt \\ &= \frac{1}{2} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t_2) dx - \frac{1}{2} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t_1) dx \end{aligned}$$

Thus, the identity is simply the difference between these equations. $\square$

The second lemma is a useful inequality for the acoustic wave equation. This inequality shows a control of the initial condition on the derivative by the solution to the acoustic wave equation and the boundary condition. The proof of this lemma is an application of Lemma 16.

Lemma 17. Let $\alpha \in [0, T]$. Let $F \in L^2(\Omega \times]0, 2T[)$, $\Phi_1 \in L^2(\Omega)$ and $h \in L^2(\partial\Omega \times]0, 2T[)$. Let $\Phi \in H^2(\Omega \times]0, 2T[)$ be solution to

$$\begin{cases} \frac{\partial^2 \Phi}{\partial t^2}(x, t) - c^2(x) \Delta \Phi(x, t) = F(x, t) & \text{in } \Omega \times]0, 2T[, \\ \Phi(x, T) = 0, \quad \frac{\partial \Phi}{\partial t}(x, T) = \Phi_1(x) & \text{in } \Omega, \\ \frac{\partial \Phi}{\partial n}(x, t) = h(x, t) & \text{on } \partial\Omega \times]0, 2T[. \end{cases}$$

Then the following inequality holds

$$\begin{aligned} \|\Phi_1\|_{L^2(\Omega)}^2 &\leq \frac{1}{2c_0^2(T-\alpha)} \left\| \frac{\partial \Phi}{\partial t} \right\|_{L^2(\Omega \times]\alpha, 2T-\alpha[)}^2 + \frac{1}{2(T-\alpha)} \|\nabla \Phi\|_{L^2(\Omega \times]\alpha, 2T-\alpha[)}^2 \\ &\quad + \|h\|_{L^2(\partial\Omega \times]0, 2T[)}^2 + \left\| \frac{\partial \Phi}{\partial t} \right\|_{L^2(\partial\Omega \times]0, 2T[)}^2 \\ &\quad + \frac{2}{c_0^2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| F(x, t) \frac{\partial \Phi}{\partial t}(x, t) \right| dx dt. \end{aligned}$$

Proof. We apply Lemma 16 to Φ at $t_1 = T$ and $t_2 = t$ an arbitrary time in $[\alpha, 2T - \alpha]$:

$$\begin{aligned} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t) + |\nabla \Phi(x, t)|^2 dx &= \int_{\Omega} \frac{1}{c^2(x)} |\Phi_1|^2(x) dx \\ &\quad + 2 \int_T^t \int_{\Omega} \frac{1}{c^2(x)} \frac{\partial \Phi}{\partial t}(x, \tau) F(x, \tau) dx d\tau + 2 \int_T^t \int_{\partial\Omega} \frac{\partial \Phi}{\partial t}(x, \tau) h(x, \tau) d\sigma(x) d\tau. \end{aligned}$$

Since $c_0 \leq c(x) \leq 1$, it yields

$$\begin{aligned} \|\Phi_1\|_{L^2(\Omega)}^2 &\leq \frac{1}{c_0^2} \int_{\Omega} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t) dx + \int_{\Omega} |\nabla \Phi(x, t)|^2 dx \\ &\quad + \frac{2}{c_0^2} \left| \int_T^t \int_{\Omega} \left| \frac{\partial \Phi}{\partial t}(x, \tau) F(x, \tau) \right| dx d\tau \right| + 2 \left| \int_T^t \int_{\partial\Omega} \left| \frac{\partial \Phi}{\partial t}(x, \tau) \right| |h(x, \tau)| d\sigma(x) d\tau \right|. \end{aligned}$$

Then, we obtain

$$\begin{aligned} \|\Phi_1\|_{L^2(\Omega)}^2 &\leq \frac{1}{c_0^2} \int_{\Omega} \left| \frac{\partial \Phi}{\partial t} \right|^2(x, t) dx + \int_{\Omega} |\nabla \Phi(x, t)|^2 dx \\ &\quad + \frac{2}{c_0^2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| F(x, \tau) \frac{\partial \Phi}{\partial t}(x, \tau) \right| dx d\tau + 2 \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial \Phi}{\partial t}(x, \tau) \right| |h(x, \tau)| d\sigma(x) d\tau. \end{aligned}$$

Integrating over $t \in [\alpha, 2T - \alpha]$ and using Cauchy-Schwarz's inequality, it yields

$$\begin{aligned} 2(T-\alpha) \|\Phi_1\|_{L^2(\Omega)}^2 &\leq \frac{1}{c_0^2} \left\| \frac{\partial \Phi}{\partial t} \right\|_{L^2(\Omega \times]\alpha, 2T-\alpha[)}^2 + \|\nabla \Phi\|_{L^2(\Omega \times]\alpha, 2T-\alpha[)}^2 \\ &\quad + 4(T-\alpha) \|h\|_{L^2(\partial\Omega \times]0, 2T[)} \left\| \frac{\partial \Phi}{\partial t} \right\|_{L^2(\partial\Omega \times]0, 2T[)} \\ &\quad + \frac{4(T-\alpha)}{c_0^2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| F(x, t) \frac{\partial \Phi}{\partial t}(x, t) \right| dx dt. \end{aligned}$$

We conclude the proof using the typical inequality $2ab \leq a^2 + b^2$. $\square$

After these general lemmas, we first prove two energy inequalities for the defined v . These inequalities give upper bounds of the first-order derivatives of v and $\frac{\partial v}{\partial t}$. The proof is an application of the energy identity from Lemma 16 on v and $\frac{\partial v}{\partial t}$ using that v verifies (21) and $\frac{\partial v}{\partial t}$ verifies (22). The results in Lemma 18 are the essential adaptation we made from [10].

Lemma 18. *Let v be the solution to Problem (21) as defined.*

Firstly, g and $\frac{\partial g}{\partial t} \in L^2(]0, 2T[, L^\infty(\Omega))$ and we denote by

$$\|g\|^2 = \int_0^{2T} \|g(t)\|_{L^\infty(\Omega)}^2 dt + \int_0^{2T} \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^\infty(\Omega)}^2 dt.$$

Then there exist two constants $C_1 > 0$ and $C_2 > 0$ such that

$$\int_0^{2T} \int_\Omega \left| \frac{\partial v}{\partial t} \right|^2(x, t) + |\nabla v|^2(x, t) dx dt \leq C_1 \|f\|_{L^2(\Omega)}^2 + 16T \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2(x, t) d\sigma(x) dt,$$

$$\int_0^{2T} \int_\Omega \left| \frac{\partial^2 v}{\partial t^2} \right|^2(x, t) + \left| \nabla \frac{\partial v}{\partial t} \right|^2(x, t) dx dt \leq C_2 \|f\|_{L^2(\Omega)}^2 + 16T \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial^2 v}{\partial t^2} \right|^2(x, t) d\sigma(x) dt.$$

Furthermore, we have

$$C_1 = \frac{16T^2 \|g\|^2}{c_0^2} \quad \text{and} \quad C_2 = C_1 + \frac{4T \|\Delta u_0\|_{L^\infty(\Omega)}^2}{c_0^2}.$$

Proof. First, we prove the assertion on g . This property is the result of the assumption that $u_* \in W^{3,\infty}(\Omega \times]0, T[)$. Then almost everywhere

$$|g(x, t)| = |\Delta U_*(x, t - T)| \leq \|\Delta u_*\|_{L^\infty(\Omega \times]0, T[)}.$$

Thus, for almost every $t \in]0, 2T[$, $g(t) \in L^\infty(\Omega)$ and

$$\|g(t)\|_{L^\infty(\Omega)}^2 \leq \|\Delta u_*\|_{L^\infty(\Omega \times]0, T[)}^2 \quad \text{for almost every } t \in]0, 2T[.$$

By integrating over $]0, 2T[$, it yields that $g \in L^2(]0, 2T[, L^\infty(\Omega))$. The analogous reasoning on $\frac{\partial g}{\partial t}$ gives the same with $\left\| \frac{\partial \Delta u_*}{\partial t} \right\|_{L^\infty(\Omega \times]0, T[)}$, since $u_* \in W^{3,\infty}(\Omega \times]0, T[)$.

Secondly, we consider the first inequality. We apply the identity from Lemma 16 to v with $t_1 = T$ and $t \in]0, 2T[$. Since v and its derivatives at $t = T$ are homogeneous and that $F(x, t) = f(x)g(x, t)$, it yields

$$\begin{aligned} \int_\Omega \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2(x, t) + |\nabla v(x, t)|^2 dx &= 2 \int_T^t \int_\Omega \frac{1}{c^2(x)} \frac{\partial v}{\partial t}(x, t) f(x) g(x, t) dx dt \\ &\quad - 2 \int_{\Sigma_{abs,1} \cap \partial\Omega \times]T, t[} \left| \frac{\partial v}{\partial t} \right|^2(x, t) d\sigma(x) dt \\ &\quad + 2 \int_{\Sigma_{abs,2} \cap \partial\Omega \times]T, t[} \left| \frac{\partial v}{\partial t} \right|^2(x, t) d\sigma(x) dt \end{aligned}$$

Then, a simple upper bound yields

$$\begin{aligned} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) + |\nabla v(x, t)|^2 dx &\leq 2 \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) |f(x)g(x, t)| dx dt \\ &\quad + 4 \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) d\sigma(x) dt. \end{aligned}$$

Using the following typical inequality

$$\forall a, b \geq 0, \quad 2ab \leq a^2 + b^2,$$

with $a = \sqrt{\varepsilon} \left| \frac{\partial v}{\partial t} \right| (x, t)$ and $b = \frac{|f(x)g(x, t)|}{\sqrt{\varepsilon}}$ for $\varepsilon > 0$, we obtain

$$\begin{aligned} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) + |\nabla v(x, t)|^2 dx &\leq \int_0^{2T} \int_{\Omega} \frac{1}{\varepsilon c^2(x)} f^2(x) g^2(x, t) dx dt \\ &\quad + \varepsilon \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) dx dt \\ &\quad + 4 \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) d\sigma(x) dt \\ &\leq \frac{1}{\varepsilon c_0^2} \int_0^{2T} \int_{\Omega} f^2(x) g^2(x, t) dx dt \\ &\quad + \varepsilon \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) dx dt \\ &\quad + 4 \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) d\sigma(x) dt \end{aligned}$$

Furthermore, using the property on g , it yields

$$\int_0^{2T} \int_{\Omega} f^2(x) g^2(x, t) dx dt \leq \int_{\Omega} f^2(x) \int_0^{2T} \|g(t)\|_{L^\infty(\Omega)}^2 dt dx \leq \|g\|^2 \|f\|_{L^2(\Omega)}^2.$$

Then, this last inequality and an integration with respect to t give

$$\begin{aligned} \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) + |\nabla v(x, t)|^2 dx dt &\leq \frac{2T \|g\|^2}{\varepsilon c_0^2} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 8T \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) d\sigma(x) dt \\ &\quad + 2T\varepsilon \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) dx dt. \end{aligned}$$

For $\varepsilon = \frac{1}{4T}$, grouping the last term with the left-hand side, it yields

$$\begin{aligned} \frac{1}{2} \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) + |\nabla v(x, t)|^2 dx dt &\leq \frac{8T^2 \|g\|^2}{c_0^2} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 8T \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial v}{\partial t} \right|^2 (x, t) d\sigma(x) dt, \end{aligned}$$

which is exactly the first inequality.

The second inequality is proven in the same way using that $\frac{\partial v}{\partial t}$ is solution to Problem (22). We only write the calculus with the modifications.

$$\begin{aligned} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial^2 v}{\partial t^2} \right|^2 (x, t) + \left| \nabla \frac{\partial v}{\partial t} (x, t) \right|^2 dx &= \int_{\Omega} \frac{1}{c^2(x)} |f(x)g(x, T)|^2 dx \\ &+ 2 \int_T^t \int_{\Omega} \frac{1}{c^2(x)} \frac{\partial^2 v}{\partial t^2} (x, t) f(x) \frac{\partial g}{\partial t} (x, t) dx dt \\ &- 2 \int_{\Sigma_{abs,1} \cap \partial\Omega \times]T, t[} \left| \frac{\partial^2 v}{\partial t^2} \right|^2 (x, t) d\sigma(x) dt \\ &+ 2 \int_{\Sigma_{abs,2} \cap \partial\Omega \times]T, t[} \left| \frac{\partial^2 v}{\partial t^2} \right|^2 (x, t) d\sigma(x) dt \end{aligned}$$

We note that the calculus are the same with $\frac{\partial v}{\partial t}$ instead of v and with an additional term. This term is easily taken into account with

$$\int_{\Omega} \frac{1}{c^2(x)} |f(x)g(x, T)|^2 dx \leq \frac{1}{c_0^2} \int_{\Omega} |f(x)g(x, T)|^2 dx.$$

Since $g(x, T) = \Delta u_0(x)$, we can bound from above by $\|\Delta u_0\|_{L^\infty(\Omega)}$. Then the final inequality becomes

$$\begin{aligned} \frac{1}{2} \int_0^{2T} \int_{\Omega} \frac{1}{c^2(x)} \left| \frac{\partial^2 v}{\partial t^2} \right|^2 (x, t) + \left| \nabla \frac{\partial v}{\partial t} (x, t) \right|^2 dx dt &\leq \frac{8T^2 \|g\|^2}{c_0^2} \|f\|_{L^2(\Omega)}^2 + \frac{2T \|\Delta u_0\|_{L^\infty(\Omega)}^2}{c_0^2} \|f\|_{L^2(\Omega)}^2 \\ &+ 8T \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial^2 v}{\partial t^2} \right|^2 (x, t) d\sigma(x) dt, \end{aligned}$$

which is the second inequality. This concludes the proof. $\square$

After all these lemmas, we have now all the bricks to begin the proof of the inequality from Theorem 4. The next lemma is a first inequality that states a control of the solutions inside Ω by the values of the parameters and the values on the boundary. This inequality is simply an application of the Carleman estimate for such $\alpha \in]0, T[$. We remind here that $w = \frac{\partial v}{\partial t}$ and w is solution to Problem (22). Then Theorem 15 states that for such $\alpha \in]0, T[$, there exists $C > 0$ and $\lambda_* > 0$ such that for every $\lambda \geq \lambda_*$

$$\begin{aligned} \lambda \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt &\leq C \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f(x) \frac{\partial g}{\partial t} (x, t) \right|^2 dx dt \\ &+ \lambda C \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt \\ &+ \lambda \int_{[0, 2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt. \quad (23) \end{aligned}$$

In this inequality, we have simply replaced the term in the first integral on the right-hand side using Problem (22). Furthermore, we can assume that $\lambda_* \geq 1$ without a lack of generality. Then the next lemma is a simple upper bound of the right-hand side and a selection of such $\alpha \in]0, T[$. We set the parameters from Lemma 19 for the rest of the proof.

Lemma 19. *Let $w = \frac{\partial v}{\partial t}$ be the solution to Problem (22).*

Then there exist $\alpha \in]0, T[$, $0 < \kappa < 1$ and a constant $C_3 > 0$ such that for all $\lambda \geq \lambda_$ the following inequality holds*

$$\begin{aligned} \lambda \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt &\leq C_3 \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\ &+ \lambda C_3 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt \\ &+ 2C_2 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 + 16T \lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 d\sigma(x) dt. \end{aligned}$$

Proof. First, we note that for all $x \in \Omega$,

$$\psi(x, 0) = \psi(x, 2T) = \theta(x) - \gamma T^2.$$

Using the choice of γ , it yields that for all $x \in \Omega$

$$\psi(x, 0) = \psi(x, 2T) < -\delta.$$

Since ψ is continuous, there exists $\alpha \in]0, T[$ such that

$$\forall x \in \Omega, \forall t \in [0, \alpha] \cup [2T - \alpha, 2T], \quad \psi(x, t) \leq -\frac{\delta}{2}.$$

Then, by definition of φ , we obtain

$$\forall x \in \Omega, \forall t \in [0, \alpha] \cup [2T - \alpha, 2T], \quad \varphi(x, t) \leq e^{-\frac{\beta\delta}{2}} < 1.$$

We define $\kappa := e^{-\frac{\beta\delta}{2}}$ and thus $0 < \kappa < 1$.

Since α is now set, we consider λ_* as given by the Carleman estimate and we denote by C_3 the constant in the Carleman estimate (23). Let $\lambda \geq \lambda_*$ be an arbitrary coefficient for the rest of the proof.

As a consequence, the inequality simply comes from an upper bound of the last term from (23). We deal with this term as

$$\begin{aligned} \lambda \int_{[0, 2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt = \\ \lambda \int_0^{\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt \\ + \lambda \int_{2T-\alpha}^{2T} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt. \end{aligned}$$

From the definition of κ , it yields that

$$\begin{aligned} \lambda \int_{[0,2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt \leq \\ \lambda e^{2\lambda\kappa} \int_0^{2T} \int_{\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt. \end{aligned}$$

Since $\lambda \geq \lambda_* \geq 1$ and as $w = \frac{\partial v}{\partial t}$, then we obtain

$$\begin{aligned} \lambda \int_{[0,2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt \leq \\ \lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\Omega} \left(\left| \nabla \frac{\partial v}{\partial t} \right|^2 + \left| \frac{\partial^2 v}{\partial t^2} \right|^2 + \left| \frac{\partial v}{\partial t} \right|^2 + |\nabla v|^2 \right) dx dt. \end{aligned}$$

Using the two inequalities from Lemma 18, it yields

$$\begin{aligned} \lambda \int_{[0,2T] \setminus [\alpha, 2T-\alpha]} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt \leq \\ \lambda^3 e^{2\lambda\kappa} (C_1 + C_2) \|f\|_{L^2(\Omega)}^2 + 16T\lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\partial\Omega} \left(\left| \frac{\partial v}{\partial t} \right|^2 + \left| \frac{\partial^2 v}{\partial t^2} \right|^2 \right) d\sigma(x) dt. \end{aligned}$$

which concludes since $C_1 \leq C_2$ by their definition. $\square$

On the other hand, we state another inequality using Lemma 17. This inequality is proven independently from the previous lemma. The proof we propose here is slightly different from the initial proof in [10].

Lemma 20. *Let $w = \frac{\partial v}{\partial t}$ be the solution to Problem (22).*

We decompose φ as

$$\varphi(x, t) = e^{\beta\theta(x)} e^{-\beta(t-T)^2} = \rho(x)\sigma(t).$$

We define the positive constants

$$C_4 = \frac{1}{\varepsilon_0^2} \max \left(1, \frac{1}{c_0^2}, \frac{1}{T-\alpha}, \frac{1}{c_0^2(T-\alpha)} \right),$$

and

$$C_5 = 24C_4 \max \left(1, \|\varphi\|_{\infty}^2, \left\| \frac{\partial \varphi}{\partial t} \right\|_{\infty}^2, \left\| \frac{\partial \varphi}{\partial t} \right\|_{\infty}^4, \left\| \frac{\partial^2 \varphi}{\partial t^2} \right\|_{\infty}^2, \|\nabla \varphi\|_{\infty}^2, \|\nabla \varphi\|_{\infty}^4, \|\Delta \varphi\|_{\infty}^2 \right),$$

where $\|\cdot\|_{\infty}$ denotes the supremum on $\Omega \times [0, 2T]$.

Then the following inequality holds for every $\lambda > 0$

$$\begin{aligned} \|e^{\lambda\rho}f\|_{L^2(\Omega)}^2 &\leq C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\ &\quad + \lambda C_5 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) dx dt \\ &\quad + C_5 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(\left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt. \end{aligned}$$

Proof. First, we define the function

$$\forall x \in \Omega, \forall t \in]0, 2T[, \quad \Upsilon(x, t) := e^{\lambda\varphi(x, t)} w(x, t).$$

We calculate some derivatives of Υ to apply Lemma 17 on it. Since $\sigma(T) = 1$ and $w(x, T) = 0$, we have

$$\forall x \in \Omega, \quad \Upsilon(x, T) = e^{\lambda\rho(x)} w(x, T) = 0.$$

Using $\frac{\partial w}{\partial t}(x, T) = f(x)g(x, T)$, it yields

$$\begin{aligned} \forall x \in \Omega, \quad \frac{\partial \Upsilon}{\partial t}(x, T) &= e^{\lambda\rho(x)} \frac{\partial w}{\partial t}(x, T) + \lambda e^{\lambda\rho(x)} w(x, T) \frac{\partial \varphi}{\partial t}(x, T) \\ &= e^{\lambda\rho(x)} f(x)g(x, T). \end{aligned}$$

We apply the acoustic wave operator on Υ , we obtain for all $x \in \Omega$ and $t \in]0, 2T[$

$$\begin{aligned} \frac{\partial^2 \Upsilon}{\partial t^2}(x, t) - c^2(x) \Delta \Upsilon(x, t) &= e^{\lambda\varphi(x, t)} f(x) \frac{\partial g}{\partial t}(x, t) \\ &\quad + 2\lambda e^{\lambda\varphi(x, t)} \left(\frac{\partial \varphi}{\partial t}(x, t) \frac{\partial w}{\partial t}(x, t) - c^2(x) \nabla \varphi(x, t) \cdot \nabla w(x, t) \right) \\ &\quad + e^{\lambda\varphi(x, t)} w(x, t) \left(\lambda^2 \left| \frac{\partial \varphi}{\partial t} \right|^2(x, t) + \lambda \frac{\partial^2 \varphi}{\partial t^2}(x, t) \right. \\ &\quad \left. - \lambda^2 c^2(x) |\nabla \varphi(x, t)|^2 - \lambda c^2(x) \Delta \varphi(x, t) \right). \end{aligned}$$

Finally, we calculate the normal derivative of Υ as

$$\forall x \in \Omega, \forall t \in]0, 2T[, \quad \frac{\partial \Upsilon}{\partial n}(x, t) = \lambda e^{\lambda\varphi(x, t)} w(x, t) \frac{\partial \varphi}{\partial n}(x, t) + e^{\lambda\varphi(x, t)} \frac{\partial w}{\partial n}(x, t).$$

We easily check that the assumptions from Lemma 17 hold since φ and its derivatives are bounded and $w \in H^1(\Omega \times]0, 2T[)$ (moreover, f is bounded and $\frac{\partial g}{\partial t} \in L^\infty(\Omega \times]0, 2T[)$ and Ω is compact). Then it yields

$$\begin{aligned} \|e^{\lambda\rho}fg(\cdot, T)\|_{L^2(\Omega)}^2 &\leq \frac{1}{2c_0^2(T-\alpha)} \left\| \frac{\partial \Upsilon}{\partial t} \right\|_{L^2(\Omega \times]\alpha, 2T-\alpha])}^2 + \frac{1}{2(T-\alpha)} \|\nabla \Upsilon\|_{L^2(\Omega \times]\alpha, 2T-\alpha])}^2 \\ &\quad + \left\| \frac{\partial \Upsilon}{\partial t} \right\|_{L^2(\partial\Omega \times]0, 2T])}^2 + \left\| \lambda e^{\lambda\varphi} w \frac{\partial \varphi}{\partial n} + e^{\lambda\varphi} \frac{\partial w}{\partial n} \right\|_{L^2(\partial\Omega \times]0, 2T])}^2 \\ &\quad + \frac{2}{c_0^2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| \left(\frac{\partial^2 \Upsilon}{\partial t^2} - c^2(x) \Delta \Upsilon \right) \frac{\partial \Upsilon}{\partial t} \right| dx dt. \end{aligned}$$

First, we deal with the term $g(T)$ in the left-hand side. By the definition of g , then $g(x, T) = \Delta u_0(x)$. Since by assumption $\Delta u_0(x) \geq \varepsilon_0$, then $\|e^{\lambda \rho} f g(\cdot, T)\|_{L^2(\Omega)}^2 \geq \varepsilon_0^2 \|e^{\lambda \rho} f\|_{L^2(\Omega)}^2$.

Then we analyse the five integrals from the right-hand side independently. The first integral can be bounded from above using the inequality $(a + b)^2 \leq 2(a^2 + b^2)$.

$$\begin{aligned} \left\| \frac{\partial \Upsilon}{\partial t} \right\|_{L^2(\Omega \times]\alpha, 2T-\alpha])}^2 &= \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left| \lambda \frac{\partial \varphi}{\partial t} w + \frac{\partial w}{\partial t} \right|^2 dx dt \\ &\leq 2 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left(\lambda^2 \left| \frac{\partial \varphi}{\partial t} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 \right) dx dt. \end{aligned}$$

The second, third and fourth terms are bounded from above similarly

$$\begin{aligned} \|\nabla \Upsilon\|_{L^2(\Omega \times]\alpha, 2T-\alpha])}^2 &\leq 2 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} (\lambda^2 |\nabla \varphi|^2 |w|^2 + |\nabla w|^2) dx dt \\ \left\| \frac{\partial \Upsilon}{\partial t} \right\|_{L^2(\partial \Omega \times]0, 2T])}^2 &\leq 2 \int_0^{2T} \int_{\partial \Omega} e^{2\lambda \varphi} \left(\lambda^2 \left| \frac{\partial \varphi}{\partial t} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 \right) d\sigma(x) dt \\ \left\| \lambda e^{\lambda \varphi} w \frac{\partial \varphi}{\partial n} + e^{\lambda \varphi} \frac{\partial w}{\partial n} \right\|_{L^2(\partial \Omega \times]0, 2T])}^2 &\leq 2 \int_0^{2T} \int_{\partial \Omega} e^{2\lambda \varphi} \left(\lambda^2 \left| \frac{\partial \varphi}{\partial n} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial n} \right|^2 \right) d\sigma(x) dt \end{aligned}$$

The last term is first bounded as

$$\begin{aligned} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| \left(\frac{\partial^2 \Upsilon}{\partial t^2} - c^2(x) \Delta \Upsilon \right) \frac{\partial \Upsilon}{\partial t} \right| dx dt &\leq \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{\lambda \varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right| \left| \frac{\partial \Upsilon}{\partial t} \right| dx dt \\ &\quad + \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left[2\lambda e^{\lambda \varphi} \left(\left| \frac{\partial \varphi}{\partial t} \right| \left| \frac{\partial w}{\partial t} \right| + c^2(x) |\nabla \varphi| |\nabla w| \right) \right. \\ &\quad \left. + e^{\lambda \varphi} |w| \left(\lambda^2 \left| \frac{\partial \varphi}{\partial t} \right|^2 + \lambda \left| \frac{\partial^2 \varphi}{\partial t^2} \right| + \lambda^2 c^2(x) |\nabla \varphi|^2 + \lambda c^2(x) |\Delta \varphi| \right) \right] \left| \frac{\partial \Upsilon}{\partial t} \right| dx dt. \end{aligned}$$

The first integral can be bounded using Cauchy-Schwarz's inequality and the first upper bound

$$\begin{aligned} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{\lambda \varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right| \left| \frac{\partial \Upsilon}{\partial t} \right| dx dt &\leq \left(\int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \right)^{\frac{1}{2}} \\ &\quad \left(\int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| \frac{\partial \Upsilon}{\partial t} \right|^2 dx dt \right)^{\frac{1}{2}} \\ &\leq \frac{1}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\ &\quad + \frac{1}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left| \frac{\partial \Upsilon}{\partial t} \right|^2 dx dt \\ &\leq \frac{1}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\ &\quad + \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda \varphi} \left(\lambda^2 \left| \frac{\partial \varphi}{\partial t} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 \right) dx dt \end{aligned}$$

Finally, the second integral can be bounded by $c(x) \leq 1$, successive Cauchy-Schwarz's inequalities and $2ab \leq a^2 + b^2$.

$$\begin{aligned}
& \int_{\alpha}^{2T-\alpha} \int_{\Omega} \left[2\lambda e^{\lambda\varphi} \left(\left| \frac{\partial\varphi}{\partial t} \right| \left| \frac{\partial w}{\partial t} \right| + c^2(x) |\nabla\varphi| |\nabla w| \right) \right. \\
& \quad \left. + e^{\lambda\varphi} |w| \left(\lambda^2 \left| \frac{\partial\varphi}{\partial t} \right|^2 + \lambda \left| \frac{\partial^2\varphi}{\partial t^2} \right| + \lambda^2 c^2(x) |\nabla\varphi|^2 + \lambda c^2(x) |\Delta\varphi| \right) \right] \left| \frac{\partial\Upsilon}{\partial t} \right| dx dt \\
& \leq (3\lambda + 1) \left\| \frac{\partial\Upsilon}{\partial t} \right\|_{L^2(\Omega \times]\alpha, 2T-\alpha])}^2 + \lambda \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| \frac{\partial\varphi}{\partial t} \right|^2 \left| \frac{\partial w}{\partial t} \right|^2 dx dt \\
& \quad + \frac{\lambda^3}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| \frac{\partial\varphi}{\partial t} \right|^4 |w|^2 dx dt + \frac{\lambda^2}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| \frac{\partial^2\varphi}{\partial t^2} \right|^2 |w|^2 dx dt \\
& \quad + \frac{\lambda^3}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} |\nabla\varphi|^4 |w|^2 dx dt + \frac{\lambda^2}{2} \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} |\Delta\varphi|^2 |w|^2 dx dt \\
& \quad + \lambda \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} |\nabla\varphi|^2 |\nabla w|^2 dx dt
\end{aligned}$$

Using all the upper bounds and the definition of C_4 , it yields

$$\begin{aligned}
& \|e^{\lambda\rho} f\|_{L^2(\Omega)}^2 \leq C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\
& + \lambda^3 C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} |w|^2 \left(19 \left| \frac{\partial\varphi}{\partial t} \right|^2 + \left| \frac{\partial\varphi}{\partial t} \right|^4 + |\nabla\varphi|^4 + |\Delta\varphi|^2 + \left| \frac{\partial^2\varphi}{\partial t^2} \right|^2 + |\nabla\varphi|^2 \right) dx dt \\
& \quad + \lambda C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| \frac{\partial w}{\partial t} \right|^2 \left(19 + 2 \left| \frac{\partial\varphi}{\partial t} \right|^2 \right) dx dt \\
& \quad + \lambda C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} |\nabla w|^2 (1 + 2 |\nabla\varphi|^2) dx dt \\
& \quad + 2C_4 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(\lambda^2 \left| \frac{\partial\varphi}{\partial t} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 \right) d\sigma(x) dt \\
& \quad + 2C_4 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(\lambda^2 \left| \frac{\partial\varphi}{\partial n} \right|^2 |w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 \right) d\sigma(x) dt.
\end{aligned}$$

Thus, by the definition of C_5 , we obtain

$$\begin{aligned}
& \|e^{\lambda\rho} f\|_{L^2(\Omega)}^2 \leq C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| f(x) \frac{\partial g}{\partial t}(x, t) \right|^2 dx dt \\
& \quad + \lambda C_5 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left(\left| \frac{\partial w}{\partial t} \right|^2 + |\nabla w|^2 + \lambda^2 |w|^2 \right) dx dt \\
& \quad + C_5 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(\left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt.
\end{aligned}$$

which is the announced inequality. $\square$

Finally, we can now finish the proof of Theorem 4.

Proof. First, we use the inequality from Lemma 19 to bound from above the inequality from Lemma 20. It yields

$$\begin{aligned} \|e^{\lambda\rho}f\|_{L^2(\Omega)}^2 &\leq C_4 \int_{\alpha}^{2T-\alpha} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt + C_3 C_5 \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt \\ &\quad + \lambda C_3 C_5 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt \\ &\quad + 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 + 16TC_5 \lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 d\sigma(x) dt \\ &\quad + C_5 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(\left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt. \end{aligned}$$

By grouping the terms and bounding from above (assuming $C_3 \geq 1$), we obtain

$$\begin{aligned} \|e^{\lambda\rho}f\|_{L^2(\Omega)}^2 &\leq (C_4 + C_3 C_5) \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt + 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 2\lambda C_3 C_5 \int_0^{2T} \int_{\partial\Omega} e^{2\lambda\varphi} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + \lambda^2 |w|^2 \right) d\sigma(x) dt \\ &\quad + 16TC_5 \lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 d\sigma(x) dt. \end{aligned}$$

Then, using that φ is bounded, it yields

$$\begin{aligned} \|e^{\lambda\rho}f\|_{L^2(\Omega)}^2 &\leq (C_4 + C_3 C_5) \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt + 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 2\lambda^3 C_3 C_5 e^{2\lambda\|\varphi\|_{\infty}} \int_0^{2T} \int_{\partial\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 \right) d\sigma(x) dt \\ &\quad + 16TC_5 \lambda^3 e^{2\lambda\kappa} \int_0^{2T} \int_{\partial\Omega} \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 d\sigma(x) dt. \end{aligned}$$

By grouping again, we finally obtain that

$$\begin{aligned} \|e^{\lambda\rho}f\|_{L^2(\Omega)}^2 &\leq (C_4 + C_3 C_5) \int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt + 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 2\lambda^3 C_5 (C_3 e^{2\lambda\|\varphi\|_{\infty}} + 8T e^{2\lambda\kappa}) \int_0^{2T} \int_{\partial\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 \right) d\sigma(x) dt. \end{aligned}$$

First, we deal with the first integral from the right-hand side. Using the property that $\frac{\partial g}{\partial t} \in L^2([0, 2T[, L^{\infty}(\Omega))$ and the decomposition of $\varphi(x, t) = \rho(x)\sigma(t)$, it becomes

$$\int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt \leq \int_{\Omega} e^{2\lambda\rho(x)} |f(x)|^2 \int_0^{2T} e^{-2\lambda\rho(x)(1-\sigma(t))} \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^{\infty}(\Omega)}^2 dt dx.$$

Since $\rho(x) = e^{\beta\theta(x)} \geq 1$ and $\sigma(t) = e^{-\beta(t-T)^2} \leq 1$, it yields

$$\int_0^{2T} \int_{\Omega} e^{2\lambda\varphi} \left| f \frac{\partial g}{\partial t} \right|^2 dx dt \leq \int_{\Omega} e^{2\lambda\rho(x)} |f(x)|^2 \int_0^{2T} e^{-2\lambda(1-\sigma(t))} \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^\infty(\Omega)}^2 dt dx.$$

We define

$$C_\lambda := \int_0^{2T} e^{-2\lambda(1-\sigma(t))} \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^\infty(\Omega)}^2 dt.$$

We prove that $C_\lambda \xrightarrow{\lambda \rightarrow +\infty} 0$. Let

$$\forall t \in]0, 2T[, \quad \mu_\lambda(t) := e^{-2\lambda(1-\sigma(t))} \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^\infty(\Omega)}^2.$$

Since $1 - \sigma(t) > 0$ for $t \in]0, 2T[\setminus \{T\}$, then for almost every $t \in]0, 2T[$, $\mu_\lambda(t) \xrightarrow{\lambda \rightarrow +\infty} 0$.

Furthermore, for $\lambda \geq \lambda_*$ and $t \in]0, 2T[$

$$|\mu_\lambda(t)| \leq \left\| \frac{\partial g}{\partial t}(t) \right\|_{L^\infty(\Omega)}^2,$$

which is in $L^1(]0, 2T[)$.

Thus, from Lebesgue's Dominated Convergence Theorem, it yields $C_\lambda \xrightarrow{\lambda \rightarrow +\infty} 0$.

As a consequence, there exists $\lambda_1 \geq \lambda_*$ such that for all $\lambda \geq \lambda_1$,

$$0 \leq (C_4 + C_3 C_5) C_\lambda \leq \frac{1}{2}.$$

Thus, coming back to the inequality, it yields

$$\begin{aligned} \frac{1}{2} \|e^{\lambda\rho} f\|_{L^2(\Omega)}^2 &\leq 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 2\lambda^3 C_5 (C_3 e^{2\lambda\|\varphi\|_\infty} + 8T e^{2\lambda\kappa}) \int_0^{2T} \int_{\partial\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 \right) d\sigma(x) dt. \end{aligned}$$

Then $e^{\lambda\rho(x)} \geq e^\lambda$, so it becomes

$$\begin{aligned} \frac{e^{2\lambda}}{2} \|f\|_{L^2(\Omega)}^2 &\leq 2C_2 C_5 \lambda^3 e^{2\lambda\kappa} \|f\|_{L^2(\Omega)}^2 \\ &\quad + 2\lambda^3 C_5 (C_3 e^{2\lambda\|\varphi\|_\infty} + 8T e^{2\lambda\kappa}) \int_0^{2T} \int_{\partial\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 \right) d\sigma(x) dt. \end{aligned}$$

We group the left-hand side with the first term from the right-hand side. We note that the coefficient is

$$e^{2\lambda} \left(\frac{1}{2} - 2C_2 C_5 \lambda^3 e^{-2\lambda(1-\kappa)} \right).$$

Since $\kappa < 1$, the second term in the parentheses tends to 0 when $\lambda \rightarrow +\infty$. Thus, there exists $\lambda_2 \geq \lambda_1$ such that for all $\lambda \geq \lambda_2$

$$e^{2\lambda} \left(\frac{1}{2} - 2C_2 C_5 \lambda^3 e^{-2\lambda(1-\kappa)} \right) \geq \frac{e^{2\lambda}}{3}.$$

Then the inequality becomes

$$\frac{e^{2\lambda}}{3} \|f\|_{L^2(\Omega)}^2 \leq 2\lambda^3 C_5 \left(C_3 e^{2\lambda\|\varphi\|_\infty} + 8T e^{2\lambda\kappa} \right) \int_0^{2T} \int_{\partial\Omega} \left(|\nabla w|^2 + \left| \frac{\partial w}{\partial t} \right|^2 + |w|^2 \right) d\sigma(x) dt.$$

To conclude the proof, we have to go back to the notations from the theorem. We remind that $w = \frac{\partial v}{\partial t}$. Thus,

$$\begin{aligned} \int_0^{2T} \int_{\partial\Omega} |w(x, t)|^2 d\sigma(x) dt &= \int_0^T \int_{\partial\Omega} \left| \frac{\partial U - U_*}{\partial t}(x, t) \right|^2 d\sigma(x) dt + \int_{-T}^0 \int_{\partial\Omega} \left| \frac{\partial U - U_*}{\partial t}(x, t) \right|^2 d\sigma(x) dt \\ &= \int_0^T \int_{\partial\Omega} \left| \frac{\partial u - u_*}{\partial t}(x, t) \right|^2 d\sigma(x) dt + \int_{-T}^0 \int_{\partial\Omega} \left| -\frac{\partial u - u_*}{\partial t}(x, -t) \right|^2 d\sigma(x) dt \\ &= 2 \int_0^T \int_{\partial\Omega} \left| \frac{\partial u}{\partial t}(x, t) - \frac{\partial u_*}{\partial t}(x, t) \right|^2 d\sigma(x) dt \\ &= 2 \left\| \frac{\partial u}{\partial t} - \frac{\partial u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)}^2. \end{aligned}$$

So, it yields

$$\begin{aligned} \|f\|_{L^2(\Omega)}^2 &\leq 12\lambda^3 C_5 \left(C_3 e^{2\lambda(\|\varphi\|_\infty - 1)} + 8T e^{2\lambda(\kappa - 1)} \right) \left(\left\| \frac{\partial u}{\partial t} - \frac{\partial u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 \right. \\ &\quad \left. + \left\| \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u_*}{\partial t^2} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 + \left\| \frac{\partial \nabla u}{\partial t} - \frac{\partial \nabla u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 \right). \end{aligned}$$

On the other hand, we have

$$\begin{aligned} \|f\|_{L^2(\Omega)}^2 &= \int_{\Omega} (c^2(x) - c_*^2(x))^2 dx \\ &= \int_{\Omega} (c(x) - c_*(x))^2 (c(x) + c_*(x))^2 dx \\ &\geq 4c_0^2 \int_{\Omega} (c(x) - c_*(x))^2 dx \end{aligned}$$

Since for $a, b, c \geq 0$, $a^2 + b^2 + c^2 \leq (a + b + c)^2$, it yields

$$\begin{aligned} \|c - c_*\|_{L^2(\Omega)}^2 &\leq 3\lambda^3 \frac{C_5}{c_0^2} \left(C_3 e^{2\lambda(\|\varphi\|_\infty - 1)} + 8T e^{2\lambda(\kappa - 1)} \right) \left(\left\| \frac{\partial u}{\partial t} - \frac{\partial u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 \right. \\ &\quad \left. + \left\| \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u_*}{\partial t^2} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 + \left\| \frac{\partial \nabla u}{\partial t} - \frac{\partial \nabla u_*}{\partial t} \right\|_{L^2(\partial\Omega \times]0, T[)}^2 \right)^2. \end{aligned}$$

Then, $C := \sqrt{3\lambda^3 \frac{C_5}{c_0^2} \left(C_3 e^{2\lambda(\|\varphi\|_\infty - 1)} + 8T e^{2\lambda(\kappa - 1)} \right)}$ is suitable for the inequality from Theorem 4. This ends the proof. $\square$

To conclude this section, we can now state that C depends on

$$\Omega, T, c_0, r_0, x_0, u_0, \varepsilon_0, \gamma, \delta, \beta, \text{ and } u_*.$$

I Calculus for the formal optimisation approach

In this appendix, we present some calculations to prove the results described in Subsection 3.2. First, we explicit the weak formulation of Problem (1) to explain the definition of $\mathcal{G}$. Then, we calculate the derivatives of $\mathcal{L}$.

I.1 Weak formulation of Problem (1)

In this subsection, we prove the weak formulation of Problem (1). The main objective is to show that if $u \in H^2(\Omega \times]0, T[)$ is solution to Problem (1) for $a \in \mathcal{C}^2(\overline{\Omega}, [1, M])$, then $\mathcal{G}(u, a) = 0$.

Firstly, if $u \in H^2(\Omega \times]0, T[)$ is solution to Problem (1), then $u \in H_u^1$ by definition of the initial condition. Furthermore, for all $\lambda \in H_\lambda^1$, integrating the wave equation, it yields

$$\int_0^T \int_\Omega a(x) \frac{\partial^2 u}{\partial t^2} \lambda \, dx \, dt - \int_0^T \int_\Omega \Delta u \lambda \, dx \, dt = 0.$$

By integration by part, we obtain

$$\int_0^T \int_\Omega a(x) \frac{\partial^2 u}{\partial t^2} \lambda \, dx \, dt = \left[\int_\Omega a(x) \frac{\partial u}{\partial t} \lambda \, dx \right]_0^T - \int_0^T \int_\Omega a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} \, dx \, dt.$$

As $\lambda(t = T) = 0$ since $\lambda \in H_\lambda^1$, we have

$$\int_0^T \int_\Omega a(x) \frac{\partial^2 u}{\partial t^2} \lambda \, dx \, dt = - \int_\Omega a(x) v_0(x) \lambda(x, 0) \, dx - \int_0^T \int_\Omega a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} \, dx \, dt.$$

Applying the Green Formula from Theorem 9 and using the boundary condition, we get

$$\int_0^T \int_\Omega \Delta u \lambda \, dx \, dt = \int_0^T \int_{\partial_1 \Omega} p(t) \lambda(x, t) \, d\sigma(x) \, dt - \int_0^T \int_\Omega \nabla u \cdot \nabla \lambda \, dx \, dt.$$

Using these last two equations, the first equation becomes

$$\begin{aligned} \int_0^T \int_\Omega \nabla u \cdot \nabla \lambda \, dx \, dt - \int_0^T \int_\Omega a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} \, dx \, dt \\ - \int_\Omega a(x) v_0(x) \lambda(x, 0) \, dx - \int_0^T \int_{\partial_1 \Omega} p(t) \lambda(x, t) \, d\sigma(x) \, dt = 0. \end{aligned} \quad (24)$$

Since (24) holds for every $\lambda \in H_\lambda^1$, it yields

$$\mathcal{G}(u, a) = 0.$$

As a consequence, $\mathcal{G}$ encodes the weak formulation of Problem (1).

I.2 Derivative of $\mathcal{L}$ according to λ and a

In this subsection, we derive the Lagrangian according to λ and a . The Fréchet derivative of $\mathcal{L}$ according to λ can be calculated as

$$\forall \lambda, \delta\lambda \in H_\lambda^1, \quad \mathcal{L}(u, a, \lambda + \delta\lambda) - \mathcal{L}(u, a, \lambda) = \mathcal{G}(u, a)(\delta\lambda).$$

Since $\mathcal{G}(u, a)$ is a continuous linear functional, then

$$\frac{\partial \mathcal{L}}{\partial \lambda}(u, a, \lambda)(\delta\lambda) = \mathcal{G}(u, a)(\delta\lambda).$$

Thus, $\frac{\partial \mathcal{L}}{\partial \lambda}(u_*, a_*, \lambda_*) = \mathcal{G}(u_*, a_*) = 0$ is the weak formulation of Problem (1). For the derivative according to a , it yields

$$\begin{aligned} \mathcal{L}(u, a + \delta a, \lambda) - \mathcal{L}(u, a, \lambda) &= \frac{\gamma}{2} \int_{\Omega} |a + \delta a - a_0|^2 dx - \frac{\gamma}{2} \int_{\Omega} |a - a_0|^2 dx \\ &\quad - \int_0^T \int_{\Omega} \delta a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt - \int_{\Omega} \delta a(x) v_0(x) \lambda(x, 0) dx \\ &= \frac{\gamma}{2} \int_{\Omega} \delta a^2 dx + \gamma \int_{\Omega} \delta a (a - a_0) dx \\ &\quad - \int_0^T \int_{\Omega} \delta a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt - \int_{\Omega} \delta a(x) v_0(x) \lambda(x, 0) dx \end{aligned}$$

By isolating the linear terms and checking that the quadratic term is a $O(\delta a^2)$, it yields to

$$\frac{\partial \mathcal{L}}{\partial a}(u, a, \lambda)(\delta a) = \gamma \int_{\Omega} \delta a (a - a_0) dx - \int_0^T \int_{\Omega} \delta a(x) \frac{\partial u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt - \int_{\Omega} \delta a(x) v_0(x) \lambda(x, 0) dx.$$

Thus the gradient according to a is

$$\nabla_a L(u, a, \lambda)(x) = \gamma(a(x) - a_0(x)) - \int_0^T \frac{\partial u}{\partial t}(x, t) \frac{\partial \lambda}{\partial t}(x, t) dt - v_0(x) \lambda(x, 0).$$

I.3 Formulation of the adjoint problem

Finally, it remains to calculate the derivative according to u . A short remark should be done concerning the space H_u^1 which is an affine space. Thus, to derive, we consider $u \in H_u^1$ and $\delta u \in H^1(\Omega \times]0, T[)$ such that $u(x, 0) = 0$ almost everywhere in Ω . As it was done for a , we calculate

$$\begin{aligned} \mathcal{L}(u + \delta u, a, \lambda) - \mathcal{L}(u, a, \lambda) &= \frac{1}{2} \int_0^T \int_{\Gamma} |u + \delta u - \tilde{u}|^2 dx dt - \frac{1}{2} \int_0^T \int_{\Gamma} |u - \tilde{u}|^2 dx dt \\ &\quad + \int_0^T \int_{\Omega} \nabla \delta u \cdot \nabla \lambda dx dt - \int_0^T \int_{\Omega} a(x) \frac{\partial \delta u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt \\ &= \frac{1}{2} \int_0^T \int_{\Gamma} \delta u^2 dx dt + \int_0^T \int_{\Gamma} \delta u (u - \tilde{u}) dx dt \\ &\quad + \int_0^T \int_{\Omega} \nabla \delta u \cdot \nabla \lambda dx dt - \int_0^T \int_{\Omega} a(x) \frac{\partial \delta u}{\partial t} \frac{\partial \lambda}{\partial t} dx dt. \end{aligned}$$

Thus, it yields

$$\frac{\partial \mathcal{L}}{\partial u}(u, a, \lambda)(\delta u) = \int_0^T \int_{\Gamma} \delta u (u - \tilde{u}) \, dx \, dt + \int_0^T \int_{\Omega} \nabla \delta u \cdot \nabla \lambda \, dx \, dt - \int_0^T \int_{\Omega} a(x) \frac{\partial \delta u}{\partial t} \frac{\partial \lambda}{\partial t} \, dx \, dt.$$

Using a comparison with the weak formulation (24) of Problem (1), we can notice that $\frac{\partial \mathcal{L}}{\partial u}(u_*, a_*, \lambda_*) = 0$ is the weak formulation of the adjoint problem

$$\begin{cases} a_*(x) \frac{\partial^2 \lambda}{\partial t^2}(x, t) - \Delta \lambda(x, t) = 0 & \text{in } \Omega \times]0, T[, \\ \lambda(x, T) = 0, \quad \frac{\partial \lambda}{\partial t}(x, T) = 0 & \text{in } \Omega, \\ \frac{\partial \lambda}{\partial n}(x, t) = \tilde{u}(x, t) - u_*(x, t) & \text{on } \Gamma \times]0, T[, \\ \frac{\partial \lambda}{\partial n}(x, t) = 0 & \text{on } \partial\Omega \setminus \Gamma \times]0, T[. \end{cases} \quad (10)$$

This concludes the proof of the calculation of formulas from Subsection 3.2.

J Attempts to improve the method of reconstruction using boundary measurements

In this appendix, we propose a further analysis of the gradient method of reconstruction using boundary measurements. First, we present a study on the selection of the optimal parameters for the reconstruction. Then, we compare the efficiency of the simple reconstruction with the conjugate gradient algorithm. Finally, we compare the reconstruction using data from the top side with the reconstruction using data on the whole boundary.

J.1 Determination of the optimal parameters

In this subsection, we analyse the selection of the parameters for the gradient method. The functions used in the algorithms to apply the gradient descent are first `gradient_1_1` which includes the four post-processing methods without the tolerance and mean post-processing and secondly `gradient_1_final` which contains all the post-processing methods except the smoothing one. These functions have several parameters that should be tested. First, we need to select the values of γ_0 and q from Algorithm 1. Then, we also have to analyse the effect of the time duration of the measurements T . We also can analyse the selection of the parameter tol from the tolerance post-processing.

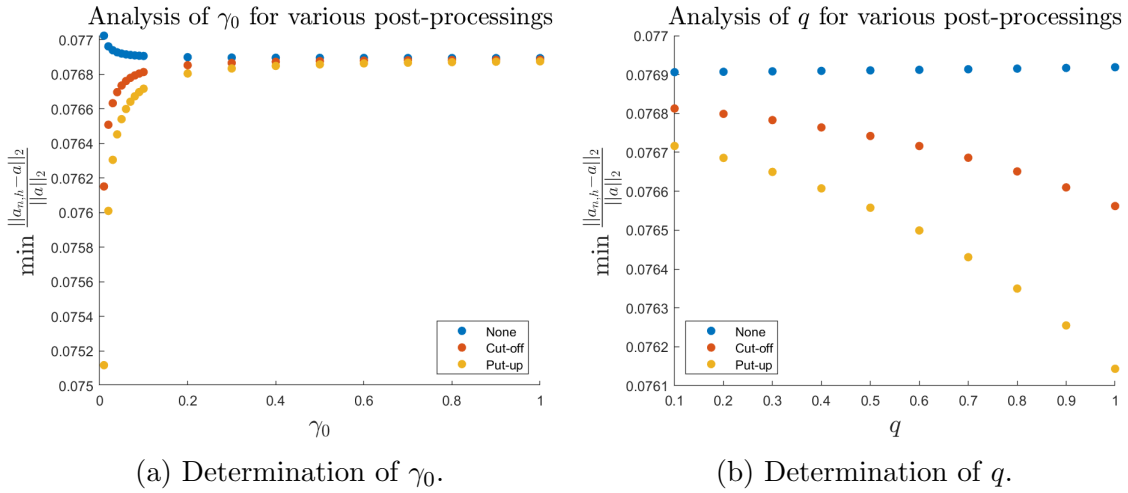


Figure 40: Analysis of the regularisation parameters for the gradient method for various post-processing methods. (a) Values of the minimal e_{quad} over the iterations for each γ_0 for various post-processing methods. (b) Values of the minimal e_{quad} over the iterations for each q for various post-processing methods.

Figure 40 presents the values of e_{quad} for various values of γ_0 and q and for the three post-processing methods (except the smoothing one). From these results, we can select the optimal parameter for each post-processing method. We note that the optimal variables for the no-post-processing method are a minimal value for q (here $q = 0.1$) and a maximal value for γ_0 (ie. $\gamma_0 = 1$) whereas it is the contrary for the other methods ($\gamma_0 = 0.01$ and $q = 1$). Thus, we select these parameters for the other figures.

Figure 41 illustrates an analysis of the effect of the value T on the reconstruction of the acoustic coefficient. We note that for the no-post-processing method, the quadratic error decreases with an increase of T as expected. On the contrary, the cut-off and put-up methods are not robust in an increase of T . We have not been able to explain this phenomenon. However, we have chosen to use only $T = 2.0$ s for the other analyses.

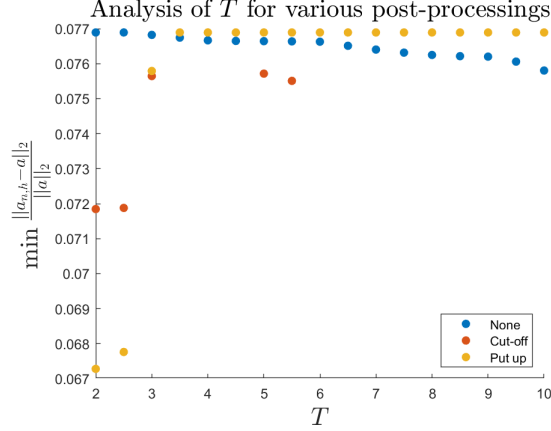


Figure 41: Analysis of the integration time for various post-processing methods. Here are the values of the minimal e_{quad} over the iterations for each integration time.

In Figure 42, we have analysed the convergence of the reconstruction method according to h . We expect a decrease of the quadratic error with h and we would like to compute the numerical order of the method. However, we note that the method does not converge with h . We observe that the reconstruction is more efficient with $h = 2^{-3}$ than for the other sizes. Furthermore, the other sizes are not in the correct order of convergence. Despite our trials, we have not been able to explain this phenomenon. However, we note that in Figures 42b and 42c, the reconstruction for $h = 2^{-5}$ and $h = 2^{-6}$ are more stable over the iterations. Thus, it seems that the reduction of h ensures a stability of the method and so a better convergence over the iterations.

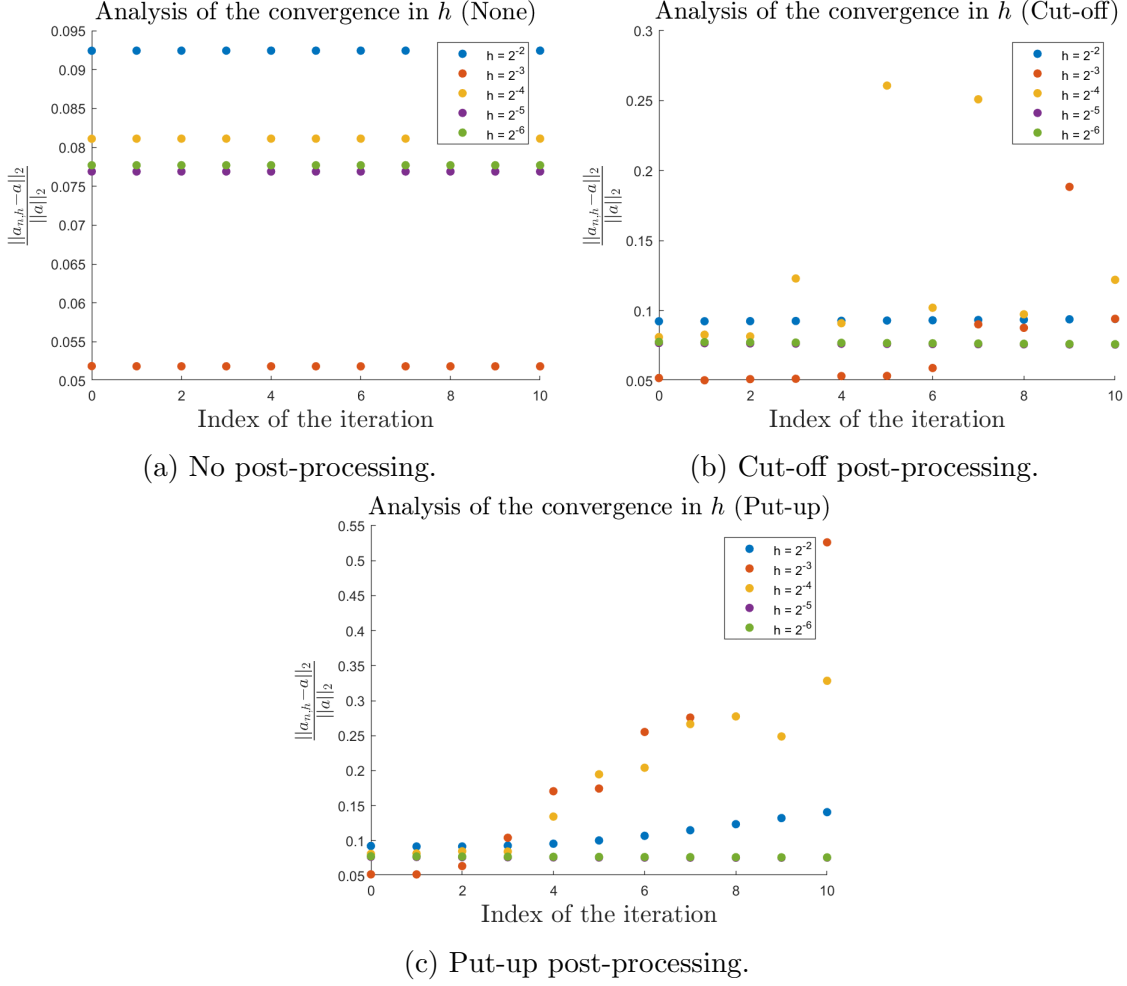


Figure 42: Analysis of the convergence according to the mesh size for various post-processing methods. (a) Values of e_{quad} over the iterations for various mesh sizes without post-processing method. (b) Values of e_{quad} over the iterations for various mesh sizes with a cut-off post-processing. (c) Values of e_{quad} over the iterations for various mesh sizes with a put-up post-processing.

Using the previous observations, we have been able to select some parameters to apply the reconstruction method on the test functions. We have described the results for Test 1 in Figure 12. Figure 43 is an additional simulation for Test 2 with the same parameters as in Figure 12. We note that we obtain qualitatively satisfying results as in Figure 12. Indeed, the methods are effective to reconstruct the two inclusions at the correct location. However, we observe some artifacts and that the inclusions are reconstructed with several peaks similarly to Figure 12.

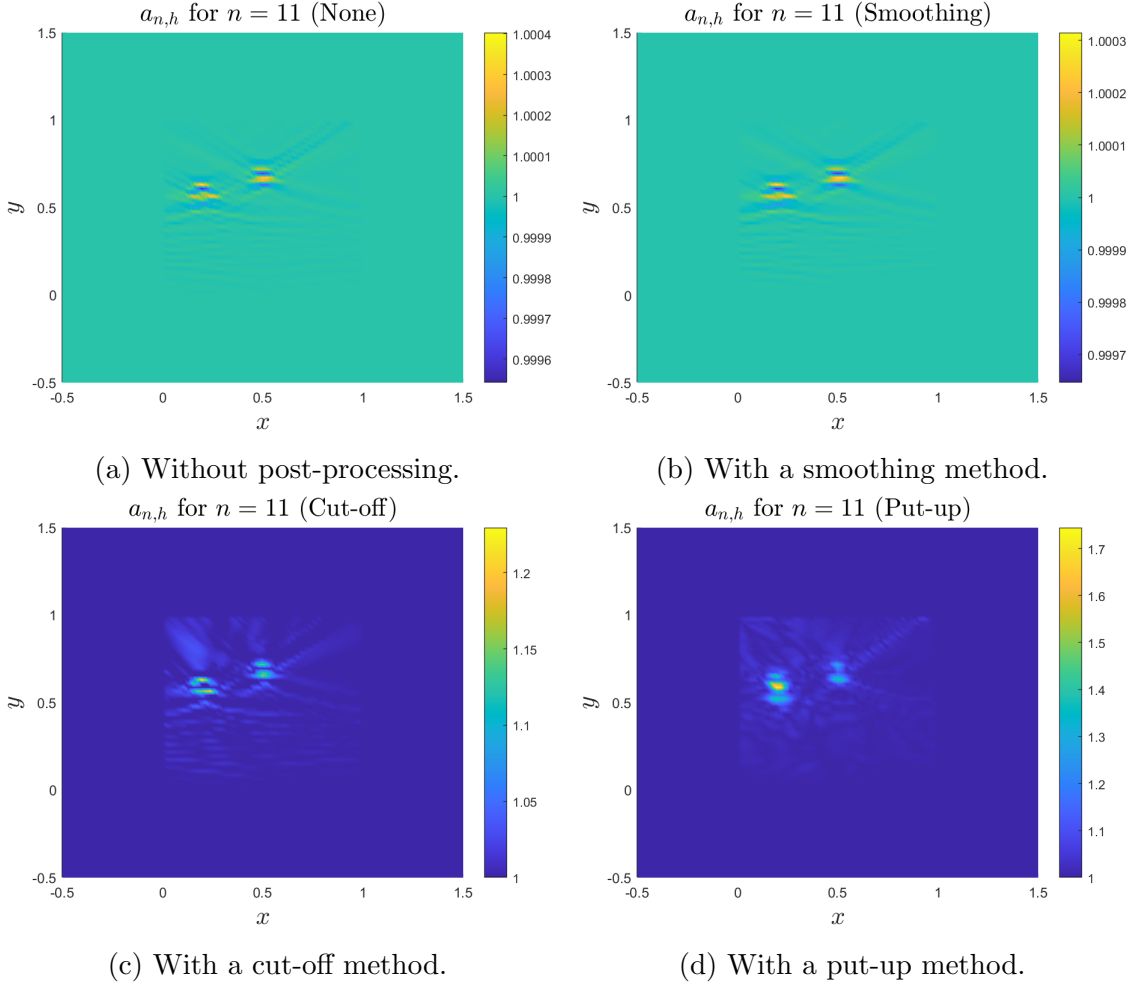


Figure 43: Reconstruction for Test 2 using data on $\partial_1\Omega$ and the gradient method with various post-processing methods. Results correspond to the 11th iteration of the gradient descent. (a) Reconstruction without post-processing. $e_{quad} = 0.137$. (b) Reconstruction with a smoothing post-processing. $e_{quad} = 0.137$. (c) Reconstruction with a cut-off post-processing. $e_{quad} = 0.131$. (d) Reconstruction with a put-up post-processing. $e_{quad} = 0.109$.

To conclude this subsection, we analyse the effect of the tolerance and mean post-processing methods. First, we determine the optimal tol for the reconstruction using the cut-off and put-up methods with or without the mean post-processing. Figure 44 shows the values of e_{quad} for the four combinations of post-processings. In this analysis, we have not been able to explain the evolution of e_{quad} with the tolerance. Thus, it seems that there is no clear tendency that comes from the effect of tol . However, we have been able to select the values for tol which give the best results. It might be useful, in a practical situation, to slowly increase the tolerance to observe the disappearance of the artifacts until a critical value in which it affects the main inclusions. Therefore, tol is a parameter that might more improve the qualitative results and not the quantitative reconstruction.

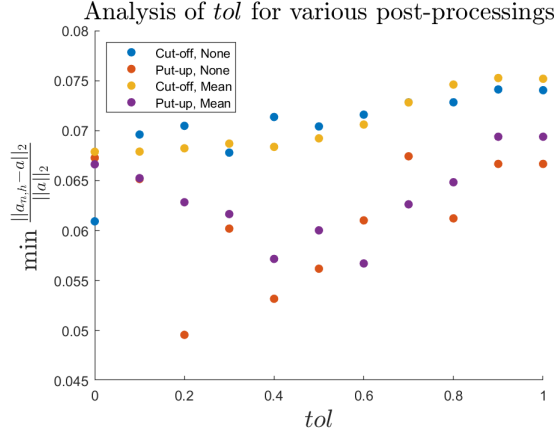


Figure 44: Analysis of the tolerance parameter for various post-processing methods. Here are the values of the minimal e_{quad} over the iterations for each tolerance parameter.

Finally, Figure 45 presents a comparison between a reconstruction with or without the mean post-processing method for the three main post-processing techniques. For each technique, we have applied the most effective tolerance parameter (with $tol = 0$ for the no-post-processing since we have not selected a convention for the values lower than 1). We note that the mean post-processing is a great improvement for the no-post-processing reconstruction. Indeed, the gradient method converges for this method whereas the error increases without the mean post-processing. A similar observation can be done for the cut-off method. The method is more stable over the iterations when a mean post-processing is applied and it seems to converge more. On the contrary, the put-up technique seems not to be improved by the mean post-processing. Despite a small improvement in stability, the error is slightly higher. However, Figure 14 shows that even if the mean post-processing does not improve in a quantitative aspect the reconstruction, in a qualitative one, it is greatly improved.

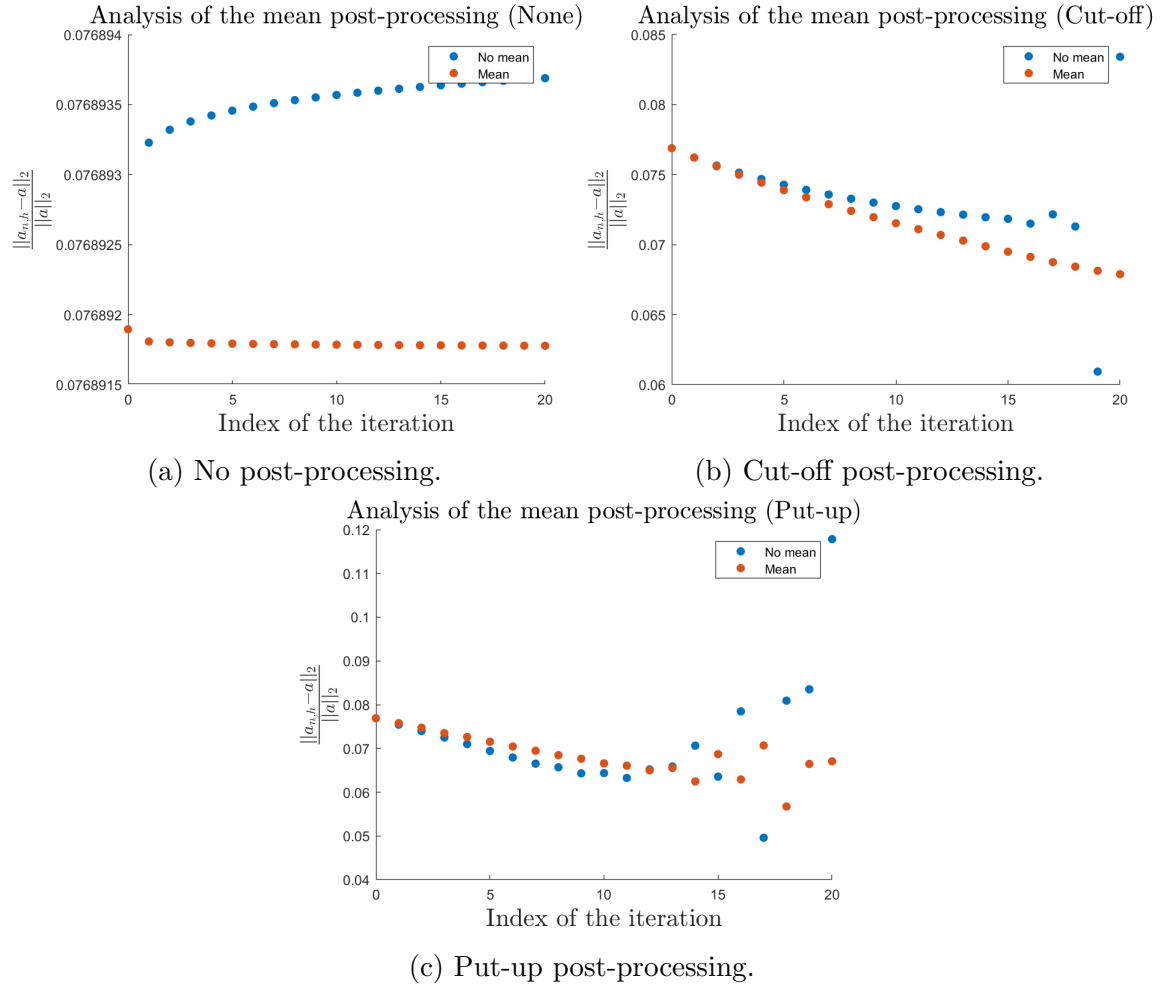


Figure 45: Analysis of the effect of the mean post-processing method for various post-processing methods. (a) Values of e_{quad} over the iterations without post-processing method. (b) Values of e_{quad} over the iterations with a cut-off post-processing. (c) Values of e_{quad} over the iterations with a put-up post-processing.

J.2 Comparison with the conjugate gradient algorithm

In this subsection, we compare the previously described gradient method with the conjugate algorithm. The principle of the conjugate gradient is similar to Algorithm 1. The only difference relies in the computation of a_{n+1} . Contrary to Algorithm 1 where a_{n+1} is compute using only the descent direction at index n , g_n , it requires here to compute another sequence $(d_n)_n$ which is the sequence of the descent direction. It is well-known that we compute

$$d_n = -g_n + \beta_n d_{n-1},$$

$$\text{where } \beta_n = \frac{\|g_n\|_2^2}{\|g_{n-1}\|_2^2}.$$

Then, we compute a_{n+1} as

$$a_{n+1} = a_n + \alpha_n d_n,$$

$$\text{where } \alpha_n = -\frac{\langle g_n | d_n \rangle_2}{\gamma_n \|d_n\|_2^2}.$$

The justification of these definition is just a well-known application of the conjugate gradient method and is also explained in [8]. Here, we do not present further explanation since the conjugate gradient method has already been largely studied in literature. As a consequence, Algorithm 1 is modified to obtain the conjugate gradient algorithm:

Algorithm 2: Conjugate gradient algorithm

Initialisation: $a_0 \equiv 1$, $d_{-1} = 0$, $g_{-1} = 1$

for $n \leftarrow 0$ **to** $N - 1$ **do**

 Solve (1) with $a_n \rightarrow u_n$,

 Solve (10) with a_n , u_n and $\tilde{u} \rightarrow \lambda_n$,

 Compute the gradient $\rightarrow g_n = \gamma_n(a_n - a_0) - \int_0^T \frac{\partial u_n}{\partial t} \frac{\partial \lambda_n}{\partial t} dt$,

 Compute the descent direction $\rightarrow d_n = -g_n + \frac{\|g_n\|_2^2}{\|g_{n-1}\|_2^2} d_{n-1}$

$a_{n+1} = a_n - \frac{\langle g_n | d_n \rangle_2}{\gamma_n \|d_n\|_2^2} d_n$.

The effectiveness of the conjugate gradient method relies on the fact that it should converge faster than the simple gradient algorithm. Figure 46 presents a comparison between the application of the simple gradient and the conjugate one. On the one hand, for the no-post-processing, we note that the conjugate gradient gives better results over the iterations than the simple method. However, the conjugate gradient does not correct the increase of the error and so the method still does not converge to the solution. In the case of the cut-off method, Figure 46b illustrates that the conjugate gradient returns less precise results on a quantitative aspect but is more stable over the iterations. Therefore, contrary to what we have expected, the conjugate gradient does not converge faster in this case but is more robust over the iterations and so we can apply it more times to get better results. Finally, the put-up method presents some surprising results. Despite a largely better quantitative reconstruction over the first iterations, the method is greatly less robust over the iterations and the artifacts grow more in the conjugate method. We have not been able to explain this

phenomenon. However, it highlights the necessity to propose and select a stopping criterion to stop the method when the error begins to grow again. In this work, we have not studied which stopping criteria could be efficient (norm of the gradient of $a_{h,n}$, norm of the relative error between $a_{h,n}$ and $a_{h,n-1}$...). This could constitute an extension of this work for further.

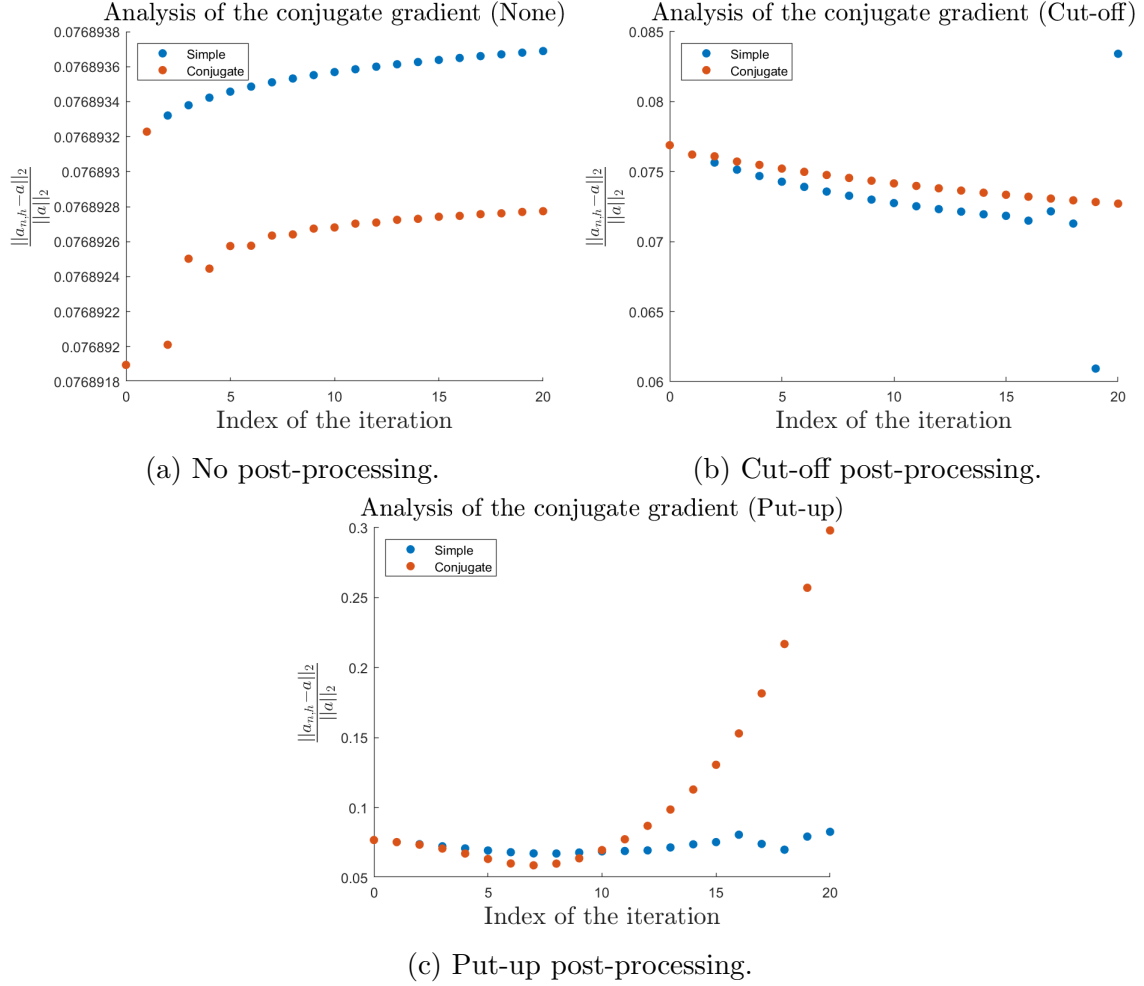


Figure 46: Comparison between the simple gradient algorithm and the conjugate gradient algorithm for various post-processing methods. (a) Values of e_{quad} over the iterations for the two gradient methods without post-processing method. (b) Values of e_{quad} over the iterations for the two gradient methods with a cut-off post-processing. (c) Values of e_{quad} over the iterations for the two gradient methods with a put-up post-processing.

Figure 47 presents some applications of the conjugate method to reconstruct the test function from Test 1. We can notice that the results obtained in Figure 47 are really similar to those obtained in Figure 12. Nonetheless, we note that these qualitatively effective reconstructions are obtained only after 7 iterations against 8 for Figure 12. Thus, the conjugate gradient method slightly reduce the required number of iterations to obtain satisfying results.

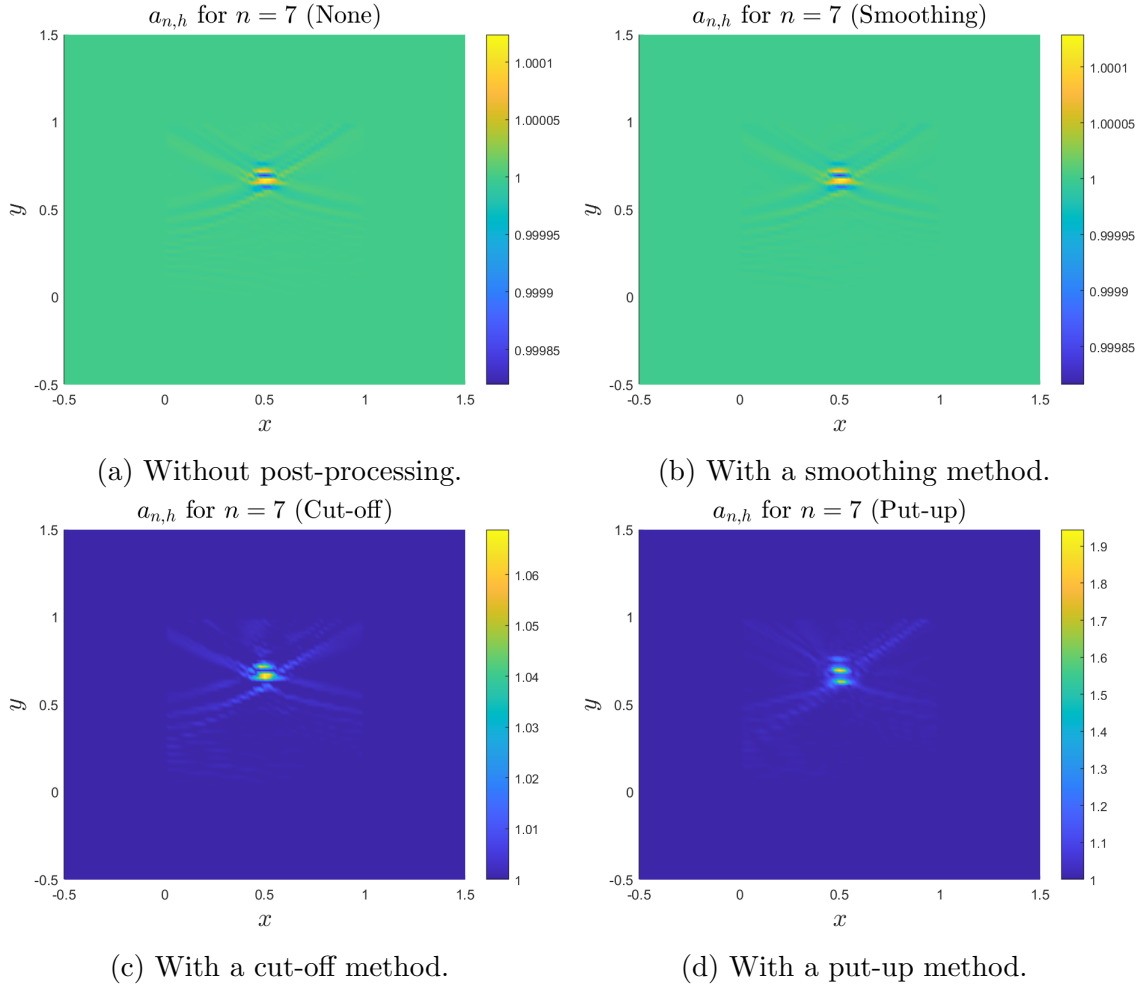


Figure 47: Reconstruction for Test 1 using data on $\partial_1\Omega$ and the conjugate gradient method with various post-processing methods. (a) Reconstruction without post-processing. $e_{quad} = 0.0769$. (b) Reconstruction with a smoothing post-processing. $e_{quad} = 0.0769$. (c) Reconstruction with a cut-off post-processing. $e_{quad} = 0.0748$. (d) Reconstruction with a put-up post-processing. $e_{quad} = 0.0588$.

J.3 Reconstruction using data on the whole boundary

In this subsection, we compare the reconstruction using data on the top side as previously described with the reconstruction using data on the whole boundary. On a theoretical aspect, the gradient descent should converge faster with data on whole $\partial\Omega$ than with data on $\partial_1\Omega$. After determining the optimal parameters for the gradient method with data on $\partial\Omega$, we have simulated some reconstructions for Test 1 with the three main post-processing methods.

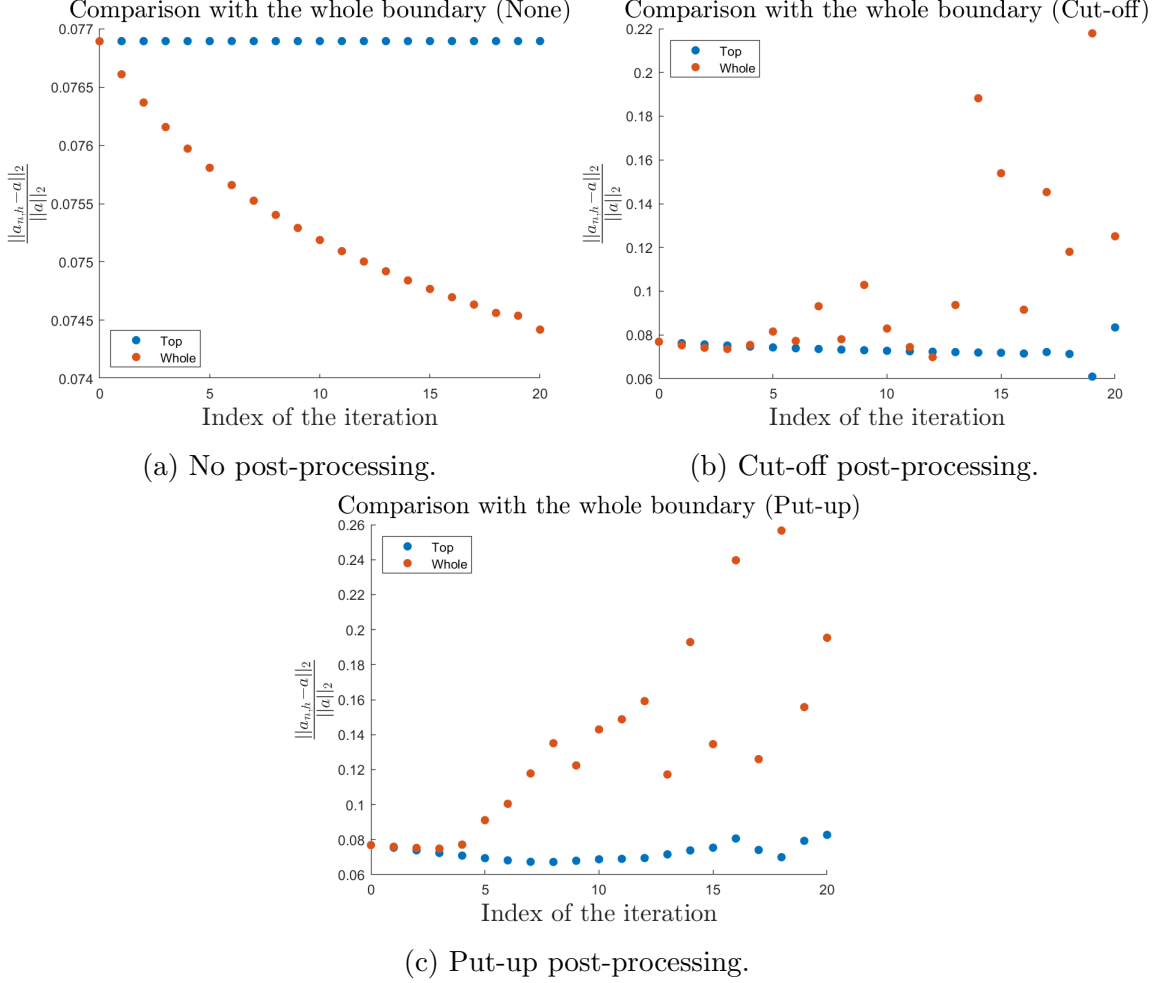


Figure 48: Comparison between the reconstruction using only measurements on the top or on the whole boundary for various post-processing methods. (a) Values of e_{quad} over the iterations without post-processing method. (b) Values of e_{quad} over the iterations with a cut-off post-processing. (c) Values of e_{quad} over the iterations with a put-up post-processing.

Figure 48 presents a quantitative comparison between the usage of data on $\partial_1\Omega$ and on $\partial\Omega$. Without post-processing method, we can note that, as expected, the gradient converge faster and the error is lower when it uses data on $\partial\Omega$. Thus, the usage of data on the whole boundary largely improve the reconstruction. However, even if on the first iterations, the error decreases slightly more, the usage of data on $\partial\Omega$ is not robust to the post-processing methods. Thus, the reconstruction creates some artifacts and thus the error skyrockets after few iterations.

Figure 49 presents some reconstructions with the various post-processing methods using data on $\partial\Omega$. On a quantitative point of view, the reconstructions in the case of no-post-processing or a smoothing, are improved to a great extent. Indeed, compared of Figure 12, the results in Figures 49a and 49b present only one inclusion and the values are mainly over 1. For the cut-off and put-up methods, we note that the reconstruction is less efficient since the height of the peak is lower even if we observe only one inclusion contrary to Figure 12. Nonetheless, despite a better shape of the inclusion, the reconstruction using data on $\partial\Omega$ presents more artifacts around the inclusion. Thus, the method might be modified in a future work to correct these artifacts.

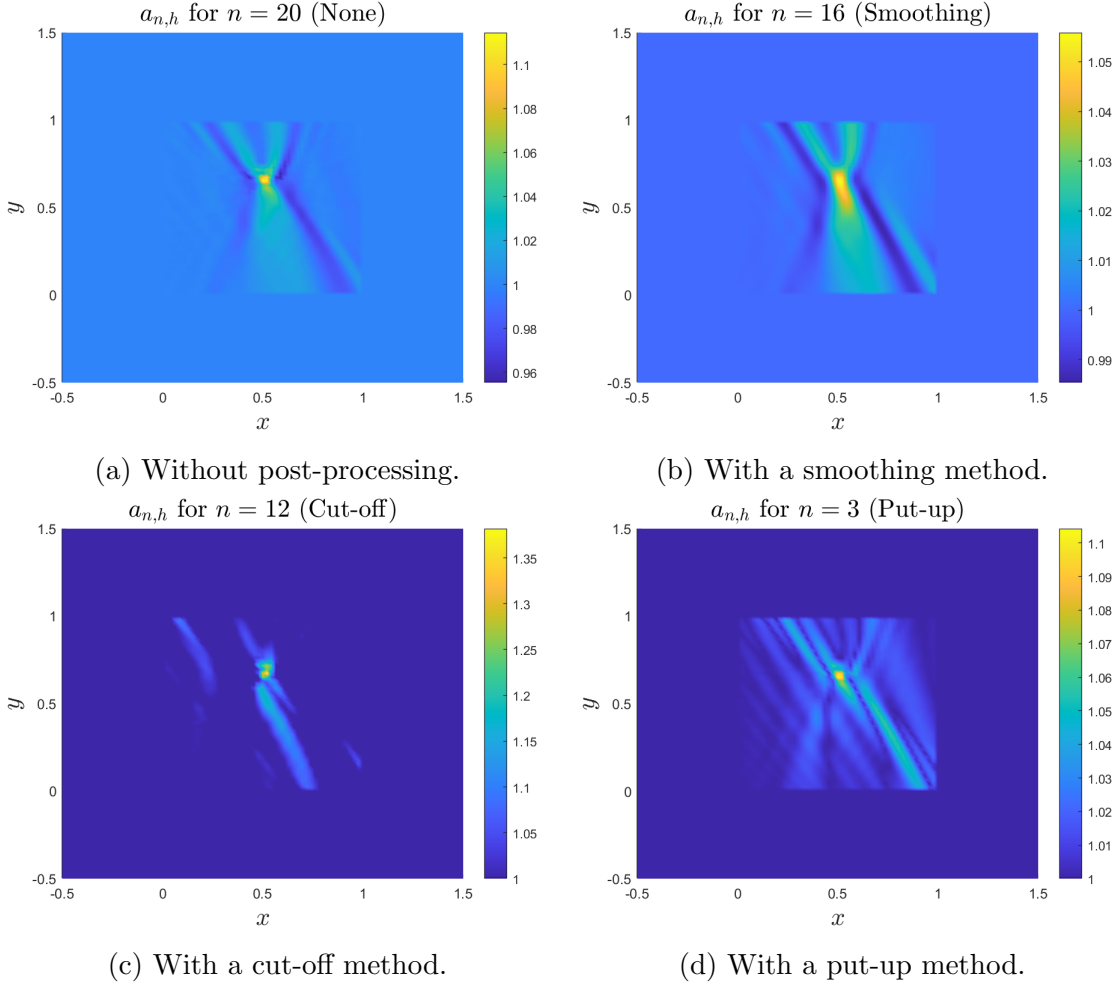


Figure 49: Reconstruction for Test 1 using data on whole $\partial\Omega$ and the gradient method with various post-processing methods. (a) Reconstruction without post-processing. $e_{quad} = 0.0744$. (b) Reconstruction with a smoothing post-processing. $e_{quad} = 0.0748$. (c) Reconstruction with a cut-off post-processing. $e_{quad} = 0.0699$. (d) Reconstruction with a put-up post-processing. $e_{quad} = 0.0749$.

K Parameters for the simulations

Figures 2 and 18:

See `Simulation_internal`.

- $a = 2a$: Test 1, $2b$: Test 2
- $h = 2^{-5}$

Figures 4, 26 and 28:

See `Analysis_simulation_1`.

- Forward solver = `forward_solver_method1`
- $a =$ Test 1
- $4b$: $s = 4.5$
- `boundary_post_processing = 4, 26b, 28`: True
- Results: $4b$: $a = 1.1503$, $b = -0.3915$

Figures 5 and 32:

See `Comparison_models`.

- $a =$ Test 1
- $h = 2^{-5}$
- $dt = 0.001$, $T = 10$
- $\delta_{Newmark} = \frac{1}{2}$, $\theta_{Newmark} = \frac{1}{2}$
- Quadrature = Trapezoidal
- `boundary_post_processing = True`

Figures 6 and 31:

See `Analysis_integration`.

- Forward solver = 31 : `forward_solver_method2_3`,
- $a =$ Test 1
- 6 : $s = 5$
- $h = 2^{-5}$

- $6b, 31$: $dt = 0.001$, $6a, 31$: $T = 10$

- $\delta_{Newmark} = \frac{1}{2}$, $\theta_{Newmark} = \frac{1}{2}$

- Quadrature = 6 : Trapezoidal

- `boundary_post_processing = True`

Figures 7, 33 and 34:

See `Simulation_internal`.

- Forward solver = 7 : `forward_solver_method2_3`, 33 : `forward_solver_method2_1`, 34 : `forward_solver_method2_2`.
- $a = 7a, 33a, 34a$: Test 1, $7b, 33b, 34b$: Test 2
- $s = 5$
- $h = 2^{-5}$
- $dt = 0.001$, $T = 10$
- $\delta_{Newmark} = \frac{1}{2}$, $\theta_{Newmark} = \frac{1}{2}$
- Quadrature = Trapezoidal
- `boundary_post_processing = True`
- Results $7a$: $e_{quad} = 0.011\,711\,810\,430\,477$, $e_{max} = 0.280\,895\,336\,090\,495$
- Results $7b$: $e_{quad} = 0.017\,945\,124\,535\,271$, $e_{max} = 0.384\,858\,684\,797\,246$
- Results $33a$: $e_{quad} = 0.010\,539\,929\,070\,140$, $e_{max} = 0.280\,965\,374\,144\,136$
- Results $33b$: $e_{quad} = 0.017\,247\,227\,806\,876$, $e_{max} = 0.385\,178\,068\,267\,210$
- Results $34a$: $e_{quad} = 0.010\,543\,511\,922\,680$, $e_{max} = 0.280\,965\,144\,893\,901$
- Results $34b$: $e_{quad} = 0.017\,249\,321\,032\,785$, $e_{max} = 0.385\,176\,278\,790\,350$

Figures 8 and 9:

See Blind_solver_v2.

- Forward solver =
9: forward_solver_method2_1
- $a = \text{Test 1}$
- $s_* = 10.2$
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{\text{Newmark}} = \frac{1}{2}, \theta_{\text{Newmark}} = \frac{1}{2}$
- $s_{\text{inf}} = 5, s_{\text{sup}} = 20, s_{\text{step}} = 0.1$
- $tol = 0.001$
- Quadrature = Trapezoidal
- boundary_post_processing = True

Figures 10 and 35:

See Blind_solver_v2.

- Forward solver =
forward_solver_method2_3
- $a = 10a, 35: \text{Test 1}, 10b: \text{Test 2}$
- $s_* = 10.2$
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{\text{Newmark}} = \frac{1}{2}, \theta_{\text{Newmark}} = \frac{1}{2}$
- $s_{\text{inf}} = 5, s_{\text{sup}} = 20, s_{\text{step}} = 0.5$
- $tol = 0.001$
- blind_inverse_solver =
10: Time, Absorbing
35a: Time, Mixed
35b: Not in time
- Quadrature = Trapezoidal

- boundary_post_processing = True
- Results 10a: $s_{\text{reconstructed}} = 10,$
 $e_{\text{quad}} = 0.041\,558\,760\,788\,021,$
 $e_{\text{max}} = 0.415\,433\,386\,020\,394$
- Results 10b: $s_{\text{reconstructed}} = 10,$
 $e_{\text{quad}} = 0.045\,289\,605\,414\,011,$
 $e_{\text{max}} = 0.587\,978\,680\,145\,733$
- Results 35a: $s_{\text{reconstructed}} = 10.5,$
 $e_{\text{quad}} = 0.055\,030\,560\,536\,815,$
 $e_{\text{max}} = 0.189\,584\,125\,402\,717$
- Results 35b: $s_{\text{reconstructed}} = 11,$
 $e_{\text{quad}} = 0.134\,429\,356\,374\,043,$
 $e_{\text{max}} = 0.317\,670\,575\,459\,823$

Figures 11, 38 and 39:

See Noise_v2.

- Forward solver = forward_solver_method2_3
- $a = 11, 38: \text{Test 1}, 39: \text{Test 2}$
- $s = 5$
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{\text{Newmark}} = \frac{1}{2}, \theta_{\text{Newmark}} = \frac{1}{2}$
- $\delta_{\text{noise}} = 3\%, \text{seed} = 1$
- Quadrature = Trapezoidal
- boundary_post_processing = True
- Results 11a: $e_{\text{quad}} = 0.011\,711\,810\,430\,477$
- Results 11b: $e_{\text{quad}} = 7.566\,500\,234\,004\,168$
- Results 11c: $e_{\text{quad}} = 0.073\,133\,950\,589\,616$
- Results 11d: $e_{\text{quad}} = 0.072\,683\,693\,732\,009$
- Results 38a: $e_{\text{quad}} = 1.583\,047\,725\,992\,278$
- Results 38b: $e_{\text{quad}} = 1.435\,625\,355\,590\,035$
- Results 39a: $e_{\text{quad}} = 0.017\,945\,124\,535\,271$

- Results 39b: $e_{quad} = 7.309\,829\,764\,803\,657$
- Results 39c: $e_{quad} = 0.129\,477\,239\,778\,562$
- Results 39d: $e_{quad} = 0.129\,464\,315\,676\,817$

Figures 12, 13 and 43:

See Simulation_boundary.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
Smoothing: gradient_1_1
None, Cut-off, Put-up: gradient_1_final
- $a = 12, 13a$: Test 1, 43, 13b: Test 2
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- Index of the iteration = 12: 8, 43: 11
- $\gamma_0 = \text{None}$, Smoothing: 1,
Cut-off, Put-up: 0.01.
- $q = \text{None}$, Smoothing: 0.1,
Cut-off, Put-up: 1.
- Results 12a: $e_{quad} = 0.076\,893\,532\,404\,584$
- Results 12b: $e_{quad} = 0.076\,892\,206\,244\,712$
- Results 12c: $e_{quad} = 0.073\,279\,227\,290\,124$
- Results 12d: $e_{quad} = 0.067\,280\,234\,927\,141$
- Results 43a: $e_{quad} = 0.136\,514\,733\,588\,040$
- Results 43b: $e_{quad} = 0.136\,514\,126\,906\,965$
- Results 43c: $e_{quad} = 0.131\,142\,481\,386\,510$
- Results 43d: $e_{quad} = 0.108\,985\,369\,347\,734$

Figure 14:

See Simulation_boundary.

- Forward solver =
forward_solver_method2_1
- Gradient solver = gradient_1_final
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = 0.01$
- $q = 1$
- $tol = \text{Cut-off: 0, Put-up: 0.6.}$
- Mean = True
- Results 14a: $e_{quad} = 0.070\,696\,103\,779\,676$
- Results 14b: $e_{quad} = 0.064\,992\,218\,016\,234$

Figure 15:

See Simulation_boundary.

- Forward solver =
forward_solver_method2_1
- Gradient solver = adaptive_gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-5}$ for the generation of the measurements
- $h = 2^{-3}$ for the initialisation
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20

- $\gamma_0 = 0.01$
- $q = 1$
- $tol = 0.95$

Figures 19, 20, 21, 22, 23 and 24:
See Simulation_internal.

- Forward solver =
19, 20: forward_solver_method2_1
21, 22: forward_solver_method2_2
23, 24: forward_solver_method2_3
- $a =$ 19, 21, 23: Test 1,
20, 22, 24: Test 2
- $h = 2^{-5}$
- $dt = 0.001, T = 10, t_1 = \frac{2\pi}{\omega}$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$

Figures 25 and 27:
See Simulation_Method_1.

- Forward solver = forward_solver_method1
- $a =$ 25a, 27a, 27b: Test 1, 25b: Test 2
- $s = 5$
- $h = 2^{-5}$
- boundary_post_processing =
25a, 25a, 27b: True
- Results 25a: $e_{quad} = 0.010\,540\,534\,046\,447$,
 $e_{max} = 0.280\,999\,165\,086\,733$
- Results 25b: $e_{quad} = 0.017\,248\,358\,395\,462$,
 $e_{max} = 0.385\,221\,070\,018\,961$
- Results 27a: $e_{quad} = 0.043\,773\,086\,131\,826$,
 $e_{max} = 1.208\,002\,656\,435\,447$
- Results 27b: $e_{quad} = 0.010\,540\,534\,046\,447$,
 $e_{max} = 0.280\,999\,165\,086\,733$

Figures 29 and 30:
See Analysis_theta.

- Forward solver =
29, 30a: forward_solver_method2_1
30b: forward_solver_method2_2
30c: forward_solver_method2_3
- $a =$ Test 1
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{Newmark} = \frac{1}{2}$
- Quadrature = Trapezoidal
- boundary_post_processing = True

Figures 36 and 37:
See Blind_solver_v1.

- Forward solver = forward_solver_method1
- $a =$ Test 1
- $s_* = 10.2$
- $h = 2^{-5}$
- $s_{inf} = 5, s_{sup} = 20, s_{step} = 0.1$
- $tol = 0.001$
- boundary_post_processing = True
- Results 36a: $\alpha = 0.654\,412$
- Results 37a: $\alpha = 0.465\,686$

Figure 40:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{40b: None: 1,}$
Cut-off, Put-up: 0.01
- $q = \text{40a: 1}$
- Whole boundary = False

Figure 41:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001$
- $T = 10$ for the measurements
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None: 1,}$
Cut-off, Put-up: 0.01
- $q = \text{None: 0.1,}$
Cut-off, Put-up: 1
- Whole boundary = False

Figure 42:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-6}$ for the measurements
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 10
- $\gamma_0 = \text{None: 1,}$
Cut-off, Put-up: 0.01
- $q = \text{None: 0.1,}$
Cut-off, Put-up: 1
- Whole boundary = False

Figure 44:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_final
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None: 1,}$
Cut-off, Put-up: 0.01
- $q = \text{None: 0.1,}$
Cut-off, Put-up: 1
- Whole boundary = False

Figure 45:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_final
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 10$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None: 1},$
Cut-off, Put-up: 0.01
- $q = \text{None: 0.1},$
Cut-off, Put-up: 1
- $tol = \text{None}, \text{Cut-off: 0},$
Put-up (no mean): 0.2
Put-up (mean): 0.6
- Whole boundary = False

Figure 46:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_1 for Simple
conjugate_gradient_1_1 for
Conjugate
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$

- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$

- Number of iterations = 20

- $\gamma_0 = \text{None: 1},$
Cut-off, Put-up: 0.01

- $q = \text{None: 0.1},$
Cut-off, Put-up: 1

- Whole boundary = False

Figure 47:

See Simulation_boundary.

- Forward solver =
forward_solver_method2_1
- Gradient solver = conjugate_gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None}, \text{Smoothing: 1},$
Cut-off, Put-up: 0.01.
- $q = \text{None}, \text{Smoothing: 0.1},$
Cut-off, Put-up: 1.
- Results 47a: $e_{quad} = 0.076\,892\,634\,833\,052$
- Results 47b: $e_{quad} = 0.076\,892\,592\,606\,917$
- Results 47c: $e_{quad} = 0.074\,762\,848\,498\,662$
- Results 47d: $e_{quad} = 0.058\,823\,319\,591\,934$

Figure 48:

See Analysis_gradient.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
gradient_1_1
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None}$: 1,
Cut-off, Put-up: 0.01,
None whole boundary: 0.3,
Cut-off whole boundary: 0.04,
Put-up whole boundary: 0.08.
- $q = \text{None}$, whole boundary: 0.1,
Cut-off, Put-up: 1

Figure 49:

See Simulation_boundary.

- Forward solver =
forward_solver_method2_1
- Gradient solver =
Smoothing: gradient_1_1
None, Cut-off, Put-up: gradient_1_final
- $a = \text{Test 1}$
- $h = 2^{-5}$
- $dt = 0.001, T = 2$
- $\delta_{Newmark} = \frac{1}{2}, \theta_{Newmark} = \frac{1}{2}$
- Number of iterations = 20
- $\gamma_0 = \text{None}$, Smoothing: 0.3,
Cut-off: 0.04,
Put-up: 0.08.
- $q = 1$
- Results 49a: $e_{quad} = 0.074\,420\,441\,748\,462$
- Results 49b: $e_{quad} = 0.074\,801\,595\,962\,756$
- Results 49c: $e_{quad} = 0.069\,863\,170\,786\,357$
- Results 49d: $e_{quad} = 0.074\,895\,743\,473\,167$